BLOCK AVERAGING AND GAUGE GEOMETRY FOR FOUR-DIMENSIONAL LATTICE YANG–MILLS THEORY

Abstract. We construct the centered contour average of SU(2) connections on periodic four-dimensional lattices. A positive flux formula gives a Wilson-energy estimate whose leading constant is one. We also give the local gauge coordinates and the differentiated averages, with bounds uniform in the number of scales on specified small-field balls. A transverse choice of fine links yields analytic coordinates and a Haar density on a fixed-average fibre.

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1. Introduction

A four-dimensional lattice SU(2) connection admits a local, gauge-covariant block average. In small curvature its Wilson energy increases by at most a factor $1+O_L(\varepsilon)$ in one step, where ε is the largest plaquette angle. Small-field coordinates give controlled derivatives of the iterated average and an analytic parametrization of the local fixed-average fibre.

See propositions 3.2 and 6.1 and theorems A and B for the precise statements.

The average of the matrix entries of several transports generally leaves the group. The contour construction replaces this operation by two averages of logarithms. The first identifies the frame at each site of a block. The second compares each fine transport across the two blocks with the central straight transport. This is the centered prescription of Bałaban [Ba87, pp. 252–253, eqs. (0.10)–(0.12)]. His earlier work develops small-curvature estimates, straightened coordinates and axial gauges for a single-comb, corner-based prescription [Ba85a, §§B–F][Ba85b, §B]. The proofs below establish the estimates for the centered prescription.

The results concern finite averaging maps, their local coordinates, differentials and measures. Constructing a constrained critical background also requires inverse estimates and a nonlinear field equation, beyond the local fibre parametrization proved here.

At the identity the endpoint terms in the link differential are a coarse gradient. The curl therefore averages the fluxes through translated squares with positive coefficients. The row and column totals of this averaging matrix have product one in dimension four. A separate local tree gauge for each square lets the same estimate control the nonlinear field. Straightening removes the endpoint terms

before the link differential is iterated. For a fixed average we instead use the plain map and solve for one fine link per coarse constraint. These are the two coordinate constructions used below.

2. Lattices and conventions

Fix an odd integer $L \geq 3$ and put $m = (L-1)/2$. The fine lattice at spacing $h > 0$ is $\Lambda_h = h\mathbb{Z}^4/(hN_1\mathbb{Z} \times \cdots \times hN_4\mathbb{Z})$. A step is allowed when L divides each N_μ and $N_\mu/L \geq 2$. The coarse lattice Λ_{Lh} consists of the images of $Lh\mathbb{Z}^4$ in the same torus. For several steps we write $h_j = L^j h_0$, and require at least two sites in every direction on the last lattice. We use the partition

$$B(y) = y + h\{-m, \dots, m\}^4, \qquad y \in \Lambda_{Lh}.$$

The block $B^j(y)$ at depth j has L^{4j} fine sites and is formed by composing these partitions. All centered paths and boxes are formed in lattice lifts before projection. Counting a repeated occurrence in a lift means counting it with its multiplicity.

A positive link $b = (x, \mu)$ joins $b_- = x$ to $b_+ = x + he_\mu$. Positive links are indexed by their base point and direction, including on a circle with two sites. The reverse $\bar{b}$ is the same edge traversed backwards. A plaquette $p = (x, \mu, \nu)$ with $\mu < \nu$ is based at $p_- = x$ and is oriented first in direction μ, then in direction ν. Unless otherwise specified, link and plaquette sums run over these independent indices. We use $[x, x']$ for a straight segment with its indicated orientation, and d for the ordinary oriented cochain boundary sum acting on link variables.

Let $G = \mathrm{SU}(2)$, $\mathfrak{g} = \mathfrak{su}(2)$ in its anti-Hermitian convention, and $G_{\mathbb{C}} = \mathrm{SL}(2, \mathbb{C})$, with Lie algebra $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}_2(\mathbb{C})$. We denote the matrix operator norm by $|\cdot|$ or $\|\cdot\|$. On $\mathfrak{g}$ this is the Euclidean norm from $\langle X, Y\rangle = -\frac{1}{2}\operatorname{tr}(XY)$. A connection U assigns a group matrix to each positive link and has $U(\bar{b}) = U(b)^{-1}$. For a path γ, $U(\gamma)$ is the product in traversal order and equals I on the empty path. A site gauge g acts by

$$U^g(b) = g(b_-)\, U(b)\, g(b_+)^{-1}.$$

Thus $U^g(\gamma) = g(\gamma_-)U(\gamma)g(\gamma_+)^{-1}$ and $(U^g)^u = U^{ug}$. We write $\mathrm{Ad}(g)X = gXg^{-1}$. We take log near I with $\log I = 0$, using the convergent series for $|U - I| < 1$. When the branch is defined, $F_p = \log U(\partial p)$ is the unscaled plaquette curvature. A link increment A means $e^A U$ in the same unscaled convention. A potential per unit length means an increment hA. Supremum norms of fields use the matrix norm above. Counting norms use counting measure, and any lattice-volume weights are specified where used. Constants subscripted by L may change from line to line and depend only on the blocking factor. Empty sums are zero and empty products are the relevant identity. The maximum of an empty family of nonnegative bounds is zero.

The affine cube-centre lattices $h_j(\mathbb{Z} + \frac{1}{2})^4$ can also be aligned simultaneously with the lattices used here. Translation by $\frac{1}{2}h_k(1,1,1,1)$ does so at every level $0 \leq j \leq k$: $h_k/h_j = L^{k-j}$ is odd, so its half differs from $1/2$ by an integer. This is one translation of the ambient torus between the two affine lattice families.

3. The centered block average

A link average must combine transports with common endpoints. We use Bałaban's symmetric contour construction [Ba87, (0.4), (0.10)–(0.12)]. The two averages in its definition serve different purposes. A mean combines alternative paths within each block. The logarithmic average of loops then compares the paths across two blocks with a single straight transport.

3.1. **Means of nearby group elements.** The series logarithm on $\|V - I\| < 1$ takes values in $\mathfrak{g}_{\mathbb{C}}$ on $G_{\mathbb{C}}$, since $\exp(\operatorname{tr}\log V) = \det V = 1$ and the two eigenvalue logarithms have imaginary parts in $(-\pi/2, \pi/2)$. It takes values in $\mathfrak{g}$ on G by unitary diagonalization. We also use the intrinsic logarithm of $V \in G \setminus \{-I\}$, obtained by assigning the eigenvalues $e^{\pm i\theta}$, $0 \leq \theta < \pi$, the logarithms $\pm i\theta$. At $V = I$ it is zero. This logarithm is continuous and respects conjugation and inversion.

For a determinant-one two-by-two matrix X, the identity $X^{-1} - I = \operatorname{adj}(X - I)$ gives

$$\|X^{-1} - I\| = \|X - I\|. \tag{3.i}$$

Indeed the adjugate of a two-by-two matrix has the same singular values in reverse order. The series logarithm therefore admits both X and X^{-1} when either is in its norm-one disk. It satisfies $\log X^{-1} = -\log X$, by the scalar identity for the holomorphic logarithm on the right half-plane.

Lemma 3.1 (Local group mean)**.** Let $V_1, \dots, V_N \in G_{\mathbb{C}}$, $N \geq 1$, and suppose $d = \max_{i,j} \|V_i V_j^{-1} - I\| < 1/40$. There is a unique $V \in G_{\mathbb{C}}$ satisfying

$$\max_i \|V_i V^{-1} - I\| \leq 1/10, \qquad \frac{1}{N} \sum_i \log(V_i V^{-1}) = 0. \tag{3.ii}$$

The mean $V = \operatorname{av}(V_i)$ is holomorphic and symmetric, and the maximum in eq. 3.ii is at most $4d$. For $V_i \in G$ it lies in G, and uniqueness holds among real candidates with that maximum less than one. For real families the mean commutes with inversion and with simultaneous left and right multiplication by elements of G. For complex families it commutes with simultaneous left and right multiplication whenever both families have diameter less than $1/40$. At a family of identities, its derivative in left logarithmic coordinates is the arithmetic mean. ■

Proof. Put $T_i = V_i V_1^{-1}$ and write the candidate as $e^X V_1$. On $\|X\| \leq 1/8$, the map

$$X \longmapsto X + \frac{1}{N} \sum_i \log(T_i e^{-X})$$

is a contraction with constant at most $2/5$. Indeed, $\|T_i e^{-X} - I\| \leq (1 + d)e^{1/8} - 1 < 1/6$. On that ball the nonlinear part of the logarithm has Lipschitz constant at most $1/5$, by summing its series after one differentiation. The same argument for the exponential gives a Lipschitz constant $e^{1/8} - 1$ for $e^{-X} - I + X$. The linear term in the displayed map is bounded by d after cancellation. The sum of these three bounds is at most

$$d + (1 + d)(e^{1/8} - 1) + \tfrac{1}{5}(1 + d)e^{1/8} < 2/5.$$

At zero the map has norm at most $d/(1 - d)$, so it takes the ball into itself and its fixed point satisfies $\|X\| \leq 2d$. The iteration starting at zero stays in the traceless algebra and, for unitary inputs, in its skew-Hermitian real form. The fixed point is holomorphic by the implicit function theorem, since the derivative of the fixed point map has norm less than one. Its relative differences are at most $e^{3d} - 1 \leq 4d$. A candidate within $1/10$ of the first family member in the relative norm has $X = -\log(V_1 V^{-1})$ of norm less than $1/8$, so is the same fixed point. This uniqueness also gives symmetry under permutations.

For real uniqueness on the larger ball, identify G with the unit quaternions, whose Euclidean norm agrees with the operator norm on their matrix representatives. For any real candidate V' in that ball, write $V_i (V')^{-1} = \cos\theta_i + \sin\theta_i \, q_i$ with q_i a unit imaginary quaternion and $0 \leq \theta_i < \pi/3$. The logarithmic equation says that $\sum_i a_i V_i (V')^{-1}$ is real, where $a_i = \theta_i / \sin\theta_i > 0$ and $a_i = 1$ at zero. Its real part is positive. Thus V' is the unit normalization of the convex combination $\sum_i a_i V_i / \sum_i a_i$. The combination is within d of V_1 and normalization moves it by at most another d. This puts V' inside the smaller uniqueness ball.

If $V_i' = u V_i v$, their normalized matrices satisfy $T_i' = u T_i u^{-1}$. Start the two fixed-point iterations at zero. Each stays in its own contraction ball, so each logarithm argument stays in the norm-$1/6$ disk. Whenever the current iterates are conjugate, the logarithm and exponential series give conjugate next iterates. Passing to the limits and multiplying by the first family member proves $\operatorname{av}(V_i') = u \operatorname{av}(V_i) v$. For inversion in the real case use $V_i^{-1} V = V^{-1} (V_i V^{-1})^{-1} V$. Finally differentiation of eq. 3.ii at the constant family gives the derivative formula. □

Complex inversion is also admitted under a quantitative condition. If $d < 1/80$ and $\|V_i\|\|V_i^{-1}\| \leq 2$ for every i, the inverse family has diameter at most $2d < 1/40$, and

$$\operatorname{av}(V_i^{-1}) = \operatorname{av}(V_i)^{-1}.$$

Indeed, if $V = \operatorname{av}(V_i)$, then $\|V_i^{-1}V - I\| \leq 2\|VV_i^{-1} - I\| \leq 8d < 1/10$. Equation 3.i gives the second inequality from the mean bound. Moreover $V_i^{-1}V = V^{-1}(V_iV^{-1})^{-1}V$. The logarithms therefore sum to zero, and uniqueness applies to the inverse family.

At any constant family $V_i = V_0$, write the inputs as $e^{A_i}V_0$. On a universal small supremum ball their mean is $e^M V_0$, where

$$\left\| M - N^{-1}\sum_i A_i \right\| \leq C \max_i \|A_i\|^2. \tag{3.iii}$$

The constant is independent of N. The contraction proof bounds M uniformly on a larger supremum ball. Cauchy's formula in two scalar directions gives a uniform second derivative on the smaller ball; the computed first derivative and Taylor's formula give the estimate.

3.2. **Contours and the three branches.** For $x \in B(y)$, let $\mathcal{G}(y,x)$ be the paths that complete all steps in one coordinate before starting the next, using each order of the nonzero coordinates once. The empty path is used when $x = y$. Define

$$T_U(y,x) = \operatorname{av}\big(U(\gamma) : \gamma \in \mathcal{G}(y,x)\big), \qquad T_U(x,y) = T_U(y,x)^{-1},$$

where the mean is defined. For a coarse link c write $D_c = B(c_-) \cup B(c_+)$, $x' = x + c_+ - c_-$ for $x \in B(c_-)$, and $S_c = U([c_-, c_+])$. On the torus the displacement in these expressions is the directed displacement of the link. All path links are in the rectangular lift of D_c. The first real branch applies if every physical fine plaquette with all its vertices in the projected D_c has defect $\|U(\partial p) - I\| < 1/(200L^2)$. At the shortest period this test can include plaquettes outside the rectangular graph. Define the analytic path formula on positive links by

$$\begin{aligned} Z_x &= T_U(c_-, x)\, U([x,x'])\, T_U(c_+, x')^{-1} S_c^{-1}, \\ \mathcal{M}_{\mathrm{an}}(U)(c) &= \exp\left(L^{-4} \sum_{x \in B(c_-)} \log Z_x \right) S_c. \end{aligned} \tag{3.iv}$$

Its row chart requires contour diameters less than $1/40$ and $\max_x \|Z_x - I\| < 1$. The first branch of the real map $\mathcal{M}$ uses $\mathcal{M}_{\mathrm{an}}$. Outside the first real branch, let

$$Z_{x,\gamma,\gamma'} = U(\gamma)U([x,x'])U(\gamma')^{-1}S_c^{-1}, \qquad (\gamma,\gamma') \in \mathcal{G}(c_-,x) \times \mathcal{G}(c_+,x').$$

If all of these elements differ from $-I$, define the real value $\mathcal{M}(U)(c)$ by replacing each $\log Z_x$ on the right side of eq. 3.iv by the arithmetic mean of the intrinsic logarithms of $Z_{x,\gamma,\gamma'}$ for that x. If any equals $-I$, use $\mathcal{M}(U)(c) = S_c$. These two rules define $\mathcal{M}$ outside its first real branch; they do not change $\mathcal{M}_{\mathrm{an}}$. A conjugation-equivariant traceless logarithm at $-I$ cannot be assigned: it would commute with all of G, hence be scalar and traceless, with exponential I.

Superscript j denotes composition through j successive lattices for each map. The analytic iterate is defined where its successive row charts are admitted. An individual row is defined locally if the charts in its backward path dependence are admitted, without conditions on unused rows. The zeroth iterates are the identity.

Proposition 3.2 (Properties of the centered map)**.** On the periodic lattices of Section 2, the three real rules define a Borel map. The value and branch on c depend only on the fine field on its projected double block. For real configurations the path formula on the reverse link gives the inverse value. For a real site gauge g,

$$\mathcal{M}(U^g)(c) = g(c_-)\mathcal{M}(U)(c)g(c_+)^{-1}.$$

The map commutes with translations and signed coordinate permutations preserving the coarse lattice and its centered blocks. Each real iterate is onto. A row of $\mathcal{M}^j$ reads only the two depth-j blocks of its endpoints.

On the row charts of eq. 3.iv, $\mathcal{M}_{\mathrm{an}}$ is holomorphic, including at real inputs outside the first real branch. Extend its positive-link values to reverse links by inversion. If the reverse path formula is admitted in its own chart, it gives the same value. The endpoint covariance identity for $\mathcal{M}_{\mathrm{an}}$ holds for complex site gauges whenever both the original and transformed fields satisfy the row-chart bounds. ■

Proof. An interchange of two successive steps in distinct coordinate directions multiplies the path transport on the left by a conjugate of an oriented plaquette. Steps in the same coordinate commute as paths, with backtracks cancelled. For unitary transports the norm of the change is bounded by the plaquette's distance from the identity. A permutation of the steps in a contour takes at most $2L^2$ adjacent interchanges, because its length is at most $2(L-1)$. Hence, writing ε for the maximum plaquette distance on D_c, each contour diameter is at most $2L^2\varepsilon < 1/100$ in the first branch.

Choose a contour γ and its translate γ'. Moving the L steps of $[x, x']$ to the front of the concatenated path takes at most $2L(L-1)$ interchanges. All interchanges stay within the double block. It follows that $\|Z_{x,\gamma,\gamma'} - I\| \le 2L^2\varepsilon$. Replacing each contour by its mean costs at most $4 \cdot 2L^2\varepsilon$ by lemma 3.1 and unitarity. Thus $\|Z_x - I\| \le 18L^2\varepsilon < 1/10$, which admits the logarithm and gives $\|\mathcal{M}(U)(c)S_c^{-1} - I\| \le CL^2\varepsilon$. All expressions within a branch are continuous on its domain, and its domain is Borel by the displayed branch tests. The series and the mean lemma give holomorphy of $\mathcal{M}_{\mathrm{an}}$ on its stated chart.

Gauge transformation conjugates the loop by the transformation at c_-. The open contour means transform at their two endpoints by lemma 3.1. In the complex case a conjugation by g increases a norm bound by at most $\|g\|\|g^{-1}\|$; the series transforms termwise as soon as the transformed arguments lie in the chart. Real endpoint conjugation preserves every branch test. Reversing a link exchanges x and x' and replaces each loop Z by $S_c^{-1}Z^{-1}S_c$. For a real field this preserves the norm test and the exclusion of $-I$. Its logarithm is $-\operatorname{Ad}(S_c^{-1}) \log Z$, which gives the inverse coarse transport. The last branch uses the reversed straight transport, also the inverse. In the complex case, when both outer logarithm charts are admitted, the same logarithm identity follows from eq. 3.i and termwise similarity covariance of the series. Translations and signed coordinate permutations permute the contour families, centered blocks and plaquette tests. The real reversal identity covers those permutations which reverse a link.

The literal paths are in the rectangular double block, and the real tests use physical plaquettes with vertices in its projection. Each depth-j block is a union of the preceding blocks. Thus both types of reads stay in the projected union of the two depth-j blocks when the map is composed.

For surjectivity fix a field on level j. Assign identity to fine links inside its depth-j blocks and its prescribed link value to every directed fine crossing of the corresponding face. Use inverses on reversed links. We claim that the same description holds after each step, with fine links replaced by links of the current level $r \le j$. For $r < j$, the centered level-r blocks below level $r+1$ are contained in the top blocks. Their internal contour paths therefore have identity transport. The two endpoints of every averaging segment are in the same top blocks as the corresponding endpoints of the central segment. In the direction of the segment a top block is an interval of L^{j-r} consecutive level-r sites. The segment is monotone and has length $Lh_r \le h_j$, so it crosses at most one top face. The transverse coordinates of the segment remain in the transverse top block of the central segment. Hence every averaging segment has the same directed face transport as the central one, or they are all internal. This is a statement on the lift. At a two-site top period the two positive coarse bonds with the same endpoint set still have distinct initial sites and distinct lifted faces. Every closed loop used by either logarithmic rule is now I. Both rules give the common segment transport. This proves the induction and gives exactly the prescribed field at level j. □

For a finite measure ν on fine connections, the pushforward $\mathcal{M}_*\nu$ preserves total mass and preserves positivity. Gauge invariance also passes to this measure because every coarse gauge extends to a fine gauge and $\mathcal{M}$ is covariant.

The path count has a general form. Reordering a path of m steps by adjacent swaps takes at most $m(m-1)/2$ swaps. If every intermediate path and elementary plaquette lies in the chosen region, each swap in distinct coordinates inserts a conjugate plaquette; a swap in one coordinate leaves the transport unchanged. For a real field with plaquette defect at most ϵ, unitary telescoping therefore bounds the change in the path transport by $m(m-1)\epsilon/2$. The containment assumption prevents applying this estimate across a missing part of the carrier.

3.3. **Local comb coordinates.** A local gauge compares each edge with the coordinate paths from a root to its endpoints. The comparison uses the curvature only inside the box.

Lemma 3.3 (Comb estimate on a box)**.** Let D be a rectangle in a lift of the fine lattice, with at most s sites in each direction and a root $y \in D$. Use its literal rectangular edges and plaquettes. For a field U on these edges, let γ_x be the path from y to x in a fixed coordinate order. Put

$$g(x) = U(\gamma_x), \qquad \varepsilon = \max_{p \subset D} \|U(\partial p) - I\|,$$

with $\varepsilon = 0$ when the box has no plaquettes. If U is real, then

$$\|U^g(b) - I\| \le Cs\varepsilon \qquad (b \subset D).$$

If also $Cs\varepsilon \le 1/2$, the logarithms $a_b = \log U^g(b)$ obey the same pointwise bound up to an absolute factor and satisfy

$$\left(\sum_{b \subset D} \|a_b\|^2\right)^{1/2} \le C_s \left(\sum_{p \subset D} \|F_p\|^2\right)^{1/2}.$$

The first constant is absolute, and C_s depends only on s. For a complex field, if every path conjugating a plaquette to the root in the comb comparison has condition number at most K, then $\|U^g(b) - I\| \le \exp(CsK\varepsilon) - 1$. All conclusions concern the lift when a periodic vertex has more than one occurrence. ■

Proof. Append b to the comb to b_-. Move this one extra step to its coordinate position and cancel an opposite step if needed. This gives the comb to b_+ in at most Cs interchanges inside D. Each interchange contributes a plaquette transported to the root by a prefix path. If the plaquette has reverse orientation, write it as the inverse of the positively oriented plaquette at its standard base point. Include any change of base point in the transporting path. The product of these transported plaquette factors is $U(\gamma_{b_-})U(b)U(\gamma_{b_+})^{-1}$.

In the real case all conjugations are isometric and the factors are unitary. Telescoping the product gives $Cs\varepsilon$. In the complex case eq. 3.i handles inverse factors, so the deviation of each transported factor is at most $K\varepsilon$. Multiplying the bounds $1 + K\varepsilon$ yields $\exp(CsK\varepsilon) - 1$. The transporting paths have length $O(s)$, since each is a prefix of a reordered comb with the extra step and at most the rebase to its plaquette.

On the norm-half disk the logarithm has bounded derivative. In the real case this bounds $\|a_b\|$ by a constant times the sum of $\|F_p\|$ over its plaquette occurrences. The finite incidence matrix has row and column counts bounded in terms of s. Cauchy–Schwarz in each row followed by the column count proves the squared-sum estimate. □

3.4. **The flat derivative and its curl.** For a fine cochain a at the identity use $a(\bar{b}) = -a(b)$ and let $a(\gamma)$ denote the sum along a path. Put

$$H_a(y) = L^{-4} \sum_{x \in B(y)} \frac{1}{|\mathcal{G}(y,x)|} \sum_{\gamma \in \mathcal{G}(y,x)} a(\gamma).$$

Let Qa be the derivative of $\log \mathcal{M}_{\mathrm{an}}(e^{ta})$ at $t = 0$, with the fine and coarse perturbations measured as raw link logarithms. The linearization of the mean and the product rule give

$$(Qa)(c) = L^{-4} \sum_{x \in B(c_-)} a([x, x']) + H_a(c_-) - H_a(c_+). \tag{3.v}$$

The derivative of the central straight transport cancels against its occurrence inside the logarithm in eq. 3.iv. In particular, if all $a(b)$ are real multiples of a fixed element of $\mathfrak{g}$ and $\|a\|_\infty$ is sufficiently small depending on L, then $\mathcal{M}(e^a) = e^{Qa}$ exactly. The products commute, the small logarithms are additive, and the defining equation of the mean reduces to an arithmetic average.

Let d and d_c be the oriented boundary sums around fine and coarse plaquettes. If $C = (y, \mu, \nu)$ is a coarse plaquette with initial vertex y and oriented coordinate directions μ, ν, let $S_x(C)$ be the L-by-L fine surface parallel to C starting at $x \in B(y)$. Then

$$(\mathrm{d}_c Qa)(C) = L^{-4} \sum_{x \in B(y)} \sum_{p \in S_x(C)} (\mathrm{d}a)(p). \tag{3.vi}$$

The terms H_a cancel in the oriented boundary sum. The four straight segments left for each x bound $S_x(C)$, and its interior links cancel in the sum of fine boundaries. The resulting positive flux weights have row sum L^2. Their column sum enters the energy estimate in eq. 4.iii. After j iterations the flux weights still have nonnegative entries, with row sum L^{2j}.

4. Curvature and Wilson energy

Write $\varepsilon(U) = \max_p |F_p|$ when all the plaquette logarithms are defined in the small branch. For real fields put

$$\mathcal{E}(U) = \sum_p \left(1 - \tfrac{1}{2} \operatorname{tr} U(\partial p)\right).$$

This sum is dimensionless and counts each plaquette once. Since the eigenvalues of $F \in \mathfrak{g}$ are $\pm i|F|$, the corresponding summand is $1 - \cos|F|$. In particular, $\mathcal{E}(U)$ is half the sum of $|U(\partial p) - I|^2$ in the operator norm.

Theorem A (One-step energy estimate)**.** There are positive constants ε_L, C_L, depending only on L, with the following property. Let U be a real connection on a torus admitting one blocking step. Suppose its fine plaquette holonomies are in the series logarithm chart and $\varepsilon(U) = \max_p \|\log U(\partial p)\| \le \varepsilon_L$. Then each output link uses the first real branch, the output plaquette holonomies lie in the series logarithm chart, and

$$\varepsilon(\mathcal{M}U) \le L^2 \varepsilon(U) + C_L \varepsilon(U)^2, \tag{4.i}$$

$$\mathcal{E}(\mathcal{M}U)^{1/2} \le (1 + C_L \varepsilon(U)) \mathcal{E}(U)^{1/2}. \tag{4.ii}$$

■□

Proof. We work in fine lattice units. Let $C = (y, \mu, \nu)$ denote a coarse plaquette. The differential of its logarithmic holonomy is the curl of the flat link differential. By eq. 3.vi, this curl is a sum of fine plaquette curls with weights L^{-4} indexed by

$$(x, i, j) \in B(y) \times \{0, \dots, L-1\}^2, \qquad p = (x + ie_\mu + je_\nu, \mu, \nu).$$

Write $w_{C,p}$ for the resulting nonnegative coefficients, retaining separate occurrences when a lift uses the same physical plaquette more than once. Each coarse row has total weight L^2. For a fixed physical p, each pair (i, j) determines the fine site x and its unique block origin y. Thus each physical column has total weight L^{-2}. For any orthogonal maps $O_{C,p}$ of $\mathfrak{g}$ and any real plaquette field f, weighted Cauchy–Schwarz gives

$$\sum_C \left| \sum_p w_{C,p} O_{C,p} f_p \right|^2 \le L^2 \sum_{C,p} w_{C,p} |f_p|^2 = \sum_p |f_p|^2. \tag{4.iii}$$

The norm here comes from the inner product $-\frac{1}{2}\operatorname{tr}(XY)$ on $\mathfrak{g}$. The product of the row and column totals is one in dimension four.

To apply this calculation at U, lift the four endpoint blocks of C together to a rectangular box D_C. It has $2L$ sites in each plaquette direction and L in each transverse direction. Take the coordinate tree gauge of lemma 3.3, rooted at y, and write the links in this gauge as e^{a_b}. For a constant depending only on L,

$$\|a\|_{\infty,D_C} \le C_L\varepsilon(U), \qquad \|a\|_{2,D_C} \le C_L\|F\|_{2,D_C}.$$

The norms count the edges and plaquettes of the rectangular lift. Its vertices have distinct projections, since each fine period has at least $2L$ sites. Extend the real gauge to the other sites. This preserves the defects of every physical plaquette, including tests across an opposite face when a direction has exactly two coarse sites. The global curvature hypothesis puts all four output links in the first real branch. Thus their plaquette product is the one obtained from the analytic path formula in the literal box. On a fixed small ball of the a_b, its contour means and logarithms are analytic and its coarse plaquette product stays in the series logarithm chart. Reducing ε_L puts the actual local logarithms in that ball.

In this gauge the fine logarithms have the form $(\mathrm{d}a)_p + r_p$, where $\|r\|_{2,D_C} \le C_L\|a\|_{\infty,D_C}\|a\|_{2,D_C}$. Indeed each plaquette product has four factors and its first derivative is the oriented boundary sum. Taylor's formula bounds the error by a constant times the maximum of the four log norms times their sum. Each edge occurs in a bounded number of faces. The coarse plaquette logarithm has the flat curl as its first derivative, with Taylor error at most $C_L\|a\|_{\infty,D_C}\|a\|_{2,D_C}$, by analyticity on a fixed smaller ball of this finite box. Substituting the fine logarithms into the flat formula and conjugating back to the original frames therefore gives

$$F_C(\mathcal{M}U) = \sum_p w_{C,p}O_{C,p}F_p(U) + r_C, \qquad |r_C| \le C_L\varepsilon(U)\|F\|_{2,D_C}. \tag{4.iv}$$

For the box gauge g_C, write the coarse Taylor error in that frame as r_C^{out}, and the fine errors as r_p. Then the terms after returning to the original frame are

$$O_{C,p} = \operatorname{Ad}(g_C(y)^{-1}g_C(p_-)), \qquad r_C = \operatorname{Ad}(g_C(y)^{-1})\left(r_C^{\mathrm{out}} - \sum_p w_{C,p}r_p\right).$$

Here p_- is its occurrence in the box. Since the nonnegative row weights sum to L^2, their squared sum has square root at most L^2. Cauchy–Schwarz bounds the second error term by $L^2\|r\|_{2,D_C}$. No cancellation between different box frames is used. For each of the six coarse plaquette orientations, a given fine plaquette base can lie in at most two translated boxes along each of the two long directions and at most one along each other direction. Thus a physical fine plaquette occurs in at most 24 boxes, including periodic multiplicities.

The first inequality follows from the row total and the finite size of each box. The family of lifted boxes has bounded overlap, counted with multiplicity on the torus. Indeed each box is a translate of one of six shapes of diameter $O(L)$ by a coarse lattice vector. Consequently $\|r\|_2 \le C_L\varepsilon(U)\|F\|_2$. Using eq. 4.iii in eq. 4.iv yields $\|F(\mathcal{M}U)\|_2 \le (1 + C_L\varepsilon(U))\|F(U)\|_2$. For the input use $1 - \cos t \ge \frac{1}{2}t^2(1 - Ct^2)$ on the small interval. For the output use $1 - \cos t \le \frac{1}{2}t^2$. The resulting factor $(1 - C\varepsilon(U)^2)^{-1/2}$ is absorbed in the right side of eq. 4.ii by reducing ε_L. □

Corollary 4.1 (Iteration)**.** There is $\alpha_L > 0$ such that, for every torus admitting $k \ge 0$ steps and every real field whose fine plaquettes admit the series logarithms and satisfy $\varepsilon(U) \le \alpha L^{-2k}$ for $0 \le \alpha \le \alpha_L$, all k steps use the first real branch and

$$\varepsilon(\mathcal{M}^jU) \le 2\alpha L^{-2(k-j)}, \qquad \mathcal{E}(\mathcal{M}^jU) \le 2\mathcal{E}(U) \quad (0 \le j \le k). \tag{4.v}$$

■□

Proof. Let $\varepsilon_j = \varepsilon(\mathcal{M}^j U)$. As long as $\varepsilon_i \le 2\alpha L^{-2(k-i)}$ for $i < j$, their sum is at most $2\alpha/(L^2-1)$. Choose $2\alpha_L \le \varepsilon_L$, so that every step through j is defined. Dividing eq. 4.i by the factor L^2 at each step and using $1 + s \le e^s$ gives

$$L^{-2j}\varepsilon_j \le \varepsilon_0 \exp\left(C_L L^{-2} \sum_{i<j} \varepsilon_i\right) \le 2\alpha L^{-2k}$$

after reducing α_L. This proves the induction, including the case $\varepsilon_0 = 0$. The product of the factors in eq. 4.ii is at most $\exp(2C_L\alpha/(L^2-1))$. Choose α_L so that its square is at most two. □

Corollary 4.2 (Curvature on complete local stocks)**.** On a torus admitting k steps, let Y be a union of depth-k blocks. Let U be real, and suppose every physical fine plaquette with all its vertices in Y has its series logarithm with norm at most αL^{-2k}, where $0 \le \alpha \le \alpha_L$. For $0 \le j \le k$, the level-j plaquettes with vertices in Y have series logarithms of norm at most $2\alpha L^{-2(k-j)}$. Every level-j averaging row with endpoints in Y, $1 \le j \le k$, uses the first real branch and depends only on fine links with endpoints in Y. ■□

Proof. At every level at most k, the block of a site in Y is contained in Y by alignment. For a row whose endpoints are in Y, all its paths and all its real test plaquettes at the preceding level therefore have vertices in Y. This includes projected opposite-face tests. The rectangular lift for any plaquette with corners in Y projects into the four endpoint blocks, which also lie in Y. The pointwise argument of theorem A thus applies using only the largest defect at the preceding level on Y. At steps below k, the inductive bound keeps the real tests inside their cutoff after a fixed decrease of α_L. The same curvature recurrence and geometric sum as in corollary 4.1 give the claim. This argument is local and does not require Y to be connected. □

4.1. **A planar interface.** The exact flux formula also describes the loss when the curvature hypothesis changes at a block face. In fine units put the tail centre at 0, take C in directions $1, 2$, and set $a_2(x) = f(x_1)X$, with $|X| = 1$ in $\mathfrak{g}$, and $a_\mu = 0$ for $\mu \ne 2$. Choose the first difference of f to equal ϕ on $-m, \dots, m-1$, $-\phi$ on the preceding $L-1$ sites and zero elsewhere. This has a periodic primitive on a sufficiently large torus. Take $|\phi|$ small enough depending on L so all the logarithms are on their analytic branches. All factors commute, and the coarse curvature is exactly the linear flux in eq. 3.vi.

The first direction of a surface starting at $x \in B(0)$ contributes $m - x_1$ copies of ϕ and no negative copies. Its second direction contributes a factor L. Averaging over x_1 gives the signed coarse curvature coefficient $L(L-1)\phi/2$, although the curvature already vanishes in the head block. If instead the fine curvature is constant on the full flux support, the signed coarse curvature coefficient is $L^2\phi$. This second configuration can also be realized with a periodic primitive by compensating the first differences outside the support. These examples locate the curvature cost of using the full read of a row across a face.

5. Relative axial gauges and straightening

A field on one block can be put in a fixed gauge by specifying how each site is transported to the block centre. A second choice fixes the value of the gauge transformation at that centre. For composing averages, the useful choice prescribes a transported average of the gauge transformation.

5.1. **Local gauges and their normalization.** On a tree with root y, the gauge $g(x) = W(\gamma_{yx})$, with γ_{yx} the tree path, sets every tree link to 1. It is unique among gauges with $g(y) = 1$. Indeed each tree edge from a parent x to a child z gives the recursion $g(z) = g(x)W(x, z)$. For real fields this change of variables also preserves the product Haar measure in the following sense. Use the $g(x)$ for $x \ne y$ as coordinates for the tree links and the transformed links as coordinates off the tree. First orient each tree coordinate from parent to child; reversing an independent positive link

preserves Haar measure by inversion. In parent-before-child order each child-gauge substitution is then a fixed left translation when the previous coordinates are held fixed. Once the tree coordinates are fixed, each other link is changed by left and right translations. The resulting measure is product normalized Haar in these coordinates. The same proof works on a forest with a prescribed root gauge on each component.

For a block $B(y)$, let $T_W(y,x)$ be the mean transport in eq. 3.iv. For two fields W, V, the relative block axial condition is $T_{V^g}(y,x) = T_W(y,x)$. This condition fixes the transformation inside the block once $g(y)$ is given. If some sites are pinned, one can impose the condition on the free sites while keeping $g(y)$ fixed. At a pinned site the condition is an equation on the existing transport, so it must be included in the datum whenever it is needed. In particular the gauge $g(x) = T_V(y,x)$ fixes the mean transports to 1 with the block centre held fixed. This choice and the coordinate tree gauge are defined by their respective paths.

The following estimate explains why the relative conditions provide local coordinates even in the presence of coarse holonomy. Let D be the rectangular lift of $B(c_-) \cup B(c_+)$ along c. Write

$$\epsilon_D(W) = \max_{p \subset D} \|W(\partial p) - 1\|.$$

Here $p \subset D$ and $b \subset D$ refer to the rectangular graph. The real branch test may also read plaquettes that project across the opposite face of a short periodic lattice.

Lemma 5.1 (Relative transport estimate)**.** Let W, V be real fields defined on D and on the plaquette test stock for c. Suppose $\epsilon_D(W), \epsilon_D(V) \le \epsilon$ for sufficiently small ϵ depending on L. If their mean transports in the two blocks differ by at most δ and their averages on c differ by at most d, then

$$\max_{b \subset D} \|V(b) - W(b)\| \le d + 2\delta + C_L\epsilon.$$

■

Proof. For an edge b internal to $B(y)$, the product $T_W(y,b_-)W(b)T_W(y,b_+)^{-1}$ differs from 1 by $C_L\epsilon$. To see this, compare both mean transports with their coordinate combs by the path interchange estimate, then close the combs with b and use Lemma 3.3. For a crossing edge with $b_- \in B(c_-)$, the corresponding product

$$T_W(c_-,b_-)W(b)T_W(c_+,b_+)^{-1}$$

differs from $W([c])$ by $C_L\epsilon$. With combs in place of the means, the path and $[c]$ bound a closed walk in D of bounded length. In a coordinate tree gauge every link on this walk is within $C_L\epsilon$ of 1, giving the estimate by the unitary product inequality. The mean transport comparison restores the same bound. The same closed walk argument bounds every loop in the first two rules defining $\mathcal{M}$ by $C_L\epsilon$ in distance from I. In particular all the fallback path loops avoid $-I$. Either rule that applies gives $\|\mathcal{M}(W)(c) - W([c])\| \le C_L\epsilon$. Thus internal links are within $C_L\epsilon$ of the product of their endpoint transports, and crossing links are within $C_L\epsilon$ of that product with $\mathcal{M}(W)(c)$ inserted between them. Compare the corresponding formulas for V and telescope the three unitary factors. □

For a site function g with values in G or $G_{\mathbb{C}}$, define its one-step average relative to W by

$$R_W g(y) = g(y) \exp\left\{ L^{-4} \sum_{x \in B(y)} \log\Big[g(y)^{-1} T_W(y,x) g(x) T_W(y,x)^{-1} \Big] \right\}. \tag{5.i}$$

All logarithms are in the branch at 1 and the formula is used on its open chart. For $V = e^a W$, with a a raw link logarithm, set

$$u_W(a)(y) = \exp\left\{ L^{-4} \sum_{x \in B(y)} \log\left[T_V(y,x) T_W(y,x)^{-1} \right] \right\}. \tag{5.ii}$$

For contour covariance in complex frames, require the original and transformed contour families to lie in the mean charts of Lemma 3.1. Prescribe $R_W g(y) = s(y)$ at the centre. Suppose the relative contour logarithms exist. For complex data also require the contour families of the candidate below after gauge transformation to lie in their mean charts. The unique admitted relative block axial gauge is then

$$g(y) = s(y)u_W(a)(y)^{-1}, \qquad g(x) = T_W(y,x)^{-1}g(y)T_V(y,x). \tag{5.iii}$$

Covariance of T gives the second formula and makes each argument of the logarithm in eq. 5.i equal to $T_V(y,x)T_W(y,x)^{-1}$. Thus the prescribed average fixes the value at the centre. These formulas adapt the transported gauge average and the axial normalization of Bałaban [Ba85a, (78)–(92)][Ba85b, §B] to the mean transports.

5.2. **Straightened link coordinates.** On the admitted analytic charts the one-step plain coordinate is

$$P_W(a) = \log[\mathcal{M}_{\mathrm{an}}(e^a W)\mathcal{M}_{\mathrm{an}}(W)^{-1}].$$

Define the one-step straightened coordinate F_W by

$$e^{F_W(a)(c)}\mathcal{M}_{\mathrm{an}}(W)(c) = u_W(a)(c_-)^{-1}\mathcal{M}_{\mathrm{an}}(e^a W)(c)\, u_W(a)(c_+). \tag{5.iv}$$

Take $F_W(a)(c)$ to be the series logarithm of the right side multiplied by $\mathcal{M}_{\mathrm{an}}(W)(c)^{-1}$. Its chart requires this product to lie in the norm-one disk about I. If the relative axial gauge in eq. 5.iii exists for $s = 1$ and its transformed field lies in the analytic row chart, covariance identifies the right side with the analytic average of that transformed field. The two factors of u_W subtract the infinitesimal block-transport terms from the plain linearization. Set $J_W = D_a F_W(0)$. For a fine Lie algebra site field λ, define the site differential by

$$q_W\lambda = \frac{d}{dt}R_W(e^{t\lambda})\Big|_{t=0}.$$

Its explicit row is computed in eq. 7.iv.

For later estimates use row and column sums of the norms of colour matrices. These are counting-coordinate sums. Define the covariant straight-segment row by

$$(S_W a)(c) = L^{-4}\sum_{x\in B(c_-)}\sum_{b\in[x,x']}\mathrm{Ad}(T_W(c_-,x)W([x,b_-]))a(b), \qquad x' = x + (c_+ - c_-). \tag{5.v}$$

Here the formula is written on positive c, with segment edges oriented along it. If H is a field on oriented links, its raw multiplicative tangent at H extends from positive edges by

$$a(\bar{b}) = -\,\mathrm{Ad}(H(b)^{-1})a(b). \tag{5.vi}$$

This follows by inverting $e^{a(b)}H(b)$. It specifies the induced action of all lattice symmetries, including those that reverse links. In particular $u_W(a)$ pulls back as a site field, by equivariance of the contour family. The field on the right of eq. 5.iv then pulls back as a link field, so its logarithm pulls back by eq. 5.vi at the transformed reference. Differentiation gives the same symmetry for J_W and its iterates.

Let s be a lattice isometry preserving the fine and coarse grids and the centered block partition. Write s^* for its pullback and s^*_W for its induced map on logarithmic tangents. When both the original and pulled-back path formulas and their logarithmic coordinates lie in the admitted analytic charts,

$$D(\mathcal{M}_{\mathrm{an}})_{s^*W}\, s^*_W = s^*_{\mathcal{M}_{\mathrm{an}}(W)}D(\mathcal{M}_{\mathrm{an}})_W, \qquad q_{s^*W}s^* = s^*q_W. \tag{5.vii}$$

On the open first real branch the analytic differential restricted to real tangents is the Borel map's differential. The corresponding analytic identity holds for the straightened derivative J_W. Pullbacks compose in the ordinary pullback order; differentiating that exact identity gives the composition law for the background-dependent tangent maps. For real W, a reversed edge contributes only a sign and a unitary adjoint in eq. 5.vi. The maps are therefore isometries in the counting supremum

and physical L^p norms, $1 \le p \le \infty$. This argument uses the permutation of the underlying links and their unchanged lengths.

Proposition 5.2 (One-step derivative)**.** There are positive constants ϵ_L, r_L, C_L, depending only on L, with the following property on a torus admitting one averaging step. Let W be real, with plaquette defect at most $0 \le \epsilon \le \epsilon_L$ on the determining double blocks and their projected first-branch tests. For a complex Lie algebra increment a on the fine positive links, $F_W(a)$ and $\log u_W(a)$ are holomorphic for $\|a\|_\infty < r_L$, with bounds $C_L\|a\|_\infty$. Then

$$\|S_W\|_{\mathrm{row}} \le L, \qquad \|S_W\|_{\mathrm{col}} \le L^{-3}, \qquad \|J_W - S_W\|_{\mathrm{row}} + \|J_W - S_W\|_{\mathrm{col}} \le C_L\epsilon.$$

On a smaller fixed ball, $D_aF_W(a) - J_W$ has both sums bounded by $C_L\|a\|_\infty$. The second derivative obeys

$$\sup_b \sum_{c,b'} \|D_bD_{b'}F_W(a)(c)\| \le C_L.$$

These estimates are uniform in the volume. For $W' = e^{a'}W$ with $\|a'\|_\infty < r_L$ after decreasing r_L, the same assertions hold holomorphically in (a', a) with $\epsilon + \|a'\|_\infty$ in the comparison bound and with $L(1 + C_L\|a'\|_\infty)$ and $L^{-3}(1 + C_L\|a'\|_\infty)$ in the bounds for $S_{W'}$. ■

Proof. Work on one determining double block in the coordinate tree gauge of Lemma 3.3. In this gauge W has link logarithms bounded by $C_L\epsilon$. Conjugation by this real gauge preserves the norms of the input and output entries. A fixed sufficiently small joint ball in the reference logarithms and the increments admits all path products, relative logarithms and mean equations in eq. 5.ii and eq. 5.iv. The paths have length bounded in terms of L. The mean transport and the remaining operations are holomorphic and bounded on this ball, uniformly in the choice of the row. Cauchy's formula on a smaller ball bounds each first and second partial derivative and each mixed reference and increment derivative by C_L. A row contains at most C_L fine bonds, and a fine bond belongs to at most a fixed number of determining double blocks. These two counts give the claimed sums, once the derivative at the trivial field is identified. The same chart contains $e^{a'}W$ and its increments, since $\log(e^{a'}W)$ in the local gauge has size at most $C_L\epsilon + C_L\|a'\|_\infty$. For real increments, a tested plaquette changes by at most $4(\|a'\|_\infty + \|a\|_\infty)$, by telescoping its four unitary factors. Choose the joint radii so that this change and the reference defect fit below the first-branch cutoff. The test stock includes projected opposite-face plaquettes on a shortest period. Thus the analytic formula agrees with the real map there. The same analytic argument applies on embedded covers of periodic carriers. Counting a repeated occurrence on a cover and adding its entry on descent only increases the bounds.

The derivative of $\log u_1$ is H_a, the block mean of contour sums defined before eq. 3.v. Its two endpoint terms cancel those in $DP_1(0)$, leaving $J_1 = S_1$. Both J_W and S_W are holomorphic functions of the finite set of reference links. Integrating their mixed derivative from the trivial reference in the local gauge gives the comparison bound. Integrating the second increment derivative gives the bound for $D_aF_W(a) - J_W$. The values $F_W(0) = 0$ and $\log u_W(0) = 0$ give the first-order size bounds.

Each real conjugation in eq. 5.v has norm one. There are L^4 segments of L links in one row. For a fixed input edge parallel to the row direction, its possible positions in a segment number L, and each such position determines the start and its block. Its total column weight is $L^{-4}L$. For the complex reference comparison the transported matrices differ from the corresponding real ones by $C_L\|a'\|_\infty$, by holomorphy on the same finite ball. This multiplies both counting bounds by at most $1 + C_L\|a'\|_\infty$. □

5.3. **Composition.** For a reference W on the fine lattice set $W_j = \mathcal{M}^j_{\mathrm{an}}(W)$ on the admitted iteration charts. For a field $e^{a_0}W$, define a_j and the accumulated site factors v_j by

$$v_0 = 1, \qquad a_{j+1} = F_{W_j}(a_j), \qquad v_{j+1} = (v_j|_{\text{level } j+1})u_{W_j}(a_j). \tag{5.viii}$$

Require the plain iterates and the endpoint conjugations by v_j to remain in the analytic row charts of Proposition 3.2. The small balls in Proposition 6.1 supply this admission. Covariance and eq. 5.iv then give

$$\mathcal{M}^j_{\mathrm{an}}(e^{a_0}W) = (e^{a_j}W_j)^{v_j}. \tag{5.ix}$$

Indeed applying $\mathcal{M}_{\mathrm{an}}$ to the right side restricts v_j to the next lattice. Applying eq. 5.iv there gives $e^{a_{j+1}}W_{j+1}$ with gauge $u_{W_j}(a_j)$, and the two gauge transformations compose in the order of eq. 5.viii. For real fields the same identity holds with the Borel map when the relevant rows use the first branch at each level. The local identity uses the backward determining blocks of the output, including the projected tests for the Borel map.

The factor v_1 can contribute already at first order. At $W = 1$, perturb a single edge leaving a block centre y by e^{tX}, where $X \in \mathfrak{g}$ and t is small. A coordinate path can use that edge only as its first step. The total weight of first steps, averaged over the endpoints $x \in B(y)$, is $1 - L^{-4}$. Signed coordinate permutations distribute it equally among the eight edges leaving y. Since all perturbed products commute, the contour mean averages their logarithms exactly on this chart. Hence

$$v_1(y) = \exp\bigl(t(1 - L^{-4})X/8\bigr).$$

All other block factors are 1. In particular the site terms in eq. 5.iv affect the linearized link coordinate.

6. Estimates through several scales

We measure all spacings in units of the final spacing h_k for the first two results of this section. Put $\eta = L^{-k}$ and $t_j = L^j\eta$ for $0 \le j \le k$. An input A gives a raw fine increment $a_0 = \eta A$. The coarse variables a_j in eq. 5.viii are raw increments on the level-j lattice. This normalization lets the linear amplification of a straight segment, of size L, appear explicitly at each step.

6.1. **The perturbation ball.**

Proposition 6.1 (Depth-uniform iteration)**.** On a torus admitting k steps, let A, B be complex Lie algebra fields on the fine positive links. Let W be real with $\sup_p \|W(\partial p) - I\| \le \alpha\eta^2$, and let $W' = e^{\eta B}W$ with $\|B\|_\infty \le \rho$. There are $c_L > 0$ and $C_L < \infty$, uniform in the depth and volume, such that the following holds if $\alpha, \rho, \|A\|_\infty < c_L$. Use W' as the reference in eq. 5.viii, and write $W'_j = \mathcal{M}^j_{\mathrm{an}}(W')$. The iterates and factorization are admitted on the regular analytic charts, with holomorphic dependence on B, A. They satisfy

$$\|a_j(A)\|_\infty \le 2t_j\|A\|_\infty, \qquad \|v_j(A) - I\|_\infty \le C_L t_j\|A\|_\infty. \tag{6.i}$$

Define $K_j(A) = D_A a_j(A)$ and let $Q^{\mathrm{st}}_j = K_j(0)$. With $\Delta = \alpha + \rho + \|A\|_\infty$, the counting-coordinate bounds are

$$\begin{aligned} \|K_j(A)\|_{\mathrm{row}} &\le t_j(1 + C_L\Delta), \\ \|K_j(A)\|_{\mathrm{col}} &\le t_j L^{-4j}(1 + C_L\Delta), \\ \sup_b \sum_{c,b'} \|D_{A(b')}K_j(A)(c,b)\| &\le C_L t_j^2 L^{-4j}. \end{aligned} \tag{6.ii}$$

Each coefficient is supported in the backward double block of its output link. In particular the column norm of $K_j(A) - K_j(\widetilde{A})$ is bounded by $C_L t_j^2 L^{-4j}\|A - \widetilde{A}\|_\infty$ on the ball.

The plain log coordinate

$$p_j(A)(c) = \log\,[\mathcal{M}^j_{\mathrm{an}}(e^{\eta A}W')(c)W'_j(c)^{-1}]$$

has derivative Q^{pl}_j at zero. Both Taylor remainders satisfy

$$\|a_j(A) - Q^{\mathrm{st}}_j A\|_\infty + \|p_j(A) - Q^{\mathrm{pl}}_j A\|_\infty \le C_L t_j^2\|A\|^2_\infty. \tag{6.iii}$$

The derivative of the straightened coordinate is exactly $Q_j^{\mathrm{st}} = \eta J_{W'_{j-1}} \cdots J_{W'_0}$, where $J_V = D_a F_V(0)$ as in proposition 5.2. At $B = 0$ its difference from $\eta S_{W_{j-1}} \cdots S_{W_0}$, $W_j = \mathcal{M}^j W$, has row norm at most $C_L \alpha t_j$ and column norm at most $C_L \alpha t_j L^{-4j}$. For complex B the same comparison with this real-reference product has $\alpha + \rho$ in place of α. ∎

Proof. Write $W_j = \mathcal{M}^j W$. By corollary 4.1, after decreasing c_L, the real reference plaquette defects are bounded by $C_L \alpha t_j^2$ throughout the iteration. The real reference uses the first branch at every step, so $W_j = \mathcal{M}_{\mathrm{an}}^j(W)$ as well. First take $B = 0$. By proposition 5.2, each $D_a F_{W_s}(a)$ has row and column sums at most L and L^{-3}, respectively, multiplied by $1 + C_L(\alpha t_s^2 + \|a\|_\infty)$. The estimates are valid on a fixed one-step ball. In particular, setting $r_s = \|a_s\|_\infty / t_s$ and integrating the derivative along the segment from zero to a_s gives

$$r_{s+1} \le r_s(1 + C_L(\alpha t_s^2 + t_s r_s)).$$

The geometric sums satisfy $\sum_{s<j} t_s \le t_j/(L-1)$ and $\sum_{s<j} t_s^2 \le t_j^2/(L^2-1)$. As long as $r_s \le 2\|A\|_\infty$, the product of the displayed factors is bounded by $\exp(C_L(\alpha + 2\|A\|_\infty)) < 2$. Since $r_0 = \|A\|_\infty$, induction gives the first bound in eq. 6.i. The one-step site log bound $\|\log u_{W_s}(a_s)\|_\infty \le C_L \|a_s\|_\infty$ and the geometric sum bound the accumulated site factors by $\exp(C_L t_j \|A\|_\infty) - 1$ in distance from I.

The one-step contour families, mean differences and outer logs lie inside their charts with a fixed reserve. This follows from the local real comb coordinates in the proof of proposition 5.2 by decreasing c_L. The accumulated factors have condition numbers at most $\exp(C_L t_j \|A\|_\infty)$, so the reserve admits their endpoint conjugations in proposition 3.2. Thus eq. 5.ix holds on the entire ball, including complex increments. If the increment is real, the same reserve can be chosen so that the plaquette defect at each input to $\mathcal{M}$ is below the real branch cutoff. Hence the holomorphic formulas agree with the real average there.

To treat $B \neq 0$, first apply this argument to the increment B about W. The exact factorization gives $W'_s = e^{b_s} W_s$, where $b_s = \log(W'_s W_s^{-1})$ satisfies $\|b_s\|_\infty \le C_L \rho t_s$. Indeed the ratio is a product of $e^{a_s(B)}$, one site factor, and the inverse site factor transported by the real W_s. Apply the shifted-reference clause of proposition 5.2 to W'_s. This adds $C_L \rho t_s$ to the distortion factor at step s. The same summation, with this term included, proves eq. 6.i and chart admission for W'. When A, B are real, the same one-step cutoff estimate applies with $a' = b_s$ and $a = a_s$. The accumulated real gauges preserve the test defects, so the iterates equal the Borel iterates there. This step uses the size of the reference link shift, which may change the fine curvature scale.

For the derivatives, put $E_s = D_a F_{W'_s}(a_s(A))$ and $P_{v,u} = E_{v-1} \cdots E_u$ for $0 \le u \le v \le k$, with the empty product equal to the identity. The row and column sums of a product are bounded by products of the corresponding sums. The total distortion of these products is at most

$$\exp\left(C_L \sum_{u \le s < v} [\alpha t_s^2 + (\rho + \|A\|_\infty) t_s] \right) \le 1 + C_L \Delta.$$

The chain rule gives $K_j(A) = \eta P_{j,0}$, which proves the first two bounds in eq. 6.ii. At zero all the a_s vanish. Replace each factor $J_{W'_s}$ by S_{W_s} in turn. Its error, in both norms, is bounded by $C_L(\alpha t_s^2 + \rho t_s)$ by the one-step comparisons in a real comb gauge. The preceding product bounds on the other factors give the stated errors, since these radii sum to $O_L(\alpha + \rho)$.

Differentiating the recursion twice gives

$$D_A^2 a_j(A) = \eta^2 \sum_{s<j} P_{j,s+1}\, D_a^2 F_{W'_s}(a_s)[P_{s,0}\,\cdot, P_{s,0}\,\cdot]. \tag{6.iv}$$

Fix the fine link of one input leg and sum the norms over the other input and the output. Use a column sum of $P_{s,0}$ on the fixed input and a row sum of $P_{s,0}$ on the summed input. The one-step

Hessian bound of proposition 5.2 sums its output and its second input with its first input fixed. Finally sum the output of $P_{j,s+1}$ by its column bound. The total is at most

$$C_L\eta^2\sum_{s<j}L^{-3(j-s-1)}L^{-3s}L^s\le C_Lt_j^2L^{-4j}.$$

Integrating this coefficient bound along the line between the inputs gives the column Lipschitz estimate. Each factor and one-step Hessian reads only the relevant double block, so the products and derivatives have the asserted support.

Using row bounds for all factors of eq. 6.iv gives

$$\|D_A^2a_j(A)[X,Y]\|_\infty\le C_L\eta^2\sum_{s<j}L^{j-s-1}L^{2s}\|X\|_\infty\|Y\|_\infty\le C_Lt_j^2\|X\|_\infty\|Y\|_\infty.$$

Taylor's formula proves the straightened remainder. The quadratic part is the sum in eq. 6.iv evaluated at zero with both input legs A, divided by two. At final depth k, its term from scale s is bounded by $C_Lt_s\|A\|_\infty^2$. If the reference used in the recursion is $W'=I$, this term vanishes when every link increment is a multiple of one fixed $X\in\mathfrak{g}$. All path factors then commute, so each one-step logarithm is exactly linear on that subspace.

The derivative of $\log v_j$ at zero, denoted by $\dot v_j$, is the sum of the linear parts of the successive block factors restricted to level j. The Taylor remainder of the logarithm of the factor from level s is bounded by $C_Lt_s^2\|A\|_\infty^2$, using the straightened remainder at level s and the analytic one-step site map. For products of exponentials with total log norm at most a fixed small number, the log of the product differs from the sum by at most a constant times the square of that total norm. This follows by expanding the product and the logarithm and summing terms of degree at least two. Consequently $\|\log v_j-\dot v_j\|_\infty\le C_Lt_j^2\|A\|_\infty^2$. By eq. 5.ix,

$$e^{p_j(A)(c)}=v_j(c_-)e^{a_j(A)(c)}[W_j'(c)v_j(c_+)^{-1}W_j'(c)^{-1}].$$

The references W_j' have uniformly bounded condition numbers, as shown by their shifts from W_j. The product log estimate gives the plain remainder and the exact identity

$$(Q_j^{\mathrm{pl}}A)(c)=(Q_j^{\mathrm{st}}A)(c)+\dot v_j(A)(c_-)-\mathrm{Ad}(W_j'(c))\dot v_j(A)(c_+).\tag{6.v}$$

The last two terms can be nonzero even at a flat reference. □

The plain coordinate also has a uniform first derivative: $\|D_Ap_j(A)\|_{\infty\to\infty}\le C_Lt_j$ on a smaller fixed input ball. The factorization and the site-factor bounds give $\|p_j(A)\|_\infty\le C_Lt_j$ on a larger such ball; Cauchy's formula along a unit input direction proves the assertion. Dividing by the output length and converting the input to physical units gives a bounded per-unit-length derivative.

There is also a useful positive majorant. Let S_j^+ be the per-unit-length straight row with every real parallel transport replaced by its norm one. Let $D(c)=B^j(c_-)\cup B^j(c_+)$ be the backward double block of the output row, and put $H_j|A|(c)=L^{-4j}\sum_{b\subset D(c)}|A(b)|$. For the real reference of Proposition 6.1, write $\widehat Q_j^{\mathrm{st}}=t_j^{-1}Q_j^{\mathrm{st}}$ for the derivative with both coordinates per unit length. It obeys

$$|\widehat Q_j^{\mathrm{st}}(W)A(c)|\le S_j^+|A|(c)+C_L\alpha H_j|A|(c).\tag{6.vi}$$

The comparison with the product of the S_{W_i} has each entry bounded by $C_L\alpha L^{-4j}$, by its column estimate, and support in $D(c)$. The product itself is bounded by S_j^+ by unitarity and the triangle inequality. These facts give the display.

For a prescribed external complex gauge g, take the real reference W (thus $B=0$ in Proposition 6.1). In its final-scale units the factorization becomes

$$\mathcal{M}_{\mathrm{an}}^j((e^{\eta A}W)^g)=(e^{a_j}W_j)^{(g|_{\mathrm{level}\ j})v_j}.\tag{6.vii}$$

Besides the small-field hypotheses, require the contour and logarithm charts and their conjugates to be admitted. A sufficient condition on the smaller balls used above is

$$\max_x \operatorname{cn}(g(x)) \max_{i,x} \operatorname{cn}(v_i(x)) \le 2, \qquad \operatorname{cn}(g) = \|g\|\|g^{-1}\|.$$

The proof of chart admission has a fixed reserve, so decreasing the input ceiling accommodates this factor two at every level. Endpoint covariance and eq. 5.ix then give the identity.

6.2. **Curvature and open transports.** For the real reference in proposition 6.1, let d_j be the supremum of the distance between $W_j(c)$ and the straight bare transport between the endpoints of c, over all level-j links. The bound on the loop correction in the proof of proposition 3.2 and unitary telescoping give

$$d_{j+1} \le L d_j + C_L \alpha t_j^2, \qquad d_0 = 0.$$

Hence $d_j \le C_L \alpha t_j^2$. In this comparison the error at the finer levels is multiplied only by the number of links in a straight segment.

Proposition 6.2 (Complex curvature at its own scale)**.** Under the assumptions of proposition 6.1, assume also $\sup_p \|W'(\partial p) - I\| \le \alpha' \eta^2$ with $0 \le \alpha' \le c_L$. After a fixed decrease of c_L,

$$\|\mathcal{M}^j_{\mathrm{an}}(W')(\partial C) - I\| \le 2\alpha' t_j^2$$

for each level-j plaquette C. For an individual plaquette, the additional curvature bound on W' is needed only on its complete backward blocks. The real reference hypotheses of proposition 6.1 remain in force. ■

Proof. The result at $j = 0$ is the assumption. For $j > 0$ take the box Y formed by the four order-j blocks at the corners of C. Work on a lift if necessary. Put $n = L^j$, $r = \alpha' t_j^2$. Use the comb gauge g in Y formed with the complex field W'. Each path and inverse path of at most Cn bare links has norm at most $e^{C\rho t_j}$. This follows by multiplying the bounds $e^{\eta\rho}$ on complex increments or their inverses and using unitarity of the real reference. We take $K = e^{C\rho t_j}$, increasing the absolute constant to bound all condition numbers in the comb estimate and those of g. With $W'^g = e^a$ on Y, lemma 3.3 gives $\|a\|_\infty \le C_L r/n$. For inverse plaquette factors one may use $\|P^{-1} - I\| = \|P - I\|$ for $P \in G_{\mathbb{C}}$: $P^{-1} - I$ is the adjugate of $P - I$, whose singular values are those of $P - I$ in reverse order.

Extend a by zero outside Y on a torus on which the read embeds. This permits use of proposition 6.1 at the flat reference with final scale j, since the normalized input has norm $n\|a\|_\infty \le C_L r$. The four output links read a only in Y. The extension serves to apply the flat supremum-ball estimate. Let $P_j(a) = \log \mathcal{M}^j_{\mathrm{an}}(e^a)$ and let $Q^{(j)}$ be the composition of the j raw flat link differentials eq. 3.v. Then

$$\|P_j(a)\|_\infty \le C_L r, \qquad \|P_j(a) - Q^{(j)} a\|_\infty \le C_L r^2.$$

At intermediate levels $s \le j$ the corresponding plain logarithms have size at most $C_L r L^{s-j}$. The contours on their small branch have a fixed reserve. Covariance under g^{-1} is therefore admitted at each step by proposition 3.2, since the conjugations cost at most K. This proves the relation between the four output links of the gauged field and those of W' on the read.

On fine plaquettes in Y, product expansion gives

$$\|\mathrm{d}a(p)\| \le K\alpha'\eta^2 + C_L \|a\|_\infty^2.$$

Apply the iterated version of eq. 3.vi at C. It reads only plaquettes in Y, has nonnegative weights, and its total weight is n^2. The coarse plaquette product differs from its linear term by $O_L(r^2)$. The displayed Taylor bound for the four coarse links and $n\|a\|_\infty \le C_L r$ therefore give

$$\|\mathcal{M}^j_{\mathrm{an}}(e^a)(\partial C) - I\| \le Kr + C_L r^2.$$

Conjugating back gives at most $K^2 r + C_L K r^2 \le 2r$ for the stated ceilings. The path and flux identities on a cover descend with the same directed links and multiplicities. □

6.3. **Rows on a layered region.** We return to physical spacings $h_j = L^j h_0$. For a level-j link c, let $D(c) = B^j(c_-) \cup B^j(c_+)$. A straightened row uses the raw fine perturbation $h_0 A$, and has output $q_c(A) = a_j(c)/h_j$, per unit length. Fix a family of rows without repetitions and assign a length $\xi(b)$ to each fine link in their carriers. Assume that $h_j/L \le \xi(b) \le L h_j$ for $b \subset D(c)$. For rows on adjacent layers this allows the smaller spacing on the part of the carrier outside the inner region. A fixed fine link can then enter rows of at most three levels, since their spacings lie in $[\xi(b)/L, L\xi(b)]$. At each level any such row is incident to the unique block containing b_-. The degree bound gives bounded overlap.

Assume on each $D(c)$ the hypotheses of proposition 6.1 with final spacing h_j. In physical units these include a real plaquette defect bounded by $\alpha(h_0/h_j)^2$ and a bound $h_j \|A\|_{\infty, D(c)} < c_L$. With a complex reference shift $h_0 B$, impose $h_j \|B\|_{\infty, D(c)} < c_L$ there as well. The proof of proposition 6.1 applies to the row on this stock. Here $D(c)$ in the hypotheses includes every fine link and plaquette whose vertices lie in the projected union. On a torus with two sites in the row direction at level j, the real branch test can include plaquettes at the other periodic face of this union. The curvature hypothesis includes those plaquettes. The analytic derivatives use just the lifted path formula. Every intermediate double block in that formula lies in the carrier. The real curvature recurrence on four endpoint blocks bounds the plaquettes needed at each subsequent step, including those tested across a periodic face.

With A measured per length, the coefficients satisfy

$$\begin{aligned} \sum_b \|D_b q_c(A)\| &\le 2, \\ \|D_b q_c(A)\| &\le 2(h_0/h_j)^4, \\ \sup_b \sum_{b'} \|D_b D_{b'} q_c(A)\| &\le C_L h_j (h_0/h_j)^4. \end{aligned} \tag{6.viii}$$

Indeed use $h_j A$ for the normalized input of proposition 6.1 with final depth j and divide the output by h_j. Each derivative multiplies by h_j, giving the factor h_j for the second derivative. The fine-link support of both legs is $D(c)$. For example, the raw row $a_j(c)$ has weighted row sum $\sum_b \|D_b a_j(c)\|/\xi(b) \le 2L$. The sum of its derivatives over all the chosen rows at fixed input link is bounded by $C_L h_0^4/\xi(b)^3$. The single adjacent-scale comparison in the first bound costs at most L, independent of how many levels enter the row.

The same density controls a weighted squared-sum estimate. All fine sums below run over the union of the row carriers, where the positive length ξ is assigned. For any fine input X, Cauchy–Schwarz in one row and then the carrier overlap give

$$\sum_c h_j^4 h_j^{-2} |D q_c(A) X|^2 \le C_L \sum_b h_0^4 \xi(b)^{-2} |X(b)|^2, \tag{6.ix}$$

where j on the left is the level of c. In fact the factor multiplying $|X(b)|^2$ from one row is bounded by $C_L h_0^4 h_j^{-2}$ by eq. 6.viii, and there are only boundedly many rows containing b.

For an assigned row current $\omega(c)$ satisfying $|\omega(c)| \le \gamma h_j^{-3}$, the second derivative also satisfies

$$\left| \sum_c h_j^4 \langle \omega(c), D^2 q_c(A)[X, Y] \rangle \right| \le C_L \gamma \left(\sum_b h_0^4 \xi(b)^{-2} |X(b)|^2 \right)^{1/2} \left(\sum_b h_0^4 \xi(b)^{-2} |Y(b)|^2 \right)^{1/2}. \tag{6.x}$$

The pairing is the bilinear extension of the real colour pairing. Fix the first fine link and sum the absolute coefficients over the second link and the rows containing the first one. The current and row measure cost at most γh_j. Multiplying by the last bound in eq. 6.viii gives $C_L \gamma h_0^4 h_j^{-2}$ for each carrier. The scale comparison and overlap bound replace this by $C_L \gamma h_0^4 \xi(b)^{-2}$ after summation. Symmetry of the second derivative gives the same estimate with the input links exchanged. Weighted Cauchy–Schwarz then gives eq. 6.x.

6.4. **Determining rows in separated regions.** Let $\Omega_0 \supset \cdots \supset \Omega_k$ be regions on the ambient periodic lattice and put $\Omega_j = \varnothing$ for $j > k$. Use half-open lattice cells for boundary membership. Suppose each Ω_j is a union of centered depth-j blocks and Ω_{j+1} is a union of centered depth-$(j+1)$ blocks in the same hierarchy. In particular, the hypothesis holds for regions formed from aligned cubes of side $M_1 h_j$, with M_1 an odd power of L. Impose the separation

$$\operatorname{dist}_\infty(\Omega_j^c, \Omega_{j+1}) > \rho_{\mathrm{gap}} h_j \qquad (0 \le j < k), \qquad \rho_{\mathrm{gap}} \ge 4L. \tag{6.xi}$$

The distance is infinite when one of the sets is empty. For a level-j site $y \in \Omega_j \setminus \Omega_{j+1}$ call y a top of level j and $B^j(y)$ its pillar. These pillars partition the fine sites of Ω_0. Indeed, a fine site in layer j belongs to a depth-j block contained in Ω_j and disjoint from Ω_{j+1}, by alignment. Its centre is a level-j top.

Use the determining rows

$$\Gamma_j = \{c : \ c \text{ is a level-}j \text{ link with both ends in } \Omega_0, \text{ one end in } \Omega_j, \text{ and no end in } \Omega_{j+1}\}. \tag{6.xii}$$

If both ends are in Ω_j, the row is interior. If only one end is in Ω_j, the row is outward. We use one positive orientation for each link. A link from layer j into Ω_{j+1} is an inward link and is excluded from Γ_j. The scale at a fine site is the spacing of the level of its pillar. For a fine link or plaquette e, let $\xi(e)$ be the largest of these spacings at its endpoints or corners.

Lemma 6.3 (Available rows and their carriers)**.** Let $\Omega_0 \supset \cdots \supset \Omega_k$ be nested regions, each a union of aligned depth-j blocks at its respective level, satisfying eq. 6.xi. For the determining rows Γ_j of eq. 6.xii, the endpoints of every row $c \in \Gamma_j$ are tops of level j or $j-1$, with only level zero allowed when $j = 0$. If $j \ge 1$, the double block $B^j(c_-) \cup B^j(c_+)$ is contained in Ω_{j-1} and disjoint from Ω_{j+1}. For an outward row the block at the outer end is entirely in layer $j-1$. All level-$(j-1)$ links of that outer block's rectangular graph are interior determining rows at that level. Each top of positive level has an incident determining row in every coordinate direction, allowing either sign. The row carriers, also with a collar of one fine cell intersected with Ω_0, have bounded overlap independent of k and the volume. The carrier scale is comparable to all its fine site scales within a factor L. ■

Proof. An interior endpoint is in layer j by eq. 6.xii. For $j \ge 1$, the other endpoint of an outward row is at distance at most h_j from Ω_j. Its block and its one-cell collar are within $2h_j$ of Ω_j, so eq. 6.xi places them in Ω_{j-1}. Alignment makes the entire outer block disjoint from Ω_j. This proves the claim about its rows. The two endpoint blocks miss Ω_{j+1} by alignment and the absence of inward rows. The same distance estimate places both blocks in Ω_{j-1}. A one-cell collar can enter Ω_{j+1} but cannot reach Ω_{j+2}. If $j+2 \le k$, this follows from eq. 6.xi with index $j+1$; otherwise Ω_{j+2} is empty. At level zero, rows lie in Ω_0 and their collar is considered only inside Ω_0. For the same reason that collar cannot reach Ω_2. This proves scale comparability on the stated carriers.

A top y of positive level j is outside Ω_{j+1}. If $j < k$, its unique aligned depth-$(j+1)$ cell misses Ω_{j+1} and has $L \ge 3$ level-j sites in each coordinate direction. Along the line through y in that cell, at least one of the two adjacent sites is still in the cell. At the highest level $j = k$, use either neighbour without a further tiling. The chosen neighbour is in Ω_{j-1} by separation, so the incident row belongs to eq. 6.xii. For each fixed j, the row carriers have diameter $O(h_j)$, their origins lie on the level-j grid, and there are at most eight incident oriented choices per origin. They therefore have bounded overlap at that level. A fine site belongs to carriers at at most three levels by scale comparability. Periodic counting uses the rectangular lifts, with bounded multiplicity under the standing compatibility condition. □

Consider a level-j mixed plaquette, $j \ge 1$, with a corner in Ω_j. Suppose its sides are retained rows or links with both ends outside Ω_j that are completed by one averaging step. A completed side reads the path links and projected branch-test plaquettes in the union of its two endpoint blocks at the preceding lattice level, including the links joining those blocks. Each endpoint is within h_j in maximum distance of the corner in Ω_j, so the full blocks are within $2h_j$ of that region.

Separation places them in Ω_{j-1} and alignment places them outside Ω_j. The completed side is therefore the actual average of the recorded finer-layer stock, and transforms at its two endpoints. The mixed plaquette transforms by conjugation at its base. For real fields, changing any of its four link transports changes the plaquette by at most the sum of their changes, by unitary telescoping. In particular the one-step derivative estimates can be used for a completed side on its full admitted stock.

6.5. **Two-field comparison after axial alignment.** We use the real curvature bound of Section 4 in its local form. To bound an individual level-n plaquette it suffices to bound the bare plaquettes in the four complete backward blocks at its corners. At one step these blocks contain every path and every real branch test used in the four output links. They contain the flux support in eq. 4.iv as well. Applying the curvature estimate on these stocks and then proceeding through the backward blocks gives the same curvature recurrence. The local claim includes the tested periodic plaquettes whenever the projected stock has an additional face. Equivalently, the recurrence can be applied on lifts once its branch tests have been checked there.

Proposition 6.4 (Depth-uniform raw rigidity)**.** Let $\Omega_0 \supset \cdots \supset \Omega_k$ be nested regions, each a union of aligned depth-j blocks at its respective level, satisfying eq. 6.xi. Use the determining rows eq. 6.xii and the associated pillar spacings ξ. Let U and V be real fields whose bare plaquettes obey

$$\max\{|U(\partial p) - I|, |V(\partial p) - I|\} \le \alpha(h_0/\xi(p))^2 \tag{6.xiii}$$

for every physical fine plaquette whose four vertices lie in $D(c) = B^j(c_-) \cup B^j(c_+)$ for some $c \in \Gamma_j$, $1 \le j \le k$. This includes opposite-face plaquettes at a short periodic seam. Assume $0 \le \alpha \le c_L$ for a sufficiently small positive constant depending only on L. Let $d \ge 0$ and suppose the raw difference of their averages on each $c \in \Gamma_j$, $0 \le j \le k$, is at most d. There is a real gauge g equal to I at all tops such that

$$|\mathcal{M}^n(U^g)(b) - \mathcal{M}^n(V)(b)| \le C_L(d + \alpha) \tag{6.xiv}$$

for all level-n links b, $0 \le n \le k$, with both endpoints in Ω_0 and at least one in Ω_n. The constants are independent of the depth and volume. The inequality concerns link matrices in the aligned frame, before dividing by a lattice length. ∎

Proof. Write $U_n = \mathcal{M}^n(U)$ and $V_n = \mathcal{M}^n(V)$. By Lemma 6.3, every positive-depth pillar is contained in a carrier at its top level. Its bare plaquette bound admits the lower rows and supplies small curvature for each contour family by Corollary 4.2. Thus the means in the descent exist. Starting at the tops, set $g(y) = I$ there and descend to finer centres. In the block of a level-$(n+1)$ centre $y \in \Omega_{n+1}$ set

$$g(x) = T_{V_n}(y, x)^{-1} g(y) T_{U_n}(y, x). \tag{6.xv}$$

At the centre this retains the value already assigned. The blocks of such centres and the level-n tops partition the level-n sites of Ω_n. The recursion assigns a single real fine gauge, which can be extended by the identity outside Ω_0. Covariance proves that the two block contours agree in every block where the recursion was used. The averages on determining rows retain their original difference since both endpoints are tops.

For a carried level-n link b let j be the larger of its endpoint pillar levels. If $n = j$, the link or its inverse is already a determining row, so the asserted comparison holds. Assume $n < j$.

For $j \ge 1$, the depth-j ancestor of the inner endpoint is its level-j top. The other ancestor is within $h_n + h_j/2$ of Ω_j, hence lies in Ω_{j-1} by separation. Both ancestors are outside Ω_{j+1} by alignment. If distinct, they form a directed level-j determining row. If they coincide, their common pillar is contained in a level-j determining carrier by Lemma 6.3. Each block at an intermediate comparison is contained in this carrier. The same is true after projection of the complete backward blocks for its curvature reads and tests.

The carrier is contained in Ω_{j-1}, so each of its bare plaquette logarithms has norm at most $2\alpha L^2 L^{-2j}$, by eq. 6.xiii and the real small-log bound. Apply Corollary 4.2 on the carrier with parameter $2L^2\alpha$. The level-n group defects in the comparison are at most $4L^2\alpha L^{2(n-j)}$. This uses only a fixed decrease of the ceiling for α, depending on L.

For $n < j-1$, both endpoints are in Ω_{n+1}. Indeed the endpoint in Ω_j has distance greater than $\rho_{\mathrm{gap}} h_{j-1}$ from Ω_{j-1}^c and the link has length at most h_{j-1}. The comparison in the next paragraph also applies at $n = j-1$ if both endpoints are in Ω_j.

If the endpoints belong to the same depth-$(n+1)$ block, their aligned contour values determine the link up to curvature by Lemma 5.1, using just that block. Otherwise their parent centres are endpoints of a level-$(n+1)$ row. The difference of that row bounds the difference of b with coefficient one, again by Lemma 5.1, because the two endpoint contour differences vanish.

If the descent comparison reaches an inward level-$(j-1)$ link, its parent row is an outward member of Γ_j. The inner endpoint contour has been aligned. The block at the outer endpoint of that row lies in layer $j-1$, by Lemma 6.3. Each of its internal paths has length at most $2(L-1)$ and uses only its determining rows. Comparing these path products in the two fields costs at most $2(L-1)d$. Replacing a comb transport by its mean in either field costs at most $C_L\alpha$ by the small-curvature path comparison. The outer contour discrepancy is therefore at most $2(L-1)d + C_L\alpha$. The outward row itself contributes d. The proof of Lemma 5.1, using the separate contour errors and the aligned inner endpoint, bounds the inward discrepancy by $(2L-1)d + C_L\alpha$.

Thus the last comparison terminates either within a block, on a determining row, or on this inward link. Each preceding comparison adds only $C_L\alpha L^{2(n-j)}$ at its own level, since the coefficient of the parent row error is one. The sum over those levels is bounded by $C_L\alpha/(L^2-1)$, which proves eq. 6.xiv. □

6.6. **Curvature and frame changes on a fixed box.** For a coordinate comb with order $4, 3, 2, 1$ rooted at y, set $s_U(x) = U(\Gamma_{y,x})$. Use fine spacing one for the next estimate. In this frame write $\widetilde{F}_{\rho\mu}(z)$ for the logarithm of the positively oriented $\rho\mu$ plaquette at z, transported to y by $s_U(z)$, with $\rho < \mu$. Suppose U is real and the plaquette defect is at most $\epsilon < 1/2$ on the box. Assume also that for some $0 < \gamma \le 1$ and $H \ge 0$,

$$|\widetilde{F}_{\rho\mu}(z) - \widetilde{F}_{\rho\mu}(y)| \le H|z-y|_1^\gamma$$

for the plaquettes in the comparison strips, with the base plaquette present in each orientation. For $r = |x-y|_1$ and $r\epsilon$ small,

$$\left| \log U^{s_U}(x,\mu) - \sum_{\rho<\mu} (x_\rho - y_\rho)\widetilde{F}_{\rho\mu}(y) \right| \le C\big(Hr^{1+\gamma} + r^2\epsilon^2\big). \tag{6.xvi}$$

To prove the estimate, append the positive μ link to the comb and move it left past each step in a direction $\rho < \mu$. The prefix before a swap is the original comb to that point. For a negative ρ step the resulting inverse plaquette has its positive base one step farther along that same prefix. Each signed swap therefore contributes the corresponding $\pm\widetilde{F}_{\rho\mu}(z)$ in the root frame. The signed number in direction ρ is $x_\rho - y_\rho$. The logarithm of the product differs from the sum of at most r small logarithms by $Cr^2\epsilon^2$, from the exponential and logarithm series on $r\epsilon$ small. Replacing each transported curvature by its base value costs at most $Hr^{1+\gamma}$. This proves eq. 6.xvi.

Proposition 6.5 (Bounded-box complex frame)**.** Let D be a rectangular box of side lengths at most Mh_j, $M \ge 1$, with root y and a complete stock of bare bonds and plaquettes on an embedded lift. Let $\alpha_0, \alpha_1 \ge 0$, and let A' be a complex Lie algebra link field per unit length. Consider $X = e^{h_0 A'} U$, where U is real, $h_j \|A'\|_{\infty,D} \le \alpha_1$, and both fields have plaquette defect at most $\alpha_0(h_0/h_j)^2$ on this stock. Suppose $M(\alpha_0 + \alpha_1) \le c_L$, for a sufficiently small positive constant

depending only on L. Let $s_X(x) = X(\Gamma_{y,x})$ and $v = s_X^{-1}$ on the level-j sites. On level-j rows with complete backward reads in D, including projected branch-test plaquettes for real arguments,

$$\mathcal{M}^j_{\mathrm{an}}(X) = V^v, \qquad V = \mathcal{M}^j_{\mathrm{an}}(X^{s_X}), \qquad \|V - I\|_\infty \le C_L M\alpha_0. \tag{6.xvii}$$

When the real first branches are admitted throughout the reads, these identities also hold for the real Borel map. The factorization $v = v'v_r$ has $v_r = s_U^{-1}$ real and independent of A', and v' holomorphic with

$$\|v' - I\|_\infty \le CM\alpha_1, \qquad \sup_x |v(x)|\,|v(x)^{-1}| \le \exp(CM\alpha_1).$$

■

Proof. Writing $d = s_X s_U^{-1}$ expresses $d(x)$ as a product of at most $4Mh_j/h_0$ link perturbations, conjugated by real prefix transports. For a reversed link the perturbation is inverted and conjugated by its real link, with unchanged norm. Consequently

$$\|d - I\|_\infty \le e^{CM\alpha_1} - 1, \qquad \sup_x \max\{|d(x)|, |d(x)^{-1}|\} \le e^{CM\alpha_1}.$$

The same estimate bounds the condition numbers of all prefixes in the comb comparison. Lemma 3.3 in that complex frame gives $|X^{s_X}(b) - I| \le C_L M\alpha_0 h_0/h_j$ on the box. Indeed its number of plaquette factors is $O(Mh_j/h_0)$ and their defects are at most $\alpha_0(h_0/h_j)^2$. The small logarithms have the same bound. Extend these logarithms by zero on a compatible larger torus to use the flat supremum-ball estimate in Proposition 6.1 at final spacing h_j. The complete reads in D are unchanged. Their normalized fine amplitude is $O_L(M\alpha_0)$, so the output logarithms are $O_L(M\alpha_0)$. The uniform chart reserve in the proof of that proposition admits conjugation back by s_X^{-1}, because its condition numbers are bounded as above. This proves eq. 6.xvii, including its holomorphic interpretation on the corresponding admitted patch.

Finally $v = s_U^{-1} d^{-1}$, so $v' = s_U^{-1} d^{-1} s_U$. The estimates for d and d^{-1} prove the asserted bounds. A pure gauge $U = I^u$ with $A' = 0$ gives $v(x) = u(x)u(y)^{-1}$. This can be far from I, even though its condition number is one and $v' = I$. □

7. Gauge identities and site and link rows

7.1. **The effective gauge transformation.** Suppose that $e^a W$ and $(e^a W)^g = e^{a^g} W$ are in the one-step chart. For complex gauges require the conjugations of Proposition 3.2 to be admitted. With the reference W held fixed, define

$$r_{W,a}(g) = u_W(a^g)^{-1}\,(g|_{\mathrm{coarse}})\,u_W(a). \tag{7.i}$$

Substitution in eq. 5.iv gives

$$e^{F_W(a^g)}\mathcal{M}_{\mathrm{an}}(W) = \big(e^{F_W(a)}\mathcal{M}_{\mathrm{an}}(W)\big)^{r_{W,a}(g)}.$$

Suppose $V = e^a W$ is contour axial relative to W, including $a = 0$. Put

$$C_x = T_W(y,x)g(x)T_W(y,x)^{-1}.$$

Then $u_W(a) = I$, and the contour ratio in $u_W(a^g)$ is $g(y)C_x^{-1}$. For a complex gauge, also require the arguments $g(y)^{-1}C_x$ of $R_W g$ to lie in its logarithm chart. On these charts, inversion and conjugation of the logarithm give

$$-\operatorname{Ad}(g(y)^{-1})\log(g(y)C_x^{-1}) = \log(g(y)^{-1}C_x).$$

Substitution in eq. 7.i gives $r_{W,a}(g) = R_W g$. If the plain iterates are mean tower axial relative to the reference, each $u_{W_i}(a_i) = I$, by eq. 5.viii. At successive stages the same identification gives $r_j = R_{W_{j-1}} \cdots R_W g$, provided the site logarithms at those stages are admitted.

For $e^{a_0}W$ and its transform by g, let a_j and a_j^g be the two recursions eq. 5.viii, with accumulated factors v_j and v_j^g, respectively. Here a_j^g denotes the recursion starting from the gauged fine field. Iterate eq. 7.i by setting $r_0 = g$ and $r_{j+1} = r_{W_j,a_j}(r_j)$. Induction gives

$$e^{a_j^g} W_j = (e^{a_j} W_j)^{r_j}, \qquad r_j = (v_j^g)^{-1} (g|_{\text{level } j}) v_j. \tag{7.ii}$$

The second identity follows by inserting the two product recursions for v_j. It specifies the residual transformations of the straight coordinates: $r_j = 1$ suffices to leave the straightened field fixed, and at $a_0 = 0$, on the admitted site charts, it becomes the normalization $R_{W_{j-1}} \cdots R_W g = 1$. For the plain average, a gauge equal to 1 on level j preserves its value by covariance.

We record the derivative of eq. 7.i at a general increment. For a Lie algebra element z put

$$E_z b = \int_0^1 \mathrm{Ad}(e^{sz}) b \, ds.$$

The left logarithmic derivative of the exponential is E_z, as follows by differentiating its convergent series. This operator has a holomorphic inverse near $z = 0$. Write $V = e^a W$, $z_x = \log[T_V(y,x) T_W(y,x)^{-1}]$, $m = L^{-4} \sum_{x \in B(y)} z_x$ and $u = e^m$. The derivative at $t = 0$ of $r_{W,a}(e^{t\lambda})(y)$ is

$$\tau_{W,a} \lambda(y) = \mathrm{Ad}(u^{-1}) \Big[\lambda(y) - E_m L^{-4} \sum_{x \in B(y)} E_{z_x}^{-1} (\lambda(y) - \mathrm{Ad}(T_V(y,x)) \lambda(x)) \Big]. \tag{7.iii}$$

Indeed the left derivative of the relative contour ratio is the expression in the inner parentheses, by covariance of T_V. Apply $E_{z_x}^{-1}$, average, and differentiate $u_W(a^{e^{t\lambda}})^{-1} e^{t\lambda(y)} u_W(a)$. At $a = 0$ the centre evaluations cancel and

$$q_W \lambda(y) = \tau_{W,0} \lambda(y) = L^{-4} \sum_{x \in B(y)} \mathrm{Ad}(T_W(y,x)) \lambda(x). \tag{7.iv}$$

On the smaller one-step ball in Proposition 5.2, $\| \tau_{W,a} - q_W \|_{\infty \to \infty} \le C_L \|a\|_\infty$. This follows from eq. 7.iii, since E_z, E_z^{-1} and $\mathrm{Ad}(e^z)$ differ from the identity by $O(\|z\|)$, and the transport depends holomorphically on the finite link increments. For a complex reference shifted from a real one by a raw increment of size δ, $\|q_W\|_{\infty \to \infty} \le 1 + C_L \delta$ by the same one-step transport comparison. Hence the effective site derivative of the j-step transformation is $\tau_{W_{j-1},a_{j-1}} \cdots \tau_{W,a_0}$. On the depth-uniform balls of Section 6, the raw increments and the changes in the reference are summable over the levels. The product is therefore bounded uniformly, and telescoping against $q_{W_{j-1}} \cdots q_W$ bounds the difference by $C_L \sum_{r<j} \|a_r\|_\infty$. In particular, the effective derivative in a fixed chart generally depends on its increment.

For a field V on links of length h, use the covariant difference with the sign fixed by $g = e^{t\lambda}$,

$$(D_V \lambda)(b) = h^{-1} [\lambda(b_-) - \mathrm{Ad}(V(b)) \lambda(b_+)]. \tag{7.v}$$

It is the generator in left multiplicative link coordinates per unit length. In the exponential coordinate $e^{hA} W$, the exact generator at A is $E_{hA}^{-1} D_{e^{hA} W}$, acting pointwise on links after the difference is taken. This follows by differentiating $e^{hA^g} W = (e^{hA} W)^g$. Differentiating eq. 7.ii therefore gives the nonlinear Ward identity. In raw output coordinates it reads

$$D_A a_j(A) \, E_{h_0 A}^{-1} D_{e^{h_0 A} W} = h_j E_{a_j(A)}^{-1} D_{e^{a_j(A)} W_j} \, \tau_{W_{j-1},a_{j-1}} \cdots \tau_{W,a_0}. \tag{7.vi}$$

The formula uses $a_0 = h_0 A$, so the input generator is per unit length and the output generator is raw. It extends by holomorphy on the same charts.

At zero increment set $Q_j'(W) = q_{W_{j-1}} \cdots q_W$, with $Q_0' = 1$. Denote by $\widehat{Q}_j^{\mathrm{st}}$ and $\widehat{Q}_j^{\mathrm{pl}}$ the straight and plain link derivatives with both input and output per unit length. They equal Q_j^{st}/t_j and Q_j^{pl}/t_j from Section 6. The physical input A becomes $h_k A$ in units of the final spacing, and the raw

output is divided by h_j. Their difference in eq. 6.v is the covariant difference of the corresponding linear site factor. Their gauge laws are

$$\widehat{Q}_j^{\mathrm{st}} D_W = D_{W_j} Q_j'(W), \qquad \widehat{Q}_j^{\mathrm{pl}} D_W = D_{W_j}(\,\cdot\,|_{\text{level } j}). \tag{7.vii}$$

The first is eq. 7.vi at zero. For the second, differentiate the endpoint covariance of $\mathcal{M}_{\mathrm{an}}^j$. In particular the straightened row annihilates $D_W \ker Q_j'$.

For a family of per-length straightened outputs let $K(c,b)$ be the derivative matrix at the admitted increment. The colour transpose ^{T} is taken in the invariant bilinear pairing. The transpose for physical link measures is

$$(K^*\omega)(b) = h_0^{-4} \sum_c h_{j(c)}^4 K(c,b)^{\mathsf{T}} \omega(c), \tag{7.viii}$$

where $j(c)$ is the level of c. This is the result of moving the kernel across the coarse weighted sum and factoring out the fine measure. For rows on one scale, the factor h_0^{-4} is canceled by the column density from eq. 6.viii. In particular the transpose has bounded supremum norm and its variation from zero to A is bounded by $C_L h_j \|A\|_\infty$, using the fixed-leg second derivative in that equation. The same proof holds for the admitted layered rows by their bounded overlap. The bilinear transpose is holomorphic in the background. For complex colours, transposing a coefficient costs at most a fixed factor from the equivalence of norms on the three-dimensional colour space; this factor is included in C_L. For raw output coordinates the derivative has the additional factor $h_{j(c)}$ on each row, which also multiplies the corresponding summand in eq. 7.viii.

On one physical scale h_j, write K_A for the per-length derivative. The same estimates give the joint bound

$$\|K_A^*\omega - K_{\widetilde{A}}^*\widetilde{\omega}\|_\infty \le C_L \|\omega - \widetilde{\omega}\|_\infty + C_L h_j \|A - \widetilde{A}\|_\infty \, \|\widetilde{\omega}\|_\infty. \tag{7.ix}$$

Use $K_A^*(\omega - \widetilde{\omega}) + (K_A - K_{\widetilde{A}})^*\widetilde{\omega}$ and integrate the fixed-input-leg second derivative along the segment between the two increments. The physical transpose cancels its fine-link density. On an admitted layered collection the same argument is applied on each complete carrier; the bounded overlap gives its weighted version with that carrier's h_j.

For the nonlinear transpose $N_A^* = (K_A - K_0)^*$ the same argument retains the radius factor in both inputs:

$$\begin{aligned} \|N_A^*\omega - N_{\widetilde{A}}^*\widetilde{\omega}\|_\infty \le C_L h_j \big(&\|A\|_\infty \|\omega - \widetilde{\omega}\|_\infty \\ &+ \|A - \widetilde{A}\|_\infty \|\widetilde{\omega}\|_\infty\big). \end{aligned} \tag{7.x}$$

Split the difference in the same way and integrate $K_A - K_0$ from zero. On a radius-r ball in both variables this gives a Lipschitz constant $C_L h_j r$.

For completeness, physical quadratic forms and currents have different coordinate powers. On a finite coordinate space let $D = \mathrm{diag}(\ell_b)$ contain the link lengths and $M = \mathrm{diag}(m_b)$ the positive physical measure weights. The raw coordinate is $a = DA$. If

$$\mathcal{Q}(A) = \tfrac{1}{2} A^{\mathsf{T}} M K A, \qquad dS(A)[H] = J^{\mathsf{T}} M H,$$

then their counting-coordinate expressions are

$$\widehat{K} = D^{-1} M K D^{-1}, \qquad \widehat{J} = D^{-1} M J. \tag{7.xi}$$

These identities follow by substituting $A = D^{-1}a$ and $H = D^{-1}\delta a$. When $\ell_b = \ell$ and $m_b = \ell^4$, they give $\widehat{K} = \ell^2 K$ and $\widehat{J} = \ell^3 J$. At unequal lengths the displayed diagonal factors retain their order; no single scale replaces them. The transpose is bilinear on the complexified coordinates.

7.2. **The relative axial gauge on nested blocks.** Let $B^j(y)$ be the bare block of side L^j with centre y. It is the union of the bare blocks of side L^{j-1} whose centres lie in the one-step block below y. For a fine site $x \in B^j(y)$ the ancestors $x = x_0, x_1, \dots, x_j = y$ are uniquely determined. For $V = e^{a_0} W$, the mean tower axial condition on g is

$$T_{\mathcal{M}^i_{\mathrm{an}}(V^g)}(x_{i+1}, x_i) = T_{W_i}(x_{i+1}, x_i) \quad (0 \le i < j).$$

For the normalized mean tower axial gauge impose also $(R_{W_{j-1}} \cdots R_W g)(y) = 1$ at each level-j centre.

Proposition 7.1 (Normalized tower axialization)**.** On the depth-uniform small charts of Proposition 6.1, the mean tower axial condition with this normalization determines a unique gauge on each tower. Its level-i transported gauge average is $(g|_{\text{level } i})v_i$, and $\mathcal{M}^j_{\mathrm{an}}(V^g) = e^{a_j} W_j$. On the real ball the same equality holds for the Borel map. The gauge is a holomorphic function of the admitted increments on its determining set and is real for real fields. Its distance from 1 is at most $C_L \sum_{i<j} \|a_i\|_\infty$ on the depth-uniform ball. ■

Proof. Set $Z_i = e^{a_i} W_i$. For an admitted solution g, the level-i field has the form $Z_i^{H_i}$ with $H_i = (g|_{\text{level } i})v_i$, by covariance and eq. 5.ix. To construct these H_i, apply eq. 5.iii to Z_i with $R_{W_i} H_i = H_{i+1}$. The only possible choice is

$$H_i(x) = T_{W_i}(y, x)^{-1} H_{i+1}(y) u_{W_i}(a_i)(y)^{-1} T_{Z_i}(y, x), \qquad x \in B(y).$$

Since $H_0 = g$, the identities $R_{W_i} H_i = H_{i+1}$ ensure that H_i are precisely its iterated averages. Start at $H_j = 1$ and descend. At $x = y$ the displayed formula gives $H_i(y) = H_{i+1}(y) u_{W_i}(a_i)(y)^{-1}$. The product rule for v_i thus makes $H_i v_i^{-1}$ agree with $H_{i+1} v_{i+1}^{-1}$ at the shared sites. This verifies the restriction of the resulting fine gauge at every level. To bound $H_i - 1$, move the reference transport to the far right of the recursion. The relative contour ratio and the block factor differ from 1 by at most $C_L \|a_i\|_\infty$. Conjugation by a real reference transport is isometric. For a shifted complex reference its norm is at most $1 + C_L \delta_i$, where δ_i is the raw shift at this level. Consequently each descent step gives

$$\|H_i - 1\|_\infty \le (1 + C_L \delta_i + C_L \|a_i\|_\infty) \|H_{i+1} - 1\|_\infty + C_L \|a_i\|_\infty.$$

The summable bounds of Section 6 control the product, and $g = H_0$. At every level the prospective gauge $H_i v_i^{-1}$ thus has condition number at most $\exp(C_L \sum_{s<j} (\delta_s + \|a_s\|_\infty))$. After reducing the ball this fits inside the reserve for the contour families and logarithms used in the factorization. Covariance is therefore admitted, so the constructed gauge satisfies the required conditions and the top field equals Z_j. The descent uses finite products of holomorphic transports and site factors, all of which are real on real inputs. □

One can also prescribe H_j at the top in place of 1, with the same chart requirement for complex gauges. For a decomposition into disjoint pillars with different top levels, perform the descent separately in each pillar. If the gauge itself is required to be 1 at each top, prescribe $H_j = v_j$ there. Relative contour equations ending at sites with already prescribed gauges remain equations on the datum. In particular a seam between different pillars does not create an additional gauge parameter.

For real fields there is a second tower gauge using one prescribed coordinate comb in each block. Take the plain iterates of the real Borel map $\mathcal{M}$, with $W_i = \mathcal{M}^i(W)$ in this construction. Write $T^\Gamma_X(y, x) = X(\Gamma_{y,x})$. Prescribe the value $g(y)$ at each top and put

$$g(x) = T^\Gamma_{W_i}(y, x)^{-1} g(y) T^\Gamma_{\mathcal{M}^i V}(y, x)$$

on the next finer level, descending from the top of each pillar. The formula follows from the same endpoint transport law as eq. 5.iii. It fixes all free sites uniquely. If x is a pinned node, keep its prescribed gauge and continue the descent below it. When y and x are both pinned to 1, the

omitted equality at x is gauge invariant under those restrictions and must be supplied as part of the datum for a representative satisfying every contour equation. This is the single-comb convention for the tower gauge in [Ba85b, §B]. The identification with eq. 5.viii above uses mean transports at every level, as do its site factors and gauge normalization.

7.3. **Pinned nodes and the action on the datum.** A tower is a finite rooted forest whose vertices are sites on the successive coarse lattices. Its edges join a site to the centre of its parent block. Copies of a centre that coincide at successive levels are identified, and the resulting zero-length parent edges are omitted. Let $\mathcal{N}$ be a set of pinned vertices containing every root. These are the endpoints of the retained coarse constraints; they may include junction nodes below a root. Put

$$\mathcal{G}_{\mathcal{N}} = \{g : \text{sites} \to G : \ g(z) = I \text{ for } z \in \mathcal{N} \text{ or } z \text{ outside the forest}\}.$$

For each forest edge (y, x) choose the same transport convention used in the corresponding average. Impose the axial condition $T_{\mathcal{M}^i U}(y, x) = I$ when x is not pinned. Omit it when x is pinned. All transports in this paragraph are assumed to lie in their real domains; single-comb transports have no domain restriction.

Proposition 7.2 (Pinned axial representative)**.** Each $\mathcal{G}_{\mathcal{N}}$ orbit in this domain has exactly one representative satisfying these conditions. For a site function $v : \mathcal{N} \to G$, define $\widetilde{v}(x)$ to be its value at the first pinned ancestor of x, including x itself, and set it to I off the forest. This lift is a group homomorphism and has the prescribed values on $\mathcal{N}$. If $\mathcal{S}(U)$ denotes the axial representative, then

$$\mathcal{S}(U^{\widetilde{v}}) = \mathcal{S}(U)^{\widetilde{v}}. \tag{7.xii}$$

If the retained constraint is $\mathcal{M}^{j(c)}U(c) = V(c)$, its endpoints being pinned, the lift sends its fibre onto the fibre with datum V^v. ■

Proof. Set $g = I$ at pinned vertices and descend from the roots. At a nonpinned child the required equation is $g(y)T_{\mathcal{M}^i U}(y, x)g(x)^{-1} = I$, so it determines $g(x) = g(y)T_{\mathcal{M}^i U}(y, x)$. At a pinned child retain its fixed value and start the descent anew below it. Covariance of the averages makes these the equations for the transformed fine field. Every free vertex has one parent, so the construction exists and is unique. The same descent applied to a transformation between two representatives forces that transformation to be I at every vertex. This proves uniqueness of the representative.

The first pinned ancestor is the same for y and a nonpinned child x. Thus $\widetilde{v}(y) = \widetilde{v}(x)$ on every imposed edge, and its identity transport remains the identity. Conjugation by $\widetilde{v}$ preserves $\mathcal{G}_{\mathcal{N}}$. It consequently sends an orbit and its unique representative to the transformed orbit and representative, proving eq. 7.xii. The lift respects products pointwise. Ordinary covariance of each $\mathcal{M}^{j(c)}$ proves the last assertion. □

At a pinned child of a pinned centre, both endpoints of the omitted transport are fixed by $\mathcal{G}_{\mathcal{N}}$, so it is constant on the orbit. Suppose y is pinned and the forest includes every site of $B(y) \setminus \{y\}$ as a child of y at this level. The full block contour energy in the representative is then

$$\sum_{\substack{x \in B(y) \setminus \{y\} \\ x \in \mathcal{N}}} \left(1 - \tfrac{1}{2} \operatorname{tr} T_{\mathcal{M}^i U}(y, x)\right). \tag{7.xiii}$$

The other summands vanish in the representative. An omitted pinned transport need not be determined by the retained link datum alone.

Let $\mathfrak{P}$ be the finite group of signed coordinate permutations preserving the periods, the lattice grids, and the family of admissible carrier configurations. For a scalar quantity $H_\Gamma(U)$ defined using a frame Γ, average $H_{s\Gamma_0}(U)$ over $s \in \mathfrak{P}$, counting each group element once. If simultaneous relabelling of the field and frame preserves the construction, the average is invariant under $\mathfrak{P}$ by reindexing the sum. On a symmetric torus and carrier family this is the full signed-permutation group of order $2^4 4!$. No average of the group-valued representatives is required.

The normalization by nonlinear site averages has a different algebraic property, examined in Section 7.5. The exact effective-gauge identities concern transformations based at the current field; they do not assert a group law for the normalized set.

7.4. **Transported site averages and outward endpoints.** Let $\mathcal{T}_j(y,x)$ be the ordered product from $y = x_j$ down to $x = x_0$ of $T_{W_{i-1}}(x_i, x_{i-1})$, $i = j, j-1, \dots, 1$. Expanding eq. 7.iv at each level and using the unique ancestors gives

$$Q'_j(W)\lambda(y) = L^{-4j} \sum_{x \in B^j(y)} \mathrm{Ad}(\mathcal{T}_j(y,x))\lambda(x). \tag{7.xiv}$$

On disjoint pillars use this formula at each top y at its assigned level $j(y)$, and denote the resulting operator by q. Use the invariant bilinear colour pairing $-\frac{1}{2}\operatorname{tr}(XY)$ and measures h_0^4 on fine sites and $h_{j(y)}^4$ at the tops. Define q^* to be the transpose for these pairings. The site lift is exactly

$$q^*\psi(x) = \mathrm{Ad}(\mathcal{T}_{j(y)}(y,x))^{-1}\psi(y), \quad x \in B^{j(y)}(y), \qquad qq^* = 1. \tag{7.xv}$$

Invariance of the colour pairing and $h_0^4 L^{4j} = h_j^4$ give the transpose formula. Inserting it in eq. 7.xiv gives the identity. For real transports the lift is an isometry for these L^2 norms and has supremum norm at most one. The algebraic identities also hold on complex charts. Norm estimates there use the condition numbers of the transports. For a complex perturbation of a real reference on the depth-uniform ball, the difference of the one-step transport from its real value is bounded by $C_L\delta_i$, where δ_i is the raw relative increment at that level. A product telescoping estimate bounds the change in every $\mathcal{T}_j(y,x)$ by $C_L \sum_{i<j} \delta_i$. This sum is bounded uniformly by Section 6. Thus each coefficient of the forward row Q'_j is bounded by $C_L L^{-4j}$ and has a bounded holomorphic continuation. The coefficients of the physical transpose q^* are instead bounded by C_L, as eq. 7.xv shows. For reference derivatives use the normalized input B in $W' = e^{\eta B}W$, with $\eta = L^{-k}$ as in Proposition 6.1. Cauchy's formula on a smaller fixed B-ball bounds an order-m derivative of a forward-row coefficient by $C_{L,m}L^{-4j}\prod_{r=1}^m \|B_r\|_\infty$. Here B_r are the derivative directions. Derivatives in raw fine-link logarithms require the corresponding factors of η^{-1}.

On a single top scale, let τ be the effective site derivative in eq. 7.vi. For a prescribed vector λ_0 of fine site values the correction

$$\lambda = \lambda_0 - q^*(\tau q^*)^{-1}\tau\lambda_0 \tag{7.xvi}$$

has $\tau\lambda = 0$. Indeed $qq^* = 1$ and the uniform bound on $\tau - q$ proves invertibility of τq^* by its Neumann series on a smaller ball. When $q\lambda_0 = 0$ the correction has size at most $C_L\|\tau - q\|_{\infty\to\infty}\|\lambda_0\|_\infty$. This is an algebraic construction in the site variables.

For a level-j determining link, the right side of eq. 7.vii uses $Q'_j\lambda$ at both endpoints. If one endpoint z is in the next finer layer, assume its full backward block $B^j(z)$ is a disjoint union of the L^4 level-$(j-1)$ pillars below it. The outer endpoint is then read as

$$Q'_j(W)\lambda(z) = L^{-4} \sum_{x \in B(z)} \mathrm{Ad}(T_{W_{j-1}}(z,x))\, q\lambda(x). \tag{7.xvii}$$

This is eq. 7.iv on the last subdivision. On an interior row the two endpoint values are the assigned values of q. Thus the same equation $\widehat{Q}_j^{\mathrm{st}} D_W = D_{W_j} Q'_j$ on an outward row uses the extra averaging eq. 7.xvii before taking the covariant difference. It annihilates $D_W \ker q$ on both kinds of rows.

7.5. **Products of transported gauge averages.** The nonlinear site averages nearly preserve multiplication on slowly varying gauge transformations. Their product defect is governed by variation within the blocks, which can be much smaller than the distances of the site values from the identity. The same distinction controls the effect of multiplying by a complex gauge transformation.

Let W be a real reference and $W_j = \mathcal{M}^j W$ for $0 \le j \le k$. Write $t_j = h_j/h_k = L^{j-k}$, and set $\mathcal{R}_0 u = u$ and $\mathcal{R}_{j+1} u = R_{W_j}(\mathcal{R}_j u)$, with R_{W_j} from eq. 5.i. For a site field u_j on level j, its transported variation within a block below y is

$$\omega_j(u_j; y) = \max_{x \in B(y)} \left\| \mathrm{Ad}(T_{W_j}(y,x)) u_j(x) - u_j(y) \right\|.$$

Write $\omega_j(u_j) = \sup_y \omega_j(u_j; y)$. For a real gauge transformation u we say that u satisfies the bounds of size α if its iterates $u_j = \mathcal{R}_j u$ are defined and

$$\|u_j - I\|_\infty \le \alpha \quad (0 \le j \le k), \qquad \omega_j(u_j) \le \alpha t_{j+1} \quad (0 \le j < k). \tag{7.xviii}$$

For real inputs, the normalized tower gauge g of Proposition 7.1 at depth k satisfies eq. 7.xviii with size $C_L \|A\|_\infty$. Its level-i iterate is H_i in that proof. At a block below y, the descent gives

$$\mathrm{Ad}(T_{W_i}(y,x)) H_i(x) - H_i(y) = H_i(y)(T_{Z_i}(y,x) T_{W_i}(y,x)^{-1} - I).$$

The right side has norm at most $C_L \|a_i\|_\infty$ by the one-step contour estimate. This is at most $C_L t_i \|A\|_\infty$ by eq. 6.i. The same descent bounds $\|H_i - I\|_\infty$ by $C_L \sum_{s=i}^{k-1} \|a_s\|_\infty \le C_L \|A\|_\infty$. Define the unscaled covariant difference of a matrix-valued site field f at a level-j bond b by

$$(\nabla^{\mathrm{raw}}_{W_j} f)(b) = f(b_-) - \mathrm{Ad}(W_j(b)) f(b_+).$$

For Lie algebra fields this is $h_j D_{W_j}$ in the convention of eq. 7.v.

Proposition 7.3 (Products and perturbations of gauge averages)**.** Suppose W is real on a torus admitting k blocking steps. Put $t_j = L^{j-k}$ for $0 \le j \le k$, and assume the fine plaquette defect is at most κt_0^2. There are $c_L > 0$ and $C_L < \infty$, independent of k, mesh and volume, for which the following hold when $0 \le \kappa, \alpha, \rho < c_L$.

(1) If two real gauges u, v satisfy eq. 7.xviii, then their pointwise product satisfies the bounds of size $2\alpha + C_L \alpha^2$. In addition,

$$\mathcal{R}_j(uv) = (\mathcal{R}_j u)(\mathcal{R}_j v) e^{z_j}, \qquad \|z_j\|_\infty \le C_L \alpha^2 t_j^2 \quad (0 \le j \le k). \tag{7.xix}$$

(2) Let v be a real gauge satisfying eq. 7.xviii. For a fine complex Lie algebra field λ with

$$\|\lambda\|_\infty \le \rho, \qquad \|\nabla^{\mathrm{raw}}_W \lambda\|_\infty \le \rho t_0,$$

the quotients $f_j = \mathcal{R}_j(e^\lambda v)(\mathcal{R}_j v)^{-1}$ are defined and holomorphic in λ on the corresponding open ball. They satisfy

$$\begin{aligned} \|f_j - I\|_\infty &\le C_L \rho, & \|\nabla^{\mathrm{raw}}_{W_j} f_j\|_\infty &\le C_L \rho t_j, \\ \|\log f_j - Q_j'(W)\lambda\|_\infty &\le C_L(\alpha + \rho)\rho t_j. \end{aligned} \tag{7.xx}$$

Here the final term is the site row from eq. 7.xiv, and the logarithm is the branch at I.

■

Proof. We first compute a one-step quotient. On a block below y write $P_x = \mathrm{Ad}(T_{W_j}(y,x))$. For site fields f, v on level j, define

$$X_x = \log(f(y)^{-1} P_x f(x)), \qquad Y_x = \log(v(y)^{-1} P_x v(x)), \qquad \widetilde{Y}_x = \mathrm{Ad}(v(y)) Y_x.$$

In the one-step computation v is real and f is either real or in a fixed small complex ball about I. Take the variations ω_f, ω_v of these fields in this block small enough that $\|X_x\| \le C\omega_f$ and $\|\widetilde{Y}_x\| \le C\omega_v$. Denote the average over $x \in B(y)$ by brackets, and let $\mathcal{L}(X,Y) = \log(e^X e^Y)$. Transport is multiplicative on matrix products. Thus

$$(f(y)v(y))^{-1} P_x (f(x)v(x)) = v(y)^{-1} e^{X_x} v(y) e^{Y_x}.$$

Conjugating the defining logarithms in eq. 5.i by $v(y)$ gives

$$\begin{aligned} R_{W_j}(fv)(y)R_{W_j}(v)(y)^{-1} &= f(y)\exp(\langle\mathcal{L}(X_x,\widetilde{Y}_x)\rangle)\exp(-\langle\widetilde{Y}_x\rangle) \\ &= f(y)\exp(\langle X_x\rangle + e(y)), \qquad \|e(y)\| \le C\omega_f\omega_v. \end{aligned} \tag{7.xxi}$$

To see the bound, $\mathcal{L}(X,Y) - X - Y$ is analytic on a fixed small ball and vanishes when either input is zero. Integrating its bounded mixed derivative gives $C\|X\|\|Y\|$. The final logarithm of the product of exponentials in the first line has the same property after subtracting $\langle X_x\rangle$. Indeed, put $A = \langle X_x\rangle$, $B = \langle\widetilde{Y}_x\rangle$ and $E = \langle\mathcal{L}(X_x,\widetilde{Y}_x) - X_x - \widetilde{Y}_x\rangle$. The logarithm is $\mathcal{L}(A+B+E,-B)$. Changing E to zero costs $C\|E\|$, and $\mathcal{L}(A+B,-B) - A$ vanishes on both coordinate axes and has a bounded mixed derivative. This proves eq. 7.xxi. The argument uses only fixed matrix logarithm balls. A sufficiently small bound on $\omega_f + \omega_v$ admits all its relative logarithms.

Two consequences will be used. First, $R_{W_j}(f)(y) = f(y)e^{\langle X_x\rangle}$, so the quotient in eq. 7.xxi differs from $R_{W_j}(f)(y)$ by at most $C\omega_f\omega_v$. Second, if $F(y) = \langle P_x f(x)\rangle$, the same quotient differs from $F(y)$ by at most

$$C\omega_f(\omega_f + \omega_v). \tag{7.xxii}$$

Indeed $F(y) = f(y)\langle e^{X_x}\rangle$ and the constant and linear terms in its expansion coincide with those of $f(y)e^{\langle X_x\rangle}$.

For the first assertion, set $u_j = \mathcal{R}_j u$, $v_j = \mathcal{R}_j v$, $p_j = u_j v_j$ and $d_j = \|\mathcal{R}_j(uv) - p_j\|_\infty$ whenever the iterate is defined. Since the two fields are unitary, $\omega_j(p_j) \le 2\alpha t_{j+1}$. If $\mathcal{R}_j(uv) = g_j p_j$, then g_j is unitary, $\|g_j - I\|_\infty = d_j$ and $\omega_j(g_j) \le 2d_j$. Formula eq. 7.xxii for $f = g_j$, $v = p_j$ bounds the next quotient's distance from I by $d_j + Cd_j(d_j + \alpha t_{j+1})$. The first consequence of eq. 7.xxi, applied to u_j, v_j, then gives

$$d_{j+1} \le d_j(1 + C(d_j + \alpha t_{j+1})) + C\alpha^2 t_{j+1}^2, \qquad d_0 = 0. \tag{7.xxiii}$$

Choose K so $2KL^{-2} + C \le K$, then decrease the ceiling on α so the coefficient of d_j is at most two under $d_j \le K\alpha^2 t_j^2$. The same ceiling admits the logarithms at step $j+1$, since the variation of $g_j p_j$ is bounded by $2d_j + 2\alpha t_{j+1}$. Induction proves that every iterate is defined and $d_j \le K\alpha^2 t_j^2$. Taking the series logarithm of $p_j^{-1}\mathcal{R}_j(uv)$ proves eq. 7.xix. The distance to I is at most $2\alpha + d_j$ and the variation at most $2\alpha t_{j+1} + 2d_j$. These are eq. 7.xviii with the asserted enlarged size.

For the second assertion, set $v_j = \mathcal{R}_j v$. At level zero $f_0 = e^\lambda$. A recursive description of the quotient is

$$f_{j+1} = R_{W_j}(f_j v_j)R_{W_j}(v_j)^{-1}.$$

To estimate its covariant differences, let

$$m_j = \|f_j - I\|_\infty, \qquad d_j = \|\nabla^{\mathrm{raw}}_{W_j} f_j\|_\infty, \qquad \epsilon_j = \sup_p \|W_j(\partial p) - I\|.$$

By Corollary 4.1, $\epsilon_j \le C_L\kappa t_j^2$, since the small logarithm and group distance are comparable. For the mean transport $T_{W_j}(y,x)$, any of its coordinate paths is within $C_L\epsilon_j$ of that mean in operator norm, by the path interchange estimate and the group mean bound. Transport $f_j(x) - I$ along such a path and telescope its covariant differences. The path has at most $2L$ links, so

$$\omega_j(f_j) \le C_L(d_j + \epsilon_j m_j). \tag{7.xxiv}$$

For the arithmetic transported output $F(y) = L^{-4}\sum_{x\in B(y)} P_x f_j(x)$, one has $\|F - I\|_\infty \le m_j$. For a coarse bond $c = (y,y')$ let $x' = x + (y' - y)$. Compare the transport $T_{W_j}(y,x)W_j([x,x'])$ with $W_{j+1}(c)T_{W_j}(y',x')$. The loop correction in eq. 3.iv bounds their difference by $C_L\epsilon_j$ for each x. Thus

$$\|F(y) - \mathrm{Ad}(W_{j+1}(c))F(y')\| \le Ld_j + C_L\epsilon_j m_j.$$

In fact the change of transport is applied only to $f_j(x') - I$, which gives the second term. The first term comes from the L adjacent differences along $[x, x']$ in unitary frames, averaged over x. Set $s_j = C_L(d_j + \epsilon_j m_j)$, with C_L large enough for eq. 7.xxiv. By eq. 7.xxii,

$$\begin{aligned} m_{j+1} &\le m_j + C_L s_j(s_j + \alpha t_{j+1}), \\ d_{j+1} &\le L d_j + C_L \epsilon_j m_j + C_L s_j(s_j + \alpha t_{j+1}). \end{aligned} \tag{7.xxv}$$

The coefficient L here comes from the straight segment count, even though the individual site transports use the contour mean.

The exponential derivative bound on the ρ-ball gives $m_0 \le e^\rho \rho$, $d_0 \le e^\rho \rho t_0$. As long as $m_j \le 2\rho$ and $d_j \le 2\rho t_j$, one has $s_j \le C_L \rho t_j$. The additive terms in eq. 7.xxv are therefore at most $C_L(\kappa + \alpha + \rho)\rho t_j^2$. For m_{j+1} sum these terms directly. For d_{j+1}/t_{j+1} the term $L d_j/t_{j+1}$ is d_j/t_j, and the remaining terms sum to at most $C_L(\kappa + \alpha + \rho)\rho \sum_j t_j$. Both geometric sums are bounded independently of depth. After reducing the ceilings these estimates give the same two bounds at the next step, with strict reserve when $\rho > 0$. The small variations also admit all relative logarithms in eq. 7.xxi. When $\rho = 0$ the quotient is I at every level. Holomorphy follows from the admitted finite formulas.

It remains to compare the quotient logarithm with the linear row. At a block centre put $Z = \log f_j(y)$. The logarithms of $P_x f_j(x)$ are $\mathcal{L}(Z, X_x)$. The logarithm of the quotient at the next level is $\mathcal{L}(Z, \langle X_x \rangle + e(y))$ by eq. 7.xxi. The same mixed derivative estimate for $\mathcal{L}$ now gives

$$\| \log f_{j+1} - q_{W_j} \log f_j \|_\infty \le C_L s_j(m_j + \alpha t_{j+1}) \le C_L(\alpha + \rho)\rho t_j.$$

Here the linear terms $Z + \langle X_x \rangle$ cancel. The operator q_{W_j} in eq. 7.iv has supremum norm at most one for a real reference. Since $\log f_0 = \lambda$, iteration and $\sum_{s<j} t_s \le t_j/(L-1)$ prove the last line of eq. 7.xx. □

The bounds adapt the product and quotient estimates for a single contour in [Ba85a, §F, Props. 8–10]. In particular the defect in eq. 7.xix need not vanish when the two separately averaged gauges are I. At the trivial reference, choose two noncentral sites of one block and let the two gauges take e^X, e^{-X} and e^Y, e^{-Y} there, respectively, with all other values I. Their separate averages are I. The logarithm of the product average has leading term $L^{-4}[X, Y]$, which is nonzero for arbitrarily small noncommuting X, Y.

The effective gauge about a nonzero field also has a uniform finite Taylor estimate. Measure lengths in final-scale units, put $h = L^{-k}$, and let the real reference W have plaquette defect at most κh^2. For a complex site logarithm set $\|\lambda\|_1 = \max\{\|\lambda\|_\infty, \|D_W \lambda\|_\infty\}$, where D_W divides by h. If $\kappa, \|A\|_\infty, \|\lambda\|_1$ are sufficiently small depending on L, define the transformed fine coordinate by $(e^{hA}W)^{e^\lambda} = e^{hA^\lambda}W$ and use the effective gauge r_k in eq. 7.ii. Then

$$\| \log r_k(A, e^\lambda) - Q_k'(W)\lambda \|_\infty \le C_L(\|A\|_\infty + \|\lambda\|_1)\|\lambda\|_1. \tag{7.xxvi}$$

The map and the logarithm are holomorphic on this joint ball, uniformly in mesh, depth and volume.

Here is a direct admission and derivative argument. At a link write $\lambda_+ = \mathrm{Ad}(W(b))\lambda(b_+)$ and $\lambda_- = \lambda(b_-)$. Their difference has norm at most $h\|\lambda\|_1$. The exact relative transformed link is $e^{\lambda_-} e^{hA(b)} e^{-\lambda_+}$. Factoring out the conjugate of $e^{hA(b)}$ and integrating the exponential between λ_+ and λ_- shows that its logarithm is $O(h(\|A\|_\infty + \|\lambda\|_1))$ on a fixed complex ball. Division by h therefore yields a uniformly bounded holomorphic A^λ. Proposition 6.1 admits both fine fields and bounds their accumulated factors. The effective factors $(v_j(A^\lambda))^{-1}(e^\lambda|_{\text{level } j})v_j(A)$ are close to I at every level, uniformly in depth. Their endpoint conjugations therefore fit the one-step chart reserves. The exact formula $r_k = v_k(A^\lambda)^{-1} e^{\lambda|_{\text{coarse}}} v_k(A)$ then gives a uniformly bounded logarithm on a larger joint ball. Its value at $\lambda = 0$ is zero. Its λ derivative at $(A, \lambda) = (0, 0)$ is $Q_k'(W)$ by eq. 7.iv and composition. Cauchy's formula on the larger ball bounds its second derivatives independently of dimension. Integrating the λ derivative from zero to λ gives eq. 7.xxvi.

7.6. **The flat link rows.** At the trivial reference, every one-step straight derivative is S_1. For $n = L^j$ abbreviate $\widehat{Q}_j^{\mathrm{st}}$ by $\widehat{Q}_j$ in this subsection. Iterating eq. 5.v gives

$$(\widehat{Q}_j A)_\mu(y) = n^{-5} \sum_{\substack{u\in\mathbb{Z}^4 \\ |u|_\infty \le (n-1)/2}} \sum_{s=0}^{n-1} A_\mu(y + h_0 u + h_0 s e_\mu). \tag{7.xxvii}$$

For the induction, each coarse start and its finer offset specify one fine start in the larger block. The segment indices at the two steps combine as $n s_{\mathrm{coarse}} + s_{\mathrm{fine}}$, each occurring once. This proves the stated normalization and support. The coefficients are nonnegative, sum to one and are each at most n^{-4}, since at most n pairs of longitudinal indices give any one input. With the fine measure h_0^4, Cauchy's inequality therefore gives on its double block D_c

$$|(\widehat{Q}_j A)(c)| \le h_j^{-2} \|A_\mu\|_{L^2(D_c)}. \tag{7.xxviii}$$

For a flat row at a seam, assign length h_j/L to fine links internal to the tail block and h_j to the remaining links. In one longitudinal line there are n^2 segment occurrences, of which $\sum_{s=1}^{n-1} s = n(n-1)/2$ are internal to the tail block. The absolute weighted sum of the raw row coefficients with weights $1/\xi(b)$ is therefore

$$1 + (L-1)\frac{n-1}{2n}.$$

This exhibits the adjacent-scale cost in the weighted row bound of Section 6 using the actual flat coefficients.

The exact Fourier multiplier on $A_\mu(x) = e^{ip\cdot x} a$ is the product

$$\left(\prod_{\nu=1}^{4} \frac{1}{n} \sum_{u=-(n-1)/2}^{(n-1)/2} e^{i h_0 p_\nu u} \right) \left(\frac{1}{n} \sum_{s=0}^{n-1} e^{i h_0 p_\mu s} \right),$$

times $e^{ip\cdot y} a$. The site multiplier omits the last factor. Both formulas include zero frequencies by their finite sums.

The map $\widehat{Q}_j$ has an exact right inverse obtained by assigning $nB(c)$ to each fine edge crossing the face between the two bare blocks of c, and zero to the internal edges. Each segment crosses its designated face once, so eq. 7.xxvii sends this lift to B. With the physical link measures h_0^4 and h_j^4, the Gram operator $\widehat{Q}_j \widehat{Q}_j^*$ has diagonal coefficient $(2+n^{-2})/3$ and coefficient $(1-n^{-2})/6$ on each of its two nearest rows in direction μ. There is no overlap with transverse shifts or other components. To see the coefficients, along the segment direction the multiplicities are $1, 2, \dots, n-1, n, n-1, \dots, 1$. Their squared sum is $(2n^3+n)/3$. The overlap with the list translated by n is $\sum_{s=1}^{n-1} s(n-s) = (n^3-n)/6$. The transverse multiplicity contributes n^3, and the weighted transpose contributes n^4. With the two factors n^{-5} these reduce to n^{-3} times the two sums above. Periodization combines coefficients when neighbors coincide. In particular

$$\tfrac{1}{3} I \le \widehat{Q}_j \widehat{Q}_j^* \le I.$$

7.7. **Comparison with coordinate combs.** The one-step block analysis also applies to a single coordinate comb in place of the contour mean. To compare with the corner-based blocks of [Ba85a], identify the blocks by translating their representatives at level j from the corner to the centre. Relative to the common fine grid this displacement is $\frac{1}{2}(L^j - 1) h_0 (1,1,1,1)$. Since L is odd, each translation aligns the tiling at that level. Formula eq. 3.v uses the chosen contour sums in the plain derivative. The subtraction in eq. 5.iv removes both sums for either choice. Thus the flat straight and site coefficients agree after alignment.

For real W, let D be the projected vertex set of the two depth-j blocks of a row, with an embedded rectangular lift. Assume every physical fine plaquette with all vertices in D has defect at most αL^{-2j}, where α is sufficiently small depending on L. This includes opposite-face tests at

the shortest period. By alignment, D is a union of complete blocks at each preceding level. The local curvature argument in Corollary 4.2 therefore admits all intermediate real branch tests. The logarithm and group-distance bounds are comparable on these small charts.

The coordinate tree gauge on the rectangular lift makes its raw fine link logarithms bounded by $C_L\alpha L^{-j}$, by Lemma 3.3. Its value at the centre of the tail block is 1. The additional projected edges enter through their curvature tests; the following link-logarithm bounds concern the literal path stock. In this gauge the raw reference logarithms of either scheme at intermediate level i are bounded by $C_L\alpha L^{i-j}$. Indeed the one-step estimates of Proposition 5.2 apply to the comb scheme by the same finite-path argument. Starting at the trivial reference, the straight recursion has $\|a_{i+1}\| \le L\|a_i\|(1 + C_L\|a_i\|)$, with $\|u_i - 1\| \le C_L\|a_i\|$. The geometric sum of the raw radii and the factorization bound both the straight and plain logarithms by $C_L\alpha L^{i-j}$, as in Section 6. Restrict these computations to the backward determining blocks of the row.

In this common fine gauge a telescoping product of one-step link derivatives against S_1 gives row error at most $C_L\alpha$ and column error at most $C_L\alpha L^{-4j}$ for the per-length straight row. At each insertion the relative one-step error is $C_L\alpha L^{i-j}$ in both sums. The normalized straight product has row scale 1 and column scale L^{-4j}, so summing these errors proves the bounds. In particular each entry error has the column bound, and an input edge appears in only a fixed number of level-j rows. In the site product the transport in this gauge differs from 1 by at most $C_L \sum_{i<j} \alpha L^{i-j}$, which gives the same error scales directly from eq. 7.xiv.

Returning to the original gauge compares the rows in their different output frames. If y, z are the centre and corner of the tail block, the corner row is transported to y by $\mathrm{Ad}(W(\gamma_{yz}))$ before subtraction, where γ_{yz} is the path of this common coordinate tree. Both rows acquire the same inverse transformation at each input site or link tail, which is isometric. Thus the real comparison is local, with the stated row and column errors, in the specified output frame. It applies to the straight and site rows used in the gauge identities above.

7.8. **Smooth extension of a coarse gauge.** Take a real flat reference W on the torus, measure lengths in units of the final coarse spacing, and put $h = L^{-k}$. The fine and coarse lattices on the cover are $h\mathbb{Z}^4$ and $\mathbb{Z}^4$. Their common periods are $M_\mu e_\mu$ with M_μ integers at least two. The bounds below for a single gauge and its extension also apply to every odd mesh $h = 1/n$, $n \ge 1$.

On the covering lattice put $P(x) = W(\gamma_{0x})$, where γ_{0x} is any path from 0 to x. The path interchange identity and flatness give path independence. In particular $P(x+he_\mu) = P(x)W(x,\mu)$, so the gauge P makes W the identity on the cover. For a period vector a, periodicity gives $P(x+a) = H_a P(x)$ with $H_a = P(a)$. The H_a form a commuting family of unitary matrices. A site field, or a link increment attached to its initial point x, is represented in this frame by conjugation with $P(x)$. Its representative satisfies the deck law $f(x+a) = \mathrm{Ad}(H_a)f(x)$. All the following norms and ordinary differences are taken in this frame.

For $0 < \alpha < 1$, put

$$[f]_{\alpha,\mathrm{loc}} = \sup_{\substack{x,x'\in h\mathbb{Z}^4 \\ 0<|x-x'|\le 1}} \frac{|f(x) - f(x')|}{|x - x'|^\alpha}, \qquad \nabla_{h,\mu} f(x) = \frac{f(x + he_\mu) - f(x)}{h}.$$

The distance between sites here is Euclidean. For an array, such as the four positive link components, take the maximum over components. Let ∇_h^r consist of all products of r forward differences. Define

$$\begin{aligned} \|f\|_{X_\alpha} &= \|f\|_\infty + \|\nabla_h f\|_\infty + [\nabla_h f]_{\alpha,\mathrm{loc}}, \\ \|f\|_{Y_\alpha} &= \|f\|_\infty + \|\nabla_h f\|_\infty + \|\nabla_h^2 f\|_\infty + [\nabla_h^2 f]_{\alpha,\mathrm{loc}}. \end{aligned} \tag{7.xxix}$$

Simultaneously translating both sites by a period conjugates their values by the same unitary matrix and preserves these norms. To compute the local seminorm on period representatives, choose the first site in a fundamental domain and retain every lift of the second site at distance at most one. If several short lifts represent the same torus pair, all enter the supremum. On the torus each lift

uses parallel transport of W along that displacement. The covariant difference D_W of eq. 7.v is $-\nabla_h$ in the parallel frame.

Lemma 7.4 (Extensions of coarse site data)**.** Fix $0 < \alpha < 1$. Let W be a real flat connection on the periodic lattices of this subsection, with fine spacing $h = L^{-k}$, coarse spacing one, and integer periods $M_\mu \geq 2$. Use the parallel-frame norms in eq. 7.xxix. There is a linear operator E from coarse $\mathfrak{g}_{\mathbb{C}}$-valued site fields to fine site fields with

$$(E\nu)|_{\text{coarse}} = \nu, \qquad \|E\nu\|_{Y_\alpha} \leq C_\alpha \|\nu\|_\infty. \tag{7.xxx}$$

The column at y is supported within distance $1/4$ of y in the maximum norm, inside its centered block. The constants are uniform in the mesh, the periods, and the flat holonomies. For coarse w in a fixed logarithm chart at the identity,

$$\mathfrak{E}(w) = \exp(E \log w) \tag{7.xxxi}$$

has restriction w to the coarse sites. It is holomorphic on the complex chart and group-valued in G for real data. For the transported site average $q = Q'_k(W)$ of eq. 7.xiv, there is a smooth lift E_q with the same column supports and

$$qE_q = I, \qquad \|E_q\nu\|_{Y_\alpha} \leq C_\alpha \|\nu\|_\infty, \qquad \|E_q\nu\|_{H^1_h} \leq C\|\nu\|_{\ell^2}. \tag{7.xxxii}$$

Here the fine H^1_h norm squares and sums the field and its first differences with weight h^4, and the coarse ℓ^2 norm uses counting measure. Each sum is over one period. ∎

Proof. Choose a nonnegative smooth function χ supported in $(-1/4, 1/4)^4$ with $\chi(0) = 1$ and a positive lower bound on $[-1/8, 1/8]^4$. In the parallel frame extend the coarse site function ν to a smooth function on the cover by

$$x \longmapsto \sum_{y \in \mathbb{Z}^4} \chi(x - y)\nu(y).$$

The supports are disjoint, and translation by a period gives the same deck law as for ν. The sum restricts to ν at every coarse node. Upon sampling on the fine lattice it defines $E\nu$. In the original frame its column is $\chi(x-y)\,\mathrm{Ad}(W(\gamma_{yx})^{-1})\nu(y)$. One may take the path γ_{yx} inside the support box whenever its coefficient is nonzero.

The suprema of all continuous derivatives of the sum up to order three are at most $C\|\nu\|_\infty$, by disjointness of the supports. A product of r fine differences of the sample, $r \leq 2$, is the average of the corresponding continuous derivative over r shifts in $[0, h]$. Subtract the two such averages at x, x' to bound their difference for $r = 2$ by $C\|\nu\|_\infty |x - x'|$, using the third derivative. This proves eq. 7.xxx at every mesh, including $h = 1$, when sampling gives the coarse nodal values themselves. The convergent matrix logarithm and exponential give eq. 7.xxxi, preserving the deck law and the real form on the real chart.

At the flat reference all contour transports in the parallel frame are the identity. The site row is therefore the arithmetic mean $qf(y) = h^4 \sum_{x \in B^k(y)} f(x)$. The only bump meeting that block is the bump at y. Thus $qE = m_h I$, where $m_h = h^4 \sum_{x \in h\mathbb{Z}^4} \chi(x)$. There are at least $(8h)^{-1}$ lattice sites in each coordinate of $[-1/8, 1/8]^4$, and at most Ch^{-4} sites in the support. This gives a positive lower bound and a finite upper bound for m_h, uniform in h. Take $E_q = m_h^{-1} E$. For the energy bound, each sampled bump and each of its first differences is bounded by a fixed constant. A difference of a bump is supported on the fine links with an endpoint in its support. Such links number at most Ch^{-4} per bump, and a given link meets the supports of at most two bumps. Sum the squared values of the columns and use the measure h^4. This proves the last estimate in eq. 7.xxxii. □

Proposition 7.5 (Finite gauge in the strong norm)**.** Let W be real and flat, fix $0 < \alpha < 1$, and use the parallel-frame norms eq. 7.xxix on a periodic mesh $0 < h \leq 1$. For a positive-link $\mathfrak{g}_{\mathbb{C}}$-valued potential A per unit length and a $\mathfrak{g}_{\mathbb{C}}$-valued site field λ, define A^λ by $(e^{hA}W)^{e^\lambda} = e^{hA^\lambda}W$, using

the small logarithm on each link. There is $r_\alpha > 0$ such that for $\|A\|_{X_\alpha} + \|\lambda\|_{Y_\alpha} < r_\alpha$ this coordinate is well defined and holomorphic, and

$$\begin{aligned} A^\lambda &= A + D_W\lambda + \mathcal{B}_h(A,\lambda), \\ \|\mathcal{B}_h(A,\lambda)\|_{X_\alpha} &\le C_\alpha(\|A\|_{X_\alpha} + \|\lambda\|_{Y_\alpha})\|\lambda\|_{Y_\alpha}. \end{aligned} \tag{7.xxxiii}$$

For two inputs with norm sums at most $r < r_\alpha/2$,

$$\begin{aligned} &\|\mathcal{B}_h(A,\lambda) - \mathcal{B}_h(\widetilde{A},\widetilde{\lambda})\|_{X_\alpha} \\ &\qquad \le C_\alpha r(\|A - \widetilde{A}\|_{X_\alpha} + \|\lambda - \widetilde{\lambda}\|_{Y_\alpha}). \end{aligned} \tag{7.xxxiv}$$

The constants and radius are uniform in $0 < h \le 1$, the periods and the flat holonomies. The coordinate is real for real inputs. Its value on a link uses its field value and the two endpoint gauge values. ■

Proof. In the parallel frame, for $b = (x,\mu)$ the exact formula is

$$A^\lambda_\mu(x) = h^{-1}\log\Big(e^{\lambda(x)}e^{hA_\mu(x)}e^{-\lambda(x+he_\mu)}\Big). \tag{7.xxxv}$$

The product is precisely the gauge-transformed link in this frame. Write $p = \nabla_{h,\mu}\lambda$, so that the last exponent in eq. 7.xxxv is $-\lambda(x) - hp(x)$. This exposes the factor of h before differentiation.

For matrix fields f on the cover, a coordinate path of length at most $2|x - x'|$ gives $[f]_{\alpha,\mathrm{loc}} \le 2\|\nabla_h f\|_\infty$. The product rule $\nabla_{h,\mu}(fg) = (\nabla_{h,\mu}f)(g(\cdot + he_\mu)) + f\nabla_{h,\mu}g$ and the product inequality for the local seminorm show that

$$\|fg\|_{X_\alpha} \le C_\alpha\|f\|_{X_\alpha}\|g\|_{X_\alpha}.$$

The same bound holds with translated factors. Also $\|\lambda\|_{X_\alpha} + \|p\|_{X_\alpha} \le C_\alpha\|\lambda\|_{Y_\alpha}$, by applying the path bound to the first differences of λ.

For one component, use the complex scalar ζ in the product

$$B_\zeta = e^{\lambda}e^{\zeta A_\mu}e^{-\lambda-\zeta p}.$$

Treat A_μ, λ, p as fixed matrix fields. In the product norm above, the exponential series converge and $B_0 = I$. Termwise differentiation of the series bounds $\|\partial_\zeta B_\zeta\|_{X_\alpha}$ by $C_\alpha(\|A_\mu\|_{X_\alpha} + \|p\|_{X_\alpha})$ for $|\zeta| \le 2$ on a fixed small input ball. Hence $B_\zeta - I$ is bounded in that norm by this expression times $|\zeta|$. On a smaller fixed ball the logarithm series converges in the product norm. Thus $\zeta^{-1}\log B_\zeta$ extends analytically through zero, with X_α bound $C_\alpha(\|A_\mu\|_{X_\alpha} + \|p\|_{X_\alpha})$. Take $\zeta = h$. The resulting small logarithms are traceless, and are skew-Hermitian for real inputs. Conjugation equivariance of the product and logarithm gives the deck law and the coordinate in the original frame.

The bound on a fixed larger complex input ball gives uniform second derivatives of the coordinate on a smaller ball. For example, two applications of Cauchy's formula in scalar directions of norm one give the bilinear derivative bound with radii fixed by the separation of the two balls. The linearization at zero of eq. 7.xxxv is $A_\mu - p$. At $\lambda = 0$ the formula is exactly A_μ. Hence the nonlinear difference $\mathcal{B}_h$ has zero derivative at the origin and vanishes at every $(A, 0)$ in the ball. Integrate its λ-derivative from zero to λ and use the uniform second derivative bound. This gives eq. 7.xxxiii. On the radius-r ball the entire first derivative of $\mathcal{B}_h$ is bounded by $C_\alpha r$, which yields eq. 7.xxxiv by integration on the segment joining the inputs. □

In particular the actual infinitesimal gauge generator $\Gamma_A\lambda = \frac{d}{dt}A^{t\lambda}\Big|_{t=0}$ satisfies

$$\|(\Gamma_A - D_W)\lambda\|_{X_\alpha} \le C_\alpha\|A\|_{X_\alpha}\|\lambda\|_{Y_\alpha}.$$

This follows by differentiating the remainder with respect to λ at zero, using its second derivative bound.

The variation of the covariant difference itself has order zero. For $U = e^{hA}W$, write $X = \mathrm{Ad}(W(b))\lambda(b_+)$. At a positive link b, direct subtraction in eq. 7.v gives

$$\begin{aligned}((D_U - D_W)\lambda)(b) &= h^{-1}(I - \mathrm{Ad}(e^{hA(b)}))X \\ &= -\int_0^1 \mathrm{Ad}(e^{thA(b)})[A(b), X]\, dt.\end{aligned} \tag{7.xxxvi}$$

The integral follows by differentiation of the adjoint action. It gives $\|D_U - D_W\|_{\infty\to\infty} \le C\|A\|_\infty$ on a fixed potential ball. The same bound holds from physical site L^2 to physical link L^2, both with measure h^4: for each direction the endpoint shift preserves the norm, and there are four directions. In the flat parallel frame the product estimate in X_α also gives $\|(D_U - D_W)\lambda\|_{X_\alpha} \le C_\alpha\|A\|_{X_\alpha}\|\lambda\|_{X_\alpha}$ on a fixed strong ball. For the derivatives, use $\mathrm{Ad}(e^{hA}) = \exp(h\,\mathrm{ad}\,A)$, where $\mathrm{ad}\,A$ is the commutator with A. The m-th derivative, $m \ge 1$, of $D_U - D_W$ in directions $A_1, \dots, A_m$ has operator norm at most

$$C_m h^{m-1} \prod_{i=1}^m \|A_i\|_\infty$$

in the two indicated supremum and physical L^2 operator norms. Indeed the derivatives of the matrix exponential are bounded on the fixed ball of $h\,\mathrm{ad}\,A$, and the chain rule gives one factor h per direction against the single h^{-1} in eq. 7.xxxvi. The same argument in the product norm gives the X_α operator bound with X_α norms of the directions. These estimates apply to the holomorphic continuation.

Proposition 7.6 (Correction to the effective site kernel)**.** Let W be a real flat reference on the periodic lattices of Lemma 7.4, with $h = L^{-k}$, and fix $0 < \alpha < 1$. Let $q = Q'_k(W)$ and let E_q be its smooth right inverse in eq. 7.xxxii. Denote by τ_A the effective site derivative in eq. 7.vi, with $a_0 = hA$ and final depth k. For $\|A\|_\infty < c_L$, the operator $E_{q,A} = E_q(\tau_A E_q)^{-1}$ is a right inverse of τ_A and satisfies the two norm bounds in eq. 7.xxxii, with constants depending also on L. For a fine site field $\lambda_0 \in \ker q$, put

$$\lambda_A = \lambda_0 - E_{q,A}\tau_A\lambda_0. \tag{7.xxxvii}$$

Then $\tau_A\lambda_A = 0$ and $\|\lambda_A - \lambda_0\|_{Y_\alpha} \le C_{L,\alpha}\|A\|_\infty\|\lambda_0\|_{Y_\alpha}$. The correction is holomorphic in A on this ball. ■

Proof. The one-step site estimate following eq. 7.iv and the summable bounds of proposition 6.1 give $\|\tau_A - q\|_{\infty\to\infty} \le C_L\|A\|_\infty$. Here q is the product of the linear site rows at zero. By eq. 7.xxxii, $\tau_A E_q$ therefore differs from I by less than $1/2$ on the stated ball after decreasing c_L. Its inverse is given by the convergent Neumann series. The site factors in eq. 7.iii read only the one-step block in their site argument, so the product τ_A reads its site argument only on $B^k(y)$ at y. Since the columns of E_q lie in their respective blocks, $\tau_A E_q$ and its inverse act separately at each coarse site. The inverse is consequently bounded on coarse ℓ^2 as well as on coarse supremum norm. Both lift estimates follow from those of E_q. Applying τ_A to eq. 7.xxxvii gives zero, and $\tau_A\lambda_0 = (\tau_A - q)\lambda_0$ gives the stated error. The site formulas and the Neumann series give holomorphy. □

The extension converts the straightened mean into a prescribed literal one without solving a gauge equation. In eq. 5.ix, use $a_0 = hA$, write $W_k = \mathcal{M}^k_{\mathrm{an}}(W)$ and denote the top factor by $v_k(A)$. Set

$$\widehat{U}_A = (e^{hA}W)^{\mathfrak{E}(v_k(A)^{-1})}, \qquad \mathcal{M}^k_{\mathrm{an}}(\widehat{U}_A) = e^{a_k(A)}W_k. \tag{7.xxxviii}$$

Indeed the gauge on the coarse lattice is $v_k(A)^{-1}$, which cancels $v_k(A)$ in the exact factorization. By proposition 6.1 and eq. 7.xxx, the fine gauge logarithm has Y_α norm bounded by $C_{L,\alpha}\|A\|_\infty$. Consequently the fine potential of $\widehat{U}_A$ has X_α norm at most $C_{L,\alpha}\|A\|_{X_\alpha}$. On a smaller fixed X_α ball both fine potentials are in the admitted averaging charts. Analytic covariance gives the coarse identity on this ball. For real fields the iteration's first-branch admission also gives the identity for

the Borel map. The coefficient at a fine link reads the link and the backward blocks of the nodes whose bumps meet its endpoints, a region of bounded diameter in coarse units.

8. Local variations and changes of frame

8.1. **Compatible local frames.** Nearby real SU(2) frames can be combined without leaving the group. Suppose $g_1, \dots, g_N \in G$ obey $\|g_i g_1^{-1} - I\| < r < 1/2$, and let $\theta_i \geq 0$ with $\sum_i \theta_i = 1$. Put $X = \sum_i \theta_i g_i$. Then

$$g = X(\det X)^{-1/2} \in G, \tag{8.i}$$

where the square root is positive on these real inputs. In the quaternion representation X is a real quaternion matrix, so $X^*X = (\det X)I$ and $\det X > 0$. Moreover $\|Xg_1^{-1} - I\| < r$, which prevents $X = 0$ and gives a fixed margin for normalization. This proves the assertion. On a smaller complex neighborhood the determinant avoids the nonpositive real axis. Take its principal square root, continuing the positive real value. Formula eq. 8.i is then holomorphic, has determinant one, and commutes with common left and right multiplication by elements of $G_{\mathbb{C}}$. For position-dependent weights the assertion applies pointwise; derivative estimates must also include derivatives of the weights. It asserts no global gluing when the frames are not close on overlaps.

There is a simple algebraic readout at a retained junction. If a link with endpoints y, z has actual value V and reduced value Z, and its node factors obey $V = n(y)Zn(z)^{-1}$, then

$$n(z) = V^{-1}n(y)Z. \tag{8.ii}$$

Thus a known factor at one endpoint and the retained link determine the other factor. This identity is exact. Near-identity or complex chart admission for its factors is a separate hypothesis.

The normalized axial gauge need not have a bounded fine-scale derivative. On a flat one-color field $U(x, \mu) = e^{hA_\mu}$, take only $A_1 = X \neq 0$, with X fixed in the small logarithm chart, and a block of physical side one. In its parallel scalar direction the normalized gauge is $g(x) = e^{(x_1 - y_1)X}$, since the centred mean of $x_1 - y_1$ is zero. It makes internal transports trivial. Across the face of two blocks its logarithm has jump magnitude $(1 - h)|X|$ in the branch at zero. The difference divided by h is therefore unbounded as $h \downarrow 0$. The smooth extension in Section 7.8 has different defining conditions.

The comparison with coordinate combs uses endpoint covariance, inversion of paths, bounded path length, and a plaquette filling of the comparison loops inside the complete carrier. These give the one-step analytic bounds and their summable composition. A single prescribed coordinate order does not have the full signed-coordinate symmetry of the contour mean. Its global Borel completion also requires its own definition; neither that completion nor an inverse-operator theorem follows from equality of the flat straight rows.

8.2. **The local second variation of the Wilson action.** The geometric second variation is useful before a gauge condition or a constraint is imposed. Fix a real connection U on a periodic fine lattice of spacing h. An increment per unit length A acts by $U_A(b) = e^{hA(b)}U(b)$ on positive links. On a reversed link use $A(\bar{b}) = -\operatorname{Ad}(U(b)^{-1})A(b)$, so that this convention also describes the inverse of the varied positive link. For an oriented plaquette with edges $b_1, \dots, b_4$, write

$$P = U(b_1) \cdots U(b_4), \qquad T_{r-1} = U(b_1) \cdots U(b_{r-1}), \qquad X_r = h \operatorname{Ad}(T_{r-1})A(b_r). \tag{8.iii}$$

The empty prefix is I. Define Y_r by replacing A with B.

Lemma 8.1 (Plaquette Hessian)**.** Let U be a real connection on a periodic lattice of spacing $h > 0$, and let A, B be real link increments per unit length, extended by $A(\bar{b}) = -\operatorname{Ad}(U(b)^{-1})A(b)$ and the same rule for B. For an oriented plaquette p, use the prefixes and insertions in eq. 8.iii. The

Hessian of $a_p(U) = 1 - \frac{1}{2}\operatorname{tr} P$ in the coordinates $U_A = e^{hA}U$ satisfies

$$D^2 a_p(U)[A,B] = \left\langle \sum_r X_r, \sum_r Y_r \right\rangle + R_p(A,B),$$
$$|R_p(A,B)| \le C\|P-I\| \left(\sum_r |X_r|^2\right)^{1/2} \left(\sum_r |Y_r|^2\right)^{1/2}. \tag{8.iv}$$

Here $\langle X, Y\rangle = -\frac{1}{2}\operatorname{tr}(XY)$. The constant is absolute. The trace formula extends holomorphically to the corresponding complex link chart. ∎

Proof. The varied plaquette is $\prod_{r=1}^{4} e^{sX_r + tY_r} P$. Its mixed derivative at zero is

$$H = \sum_r \tfrac{1}{2}(X_r Y_r + Y_r X_r) + \sum_{r<q} (X_r Y_q + Y_r X_q),$$

followed by multiplication by P. Cyclicity gives

$$\operatorname{tr} H = \operatorname{tr}\left[\left(\sum_r X_r\right)\left(\sum_r Y_r\right)\right].$$

Thus $R_p = -\frac{1}{2}\operatorname{tr}(H(P-I))$. The trace estimate $|\operatorname{tr}(CD)| \le 2\|C\|\|D\|$, followed by Cauchy–Schwarz on the four indices, proves the bound. The formula is a polynomial in the insertions and the plaquette matrix; it involves no complex conjugation. □

Define the transported plaquette curl

$$(\mathcal{C}_U A)(p) = h^{-1} \sum_{r=1}^{4} \operatorname{Ad}(T_{r-1}) A(b_r).$$

Suppose each plaquette has an assigned length $\xi_p \ge h$ and $\|P - I\| \le \varepsilon h^2 \xi_p^{-2}$. Assign a length ξ_b to each positive link and require $\xi_p^{-2} \le C_0 \xi_b^{-2}$ whenever b belongs to p. Set $\|A\|_\xi^2 = \sum_b h^4 \xi_b^{-2} |A(b)|^2$ and $\|F\|_2^2 = \sum_p h^4 |F(p)|^2$. Summing the lemma gives

$$\left| D^2 \sum_p a_p(U)[A,B] - \langle \mathcal{C}_U A, \mathcal{C}_U B\rangle_2 \right| \le C C_0 \varepsilon \|A\|_\xi \|B\|_\xi. \tag{8.v}$$

Indeed $\sum_r |X_r|^2 = h^2 \sum_{b\in\partial p} |A(b)|^2$ by unitarity, and a positive link has six plaquette incidences in four dimensions, counted with multiplicity on a short torus. The two factors of h^2 give exactly the physical measure h^4.

For comparison put

$$(\nabla^+_{U,\mu} A_\nu)(x) = h^{-1}[\operatorname{Ad}(U_\mu(x)) A_\nu(x + he_\mu) - A_\nu(x)], \qquad (\mathrm{c}_U A)_{\mu\nu} = \nabla^+_{U,\mu} A_\nu - \nabla^+_{U,\nu} A_\mu.$$

Multiplication of the four prefixes, with the reverse-link convention, gives the exact identity

$$h(\mathcal{C}_U A - \mathrm{c}_U A)_{\mu\nu}(x) = (I - \operatorname{Ad} P)[\operatorname{Ad}(U_\nu(x)) A_\mu(x + he_\nu) + A_\nu(x)]. \tag{8.vi}$$

For unitary P, $\|I - \operatorname{Ad} P\| \le 2\|P - I\|$. Using $h \le \xi_p$ and the same incidence count yields $\|\mathcal{C}_U A - \mathrm{c}_U A\|_2 \le C C_0^{1/2} \varepsilon \|A\|_\xi$. Consequently, for $0 < \varepsilon < 1/2$,

$$D^2 \sum_p a_p(U)[A,A] \ge (1 - C\varepsilon)\|\mathrm{c}_U A\|_2^2 - C(1 + C_0)\varepsilon \|A\|_\xi^2. \tag{8.vii}$$

This follows by expanding the curl square in eq. 8.v and using $2ab \le \varepsilon a^2 + \varepsilon^{-1} b^2$. For positive plaquette weights w_p fixed as the connection varies, use $\|F\|_{2,w}^2 = \sum_p h^4 w_p |F(p)|^2$. The same pointwise remainders and incidence count give

$$D^2 \sum_p w_p a_p(U)[A,A] \ge (1 - C\varepsilon)\|\mathrm{c}_U A\|_{2,w}^2 - C\|w\|_\infty (1 + C_0)\varepsilon \|A\|_\xi^2.$$

Indeed the summed Hessian has leading term $\|\mathcal{C}_U A\|^2_{2,w}$, and the squared weighted curl error is at most $C\|w\|_\infty C_0\varepsilon^2\|A\|^2_\xi$. The same inequality $2ab \le \varepsilon a^2+\varepsilon^{-1}b^2$ therefore gives the displayed bound. These are local Wilson estimates. They do not identify a constrained Hessian, include multiplier terms, or prove coercivity after a gauge projection.

Covariant curl and divergence. For a real link field A on the full periodic graph, let $S_\mu f(x) = \mathrm{Ad}(U_\mu(x))f(x+he_\mu)$ and $\delta_\mu = h^{-1}(S_\mu - I)$. Thus $\mathrm{c}_U A$ has components $\delta_\mu A_\nu - \delta_\nu A_\mu$, and $D_U^* A = -\sum_\mu \delta_\mu^* A_\mu$ for the physical adjoint. Each S_μ is unitary. Expanding the two squares and subtracting the component-gradient square gives

$$\begin{aligned}\|\mathrm{c}_U A\|_2^2 + \|D_U^* A\|_2^2 - \sum_{\mu,\nu}\|\delta_\mu A_\nu\|_2^2 \\ = 2\,\mathrm{Re}\sum_{\mu<\nu}\langle A_\mu, [\delta_\mu,\delta_\nu^*]A_\nu\rangle_2.\end{aligned} \tag{8.viii}$$

Indeed the diagonal terms $\|\delta_\mu^* A_\mu\|_2^2$ and $\|\delta_\mu A_\mu\|_2^2$ agree because δ_μ is normal. In a pair of distinct directions the divergence contributes the product $\delta_\mu\delta_\nu^*$ and the curl contributes its reversed order with negative sign.

The commutator is $h^{-2}[S_\mu, S_\nu^{-1}]$. Its two paths from x to $x+he_\mu-he_\nu$ bound the backward plaquette in those directions. Their adjoint transports differ in norm by at most twice the plaquette's group defect, since both are unitary and their ratio is a based plaquette holonomy or its inverse. The two link values paired in eq. 8.viii belong to that plaquette. Under the inverse-length comparison assumed in eq. 8.v, each such term is bounded after summation by $CC_0\varepsilon\|A\|^2_\xi$. Here h^{-2} cancels the h^2 in the curvature bound, and the finite incidence count and Cauchy–Schwarz give

$$\left|\|\mathrm{c}_U A\|_2^2 + \|D_U^* A\|_2^2 - \sum_{\mu,\nu}\|\delta_\mu A_\nu\|_2^2\right| \le CC_0\varepsilon\|A\|^2_\xi. \tag{8.ix}$$

On an inner graph the adjoint depends on the boundary convention, so the periodic calculation requires separate boundary terms there.

8.3. **Local curvature coefficients.** Take $0<h\le 1$ in fixed physical units. In the flat parallel frame of section 7.8, write $U_\mu(x) = e^{hA_\mu(x)}$ for the perturbation $e^{hA}W$. The forward difference $\nabla_{h,\mu}$ divides the step in direction μ by h, and $(\mathrm{d}_h A)_{\mu\nu} = \nabla_{h,\mu}A_\nu - \nabla_{h,\nu}A_\mu$. All link components refer to the positive orientations. The parallel frame allows these formulas on a periodic lattice with nontrivial flat holonomy.

Lemma 8.2 (Curvature expansion)**.** Let $0<h\le 1$ and let W be a real flat periodic connection. Work in its parallel frame on the cover, with the boundary identifications determined by its flat holonomy. For a link potential A on a fixed small complex supremum ball set $U = e^{hA}W$ and $F_h(A)(p) = h^{-2}\log U(\partial p)$ in that frame. Then

$$F_h(A) = \mathrm{d}_h A + R_h(A), \qquad \|R_h(A)\|_\infty \le C\|A\|_\infty^2, \qquad \|\nabla_h R_h(A)\|_\infty \le C\|A\|_\infty\|\nabla_h A\|_\infty. \tag{8.x}$$

Here $(\mathrm{d}_h A)_{\mu\nu} = \nabla_{h,\mu}A_\nu - \nabla_{h,\nu}A_\mu$, and ∇_h uses forward differences divided by h. On plaquette fields it compares translates of the same orientation. The analytic ball and the constant are uniform in the mesh, volume and real flat holonomy. ∎

Proof. At $p=(x,\mu,\nu)$ the ordered link logarithms are $ha_1,\dots,ha_4$, where

$$(a_1,a_2,a_3,a_4) = (A_\mu(x), A_\nu(x+he_\mu), -A_\mu(x+he_\nu), -A_\nu(x)).$$

Their sum divided by h is $(\mathrm{d}_h A)(p)$. For a complex parameter z, the logarithm $\log\prod_{j=1}^4 e^{za_j}$ is analytic for $|z|\le 2$ on a sufficiently small fixed ball of four-tuples. Its value and first derivative at $z=0$ are 0 and $\sum_j a_j$. Subtract the linear term and divide by z^2. The result extends analytically at zero and is bounded on the fixed ball for $|z|\le 1$. Its value and differential in the four variables vanish at their common origin. Cauchy's formula and Taylor's formula give a bound $C\max_j|a_j|^2$

for the result and $C\max_j|a_j|$ for its differential, uniformly in z. Use $z=h$ for the curvature. Between two neighboring plaquettes of the same orientation the four inputs change by at most $h\|\nabla_h A\|_\infty$. Integrating the derivative bound on the segment of inputs and dividing by h proves the last estimate. The same calculation bounds the first and second derivatives of R_h in supremum norm by $C\|A\|_\infty$ and C, respectively. The first derivative of F_h in direction B is therefore bounded by $C(\|\nabla_h B\|_\infty+\|A\|_\infty\|B\|_\infty)$. □

For a plaquette logarithm h^2F, the odd Wilson factor in the skew-Hermitian convention is

$$\mathcal{S}_h(F)=\frac{e^{h^2F}-e^{-h^2F}}{2h^2}.$$

It is the matrix sine in the Hermitian convention multiplied by i. Write $\mathcal{S}_h(F)=F+S_h(F)$. For $\|h^2F\|_\infty$ bounded by a fixed constant, its series gives

$$\|S_h(F)\|_\infty\le Ch^4\|F\|_\infty^3,\qquad \|\nabla_h S_h(F)\|_\infty\le Ch^3\|F\|_\infty^3. \tag{8.xi}$$

For the second inequality subtract two values of the remainder and use the first inequality at both plaquettes before dividing by h.

The exact Wilson current. For a real connection U on the full periodic lattice, let $J_{\mathrm{left}}(U)$ represent the first variation of the Wilson energy from Section 4:

$$\frac{d}{dt}\mathcal{E}(e^{thB}U)\Big|_{t=0}=\langle J_{\mathrm{left}}(U),B\rangle_2.$$

Here B is a real positive-link field per unit length. Use the bilinear colour pairing with weight h^4 on both links and plaquettes, and let $^\sharp$ denote its transpose. For $P_p=U(\partial p)$, put $\mathcal{S}_U(p)=(P_p-P_p^{-1})/(2h^2)$, a $\mathfrak{g}$-valued plaquette field. Then

$$J_{\mathrm{left}}(U)=\mathcal{C}_U^\sharp\mathcal{S}_U=\mathrm{c}_U^\sharp\mathcal{S}_U. \tag{8.xii}$$

When $P_p=e^{h^2F(p)}$, this factor is $\mathcal{S}_U(p)=\mathcal{S}_h(F(p))$.

To prove the identity, the insertion formula eq. 8.iii gives $P_p'=h^2(\mathcal{C}_UB)(p)P_p$. The Cayley–Hamilton identity $P_p+P_p^{-1}=(\operatorname{tr}P_p)I$ and the tracelessness of $\mathcal{C}_UB$ therefore give

$$\frac{d}{dt}\mathcal{E}(e^{thB}U)\Big|_{t=0}=h^4\sum_p\langle(\mathcal{C}_UB)(p),\mathcal{S}_U(p)\rangle.$$

This proves the first equality in eq. 8.xii. For the second, pair eq. 8.vi with $\mathcal{S}_U$. Its right side pairs to zero because $\mathrm{Ad}(P_p^{-1})\mathcal{S}_U(p)=\mathcal{S}_U(p)$. The same algebra proves the identity on complex link charts, using the bilinear transpose. No plaquette logarithm is needed for the formula in terms of $\mathcal{S}_U$.

The derivative in normal coordinates at a nonzero potential uses the exponential differential E_z from eq. 7.iii. Its transpose for the invariant bilinear colour pairing, extended over $\mathbb{C}$, is

$$E_z^\sharp B=\int_0^1\mathrm{Ad}(e^{-tz})B\,dt. \tag{8.xiii}$$

This follows by moving the adjoint action across the trace pairing inside the defining integral. The normal-coordinate variation $A+sB$ gives left variation $E_{hA}B$ at $U=e^{hA}W$ when both variations are measured per unit length. Consequently, eq. 8.xii gives the Wilson current in normal coordinates as

$$J_{\mathrm{normal}}(A)=E_{hA}^\sharp\mathrm{c}_U^\sharp\mathcal{S}_U,\qquad U=e^{hA}W.$$

The same transpose rule converts any differential represented in the left variable to the normal variable. The integral formulas give

$$\|E_{hA}-I\|+\|E_{hA}^\sharp-I\|\le Ch\|A\|_\infty$$

on a fixed potential ball, and their first potential derivatives are bounded by Ch in the corresponding supremum operator norm.

The transported plaquette curl $\mathcal{C}_U$ also has local background derivatives. Treat $U = e^{hA}W$, with W fixed and real. In each coefficient of $\mathcal{C}_U$ the adjoint transport is a product of at most four factors associated with links or their inverses. For a reversed insertion, the fourth factor may be the inverse of the positive link at which the input is stored. With $U = e^{hA}W$, each varying adjoint factor is an exponential $\exp(\pm h \operatorname{ad} A(b))$, possibly preceded or followed by a fixed unitary adjoint action. Its first and second derivatives have factors h and h^2. The prefactor h^{-1} in the curl leaves the bounds

$$\begin{aligned} \|\mathcal{C}_U - \mathcal{C}_W\| &\le C\|A\|_\infty, \\ \|D_A\mathcal{C}_U[A_1]\| &\le C\|A_1\|_\infty, \\ \|D_A^2\mathcal{C}_U[A_1, A_2]\| &\le Ch\|A_1\|_\infty\|A_2\|_\infty. \end{aligned} \tag{8.xiv}$$

The first estimate follows by integration from $A = 0$. These are coefficient bounds and also operator bounds from links to plaquettes in supremum norm and in the physical L^2 norms with measure h^4. Each row involves at most four links, and each link occurs in at most six plaquette positions, counted with multiplicity. The same proof applies to complex A in the fixed small ball, since the varying factors are uniformly bounded there. For flat W, $\mathcal{C}_W = \mathrm{d}_h$ in its parallel frame.

Lemma 8.3 (Quadratic Wilson current remainder)**.** Let W be real and flat on the full periodic fine lattice, with $0 < h \le 1$, and use its parallel frame with the deck law. For $0 < \alpha < 1$, take X_α from eq. 7.xxix. The normal current of $e^{hA}W$ has a holomorphic remainder

$$\mathcal{N}_h(A) = J_{\mathrm{normal}}(A) - \mathrm{d}_h^\sharp \mathrm{d}_h A$$

on a fixed small complex X_α ball. There are $r_0 > 0$ and C such that, for $0 < r \le r_0$ and $\|A\|_{X_\alpha}, \|\widetilde{A}\|_{X_\alpha} \le r$,

$$\|\mathcal{N}_h(A)\|_\infty \le C\|A\|_{X_\alpha}^2, \qquad \|\mathcal{N}_h(A) - \mathcal{N}_h(\widetilde{A})\|_\infty \le Cr\|A - \widetilde{A}\|_{X_\alpha}. \tag{8.xv}$$

The radii and constants are uniform in the mesh, volume and real flat holonomy. ■

Proof. Put $U = e^{hA}W$. The flat curl transpose is a sum of backward differences in this frame. The formula for the forward covariant curl shows that $\|\mathrm{c}_U^\sharp - \mathrm{d}_h^\sharp\|_{\infty\to\infty} \le C\|A\|_\infty$. Each translated coefficient changes by a factor of the form $h^{-1}(\mathrm{Ad}(e^{-hA(b)}) - I)$. Write $F = F_h(A)$ and $R = R_h(A)$. The exact current identity gives

$$J_{\mathrm{left}}(U) - \mathrm{d}_h^\sharp \mathrm{d}_h A = \mathrm{d}_h^\sharp R + (\mathrm{c}_U^\sharp - \mathrm{d}_h^\sharp)F + \mathrm{c}_U^\sharp S_h(F).$$

The first two terms are bounded by $C\|A\|_{X_\alpha}^2$ by eq. 8.x. Split the last transpose as its flat value plus the same coefficient difference. Equation 8.xi bounds this term by $C(h^3 + h^4\|A\|_\infty)\|F\|_\infty^3$, which is absorbed on the small ball.

The exact normal-coordinate factor is $E_{hA}^\sharp$. It is uniformly bounded, and its difference from I is at most $Ch\|A\|_\infty$. The difference applied to $\mathrm{d}_h^\sharp \mathrm{d}_h A$ is therefore bounded by $C\|A\|_\infty\|\nabla_h A\|_\infty$, using $\|\mathrm{d}_h^\sharp \mathrm{d}_h A\|_\infty \le Ch^{-1}\|\nabla_h A\|_\infty$. This proves the quadratic bound. All these expressions and estimates hold on a fixed complex ball. Cauchy's formula in two scalar directions bounds the second derivative on a smaller ball. The quadratic bound gives $D\mathcal{N}_h(0) = 0$; integrate the derivative along segments in the radius-r ball to obtain the Lipschitz estimate. □

8.4. **Averaging penalty ranges.** Choose a real reference U admitting the row charts. Let q and Q_U^{st} be its transported site row and its straightened link derivative per unit length at zero perturbation. Denote their coarse site and link index sets by $\mathcal{Y}$ and $\mathcal{L}$. Use the transposes for the physical pairings and write $D = D_U$. For the scalar and vector penalties

$$V_s = q^\sharp q, \qquad V_v = (Q_U^{\mathrm{st}})^\sharp Q_U^{\mathrm{st}} - Dq^\sharp q D^\sharp$$

one has

$$\operatorname{rank} V_s \le 3|\mathcal{Y}|, \qquad \operatorname{rank} V_v \le 3(|\mathcal{Y}| + |\mathcal{L}|). \tag{8.xvi}$$

Indeed, the scalar range lies in the range of $q^\sharp$, and the vector range lies in the sum of the ranges of $(Q_U^{\mathrm{st}})^\sharp$ and $Dq^\sharp$. Each coarse coordinate has three colour components. The argument also applies to the bilinear complex transposes. In a translation-invariant coarse Fourier fibre with one site and four positive links per cell the two bounds are 3 and 15.

8.5. **A constant Abelian curvature.** On the covering lattice, fix $F \in \mathfrak{g}$ and $t \in \mathbb{R}$. In the $1,2$ plane consider the two potentials

	$A_1(x)$	$A_2(x)$
linear	$-tx_2F$	0
symmetric	$-\frac{1}{2}tx_2F$	$\frac{1}{2}tx_1F.$

The other two components vanish. All link logarithms lie on the same Abelian line. With $g(x) = \exp(-tx_1x_2F/2)$, the finite gauge action on the linear links $e^{hA_\mu(x)}$ gives exactly the symmetric links, since the commuting site difference adds the forward difference of $tx_1x_2F/2$ to each potential. In each gauge the $1,2$ plaquette holonomy is e^{h^2tF} by multiplication of the four commuting link factors. The factor h^2 is the plaquette area. The adjoint action of g at the site or link tail preserves every fixed coordinate support. A covariant endomorphism of the site or link space therefore transforms by unitary conjugation, as does its compression to such a support.

There is one plaquette of this orientation per cell of volume h^4. Since the eigenvalues of F are $\pm i|F|$, the Wilson contribution per unit physical volume is

$$h^{-4}(1 - \cos(h^2t|F|)) = \tfrac{1}{2}t^2|F|^2 + O(h^4t^4|F|^4). \tag{8.xvii}$$

In particular the quadratic coefficient retains its normalization when converting plaquette angles to physical curvature.

9. Coordinates on a fixed-average fibre

Use units in which the fine spacing of the step is one. Let E be a finite set of positive fine bonds and C a set of positive coarse bonds whose complete averaging paths are contained in E with either orientation. For each $c = (y, y + Le_\mu) \in C$, write D_c for the rectangular patch formed by its two blocks. On a torus we assume that the vertices of each such rectangle have distinct projections. We use the bonds of its rectangular lift for all path calculations, including when the chosen rectangle crosses a periodic seam. The path support of row c will be denoted $E_c \subset E$. In particular, E_c need not contain every physical bond joining two projected vertices of D_c.

Fix a background V on E. For each c, suppose there is a unitary gauge s_c on the rectangle such that

$$\max_{b \subset D_c} \|V^{s_c}(b) - I\| \le \epsilon. \tag{9.i}$$

The gauges for different rows can be chosen separately. By the local comb estimate of Lemma 3.3, a real field with plaquette defect at most ω in these rectangles admits eq. 9.i with $\epsilon \le CL\omega$. For a complex perturbation $V = e^X V_0$ of such a real field, the same real gauges give $\epsilon \le C(L\omega + \|X\|_\infty)$ when the right side is small. Coarse links in their original frames can have nontrivial holonomy.

Use the analytic map $\mathcal{M}_{\mathrm{an}}$ from eq. 3.iv on the paths of these rows. Put $W = \mathcal{M}_{\mathrm{an}}(V)|_C$ and use the corresponding plain coordinate on the rows in C. Write

$$P_V(A)(c) = \log\left[\mathcal{M}_{\mathrm{an}}(e^A V)(c) W(c)^{-1}\right], \qquad K_V = DP_V(0), \qquad N_V(A) = P_V(A) - K_V A. \tag{9.ii}$$

Here A takes values in $\mathfrak{g}_{\mathbb{C}}$, or in $\mathfrak{g}$ on the real chart, and each exponential multiplies its reference link on the left. Thus $P_V(0) = 0$ throughout the analytic reference domain. All averaging expressions in this section use $\mathcal{M}_{\mathrm{an}}$ until the real admission condition below is imposed; on that real chart they agree with the global Borel map $\mathcal{M}$. Norms of fields are suprema of the matrix operator norm.

In this section $C_0, C_1, \dots$ and $c_0, c_1, \dots$ are positive constants independent of the lattice, the row selection, and $L \geq 3$. Constants denoted C_L may depend on L.

For a chart of the real Borel map, suppose also that V is defined on every link used in the first-branch plaquette tests and that

$$\max_{p \text{ tested by } C} \|V(\partial p) - I\| \leq \frac{1}{400L^2}. \tag{9.iii}$$

A test on a projected double block can include plaquettes outside its rectangular lift. Links of those plaquettes outside E are held fixed. When every tested plaquette lies in the rectangular lift, eq. 9.i gives a defect at most 4ϵ on each test, by unitary telescoping. Thus eq. 9.iii follows on this stock for a sufficiently small ϵ of order L^{-5}.

9.1. **The local Taylor bounds and the pivot coefficient.**

Lemma 9.1 (Taylor bounds in physical bond logarithms)**.** If eq. 9.i holds in unitary local gauges s_c with $\epsilon \leq c_0/L$, the expression eq. 9.ii using eq. 3.iv is holomorphic for $\|A\|_\infty < c_0/L$. Each row depends only on $A|_{E_c}$ and $V|_{E_c}$. On this ball,

$$\|K_V\| \leq C_0 L, \qquad \|N_V(A)\|_\infty \leq C_0 L^2 \|A\|_\infty^2, \qquad \|DN_V(A)\| \leq C_0 L^2 \|A\|_\infty. \tag{9.iv}$$

In a local unitary frame as in eq. 9.i, the row of $K_{V^{s_c}} - K_I$ has operator norm at most $C_0 L^2 \epsilon$ on the local supremum space. ■

Proof. In the frame s_c put $V^{s_c} = e^Z$, with $\|Z\|_\infty \leq 2\epsilon$. A product of at most $2L$ factors $e^{\pm Z_b}$ or $(e^{A_b} e^{Z_b})^{\pm 1}$ is within $\exp(C_1 L(\|Z\|_\infty + \|A\|_\infty)) - 1$ of I. This follows by telescoping the product and bounding each factor by its exponential norm bound. It puts every internal contour family in the fixed local domain of the contour mean. In eq. 3.iv the subsequent operations are products of a bounded number of such near-identity matrices, local logarithms, and a normalized sum. Hence $P_{e^Z}(A)(c)$ is bounded by a universal constant on joint balls in Z, A of radius c_1/L.

Apply the scalar Cauchy formula on a smaller ball in two variables, along directions of supremum norm at most one. The circles can have radius c_2/L. This bounds $D_A^2 P_{e^Z}$ and $D_Z D_A P_{e^Z}$ by $C_2 L^2$. One circle bounds $D_A P_{e^Z}$ by $C_2 L$. Integration from zero to A and from zero to Z gives the stated bounds and the comparison with K_I.

By gauge covariance of the average, a left fluctuation transforms by $A^{s_c}(b) = \mathrm{Ad}(s_c(b_-))A(b)$. The relative logarithm transforms by $\mathrm{Ad}(s_c(y))$. These conjugations have norm one. The holomorphic formulas therefore give the same estimates in the original frame. This also proves locality, since every operation used only the path variables in the row. □

The pivot of $c = (y, y + Le_\mu)$ is the positive fine bond

$$b_c = (y + me_\mu, y + (m+1)e_\mu).$$

It is the central bond crossing from the first block to the second. An internal contour never crosses a block face. A parallel segment from x to $x + Le_\mu$, $x \in B(y)$, crosses this face exactly once. Thus the only pivot in E_c is b_c. This path argument also applies when two opposite faces have the same pair of neighbouring coarse sites on a short periodic lattice.

Let $P = \{b_c : c \in C\}$ and $E_f = E \setminus P$. The map $c \mapsto b_c$ is injective, since the direction and initial point determine c. Write π for restriction to E_f and ι for extension from E_f by zero. At the identity, only the parallel segments in the flat derivative eq. 3.v can contribute a pivot derivative. Exactly L of them contain b_c, one for each longitudinal starting coordinate with fixed transverse coordinates y_ν for $\nu \neq \mu$. Each contributes with positive sign and weight L^{-4}. Therefore, with

$$J_c(V) = D_{A(b_c)} P_V(0)(c), \qquad J_c(I) = L^{-3} I_{\mathfrak{g}_\mathbb{C}}, \tag{9.v}$$

the restriction of K_V to P is diagonal in the bond index. In a flat field on the rectangle the coefficient in the original frame is $L^{-3}\,\mathrm{Ad}(V[y, b_{c,-}])$, where $V[y, b_{c,-}]$ is the prefix along the central segment. This follows from eq. 9.v by the gauge change in Lemma 9.1.

Lemma 9.2 (Right inverse on the pivots)**.** Under eq. 9.i in unitary local gauges s_c, with $\epsilon \le c_3 L^{-5}$, each $J_c(V)$ is invertible. The maps

$$(H_V v)(b_c) = J_c(V)^{-1} v(c), \qquad (H_V v)|_{E_f} = 0, \qquad \mathsf{C}_V = \iota - H_V K_V \iota \tag{9.vi}$$

satisfy $K_V H_V = I$, $K_V \mathsf{C}_V = 0$, $\pi \mathsf{C}_V = I$, and $\|H_V\| \le 2L^3$, $\|\mathsf{C}_V\| \le C_3 L^4$. The image of C_V is $\ker K_V$. Both maps are holomorphic in the admitted backgrounds and have local rows. ■

Proof. The frame used in eq. 9.i puts $J_c(V)$ within $C_0 L^2 \epsilon$ of $L^{-3} I$, up to the unitary transformations at its source and target. Choose c_3 so that the relative perturbation in that frame is less than $1/2$. A Neumann series gives the inverse, its bound and holomorphic dependence. The path support and eq. 9.v show that each pivot inverse acts on its own row. The identities and operator bounds follow from eq. 9.vi and Lemma 9.1. Finally a vector in $\ker K_V$ is uniquely determined by its restriction to E_f, because the difference of two such vectors with the same restriction is supported on P, where K_V is invertible. □

9.2. **Exact straightening and the fibre graph.**

Theorem B (One-step fibre chart)**.** Under the hypotheses of Lemma 9.2, there is a holomorphic change of variables $A = \Psi_V(t)$ for $\|t\|_\infty < c_4 L^{-5}$, with

$$\pi \Psi_V(t) = \pi t, \qquad P_V(\Psi_V(t)) = K_V t, \qquad \|D\Psi_V(t) - I\| \le C_4 L^5 \|t\|_\infty. \tag{9.vii}$$

The inverse on the image is $A \mapsto A + H_V N_V(A)$. For $\|u\|_\infty < c_5 L^{-9}$ the map

$$\Phi_V(u) = \Psi_V(\mathsf{C}_V u) \tag{9.viii}$$

gives the unique solution of $P_V(A) = 0$, $\pi A = u$ in a full logarithm ball of radius $c_6 L^{-5}$. It obeys $\|\Phi_V(u)\|_\infty \le C_5 L^4 \|u\|_\infty$ and $\|D\Phi_V(u)\| \le C_5 L^4$. Its pivot at b_c depends only on $V|_{E_c}$ and $u|_{E_c \cap E_f}$. The chart is real when V, u are real. ■□

Proof. For t in the stated ball, solve for a field d on C by

$$d = N_V(t - H_V d), \qquad \Psi_V(t) = t - H_V d. \tag{9.ix}$$

For fixed t, restrict to $\|d\|_\infty \le C_6 L^2 \|t\|_\infty^2$ with $C_6 \ge 4C_0$. Decrease c_4 so that $\|H_V d\|_\infty \le \|t\|_\infty$ and $2\|t\|_\infty < c_0/L$. The second bound in eq. 9.iv makes this ball invariant. The last bound there gives a Lipschitz constant at most $4C_0 L^5 \|t\|_\infty < 1/2$. Iteration from zero gives a locally uniform holomorphic limit in t, V. The real subspace is preserved by the iteration.

Substitution into $P_V(A) = K_V A + N_V(A)$ and $K_V H_V = I$ prove the first two identities in eq. 9.vii. Differentiating eq. 9.ix gives

$$D_t d = (I + DN_V(\Psi_V(t)) H_V)^{-1} DN_V(\Psi_V(t)).$$

The inverse in this formula has norm at most two. This proves the derivative bound. The formula for the inverse of Ψ_V follows because $d = N_V(\Psi_V(t))$.

Choose c_6 so that $\|H_V\| \sup_{\|A\| < c_6 L^{-5}} \|DN_V(A)\| < 1/2$, and then choose c_5 so that eq. 9.viii is in this ball and $\mathsf{C}_V u$ is in the domain of Ψ_V. This is possible by Lemma 9.2 and the bounds above. Two solutions A, A' in the full ball with the same u have a pivot-supported difference. Their constraint implies $A - A' = -H_V(N_V(A) - N_V(A'))$. The strict derivative bound on this convex ball gives $A = A'$. Each equation for $d(c)$ involves only the free part of t on E_c and the single pivot b_c. This proves the stated support property.

For real arguments satisfying eq. 9.iii, the change in a tested plaquette product is at most $4\|A\|_\infty$. Indeed, telescope its four unitary factors and use $\|e^{A(b)} - I\| \le \|A(b)\|$, with the same bound on a reversed factor. Decrease c_4, c_6, and then c_5 accordingly, so that $4\|A\|_\infty < 1/(400L^2)$ on the logarithm balls used here. Then the chart lies in the first branch of the real map. The larger balls in Lemma 9.1 serve to estimate the analytic formula. □

For nearby targets write $t = \mathsf{C}_V u + H_V v$ and set

$$\Theta_V(u,v) = \Psi_V(\mathsf{C}_V u + H_V v). \tag{9.x}$$

For $\|u\|_\infty < c_7 L^{-9}$ and $\|v\|_\infty < c_7 L^{-8}$ the bound on t is $(C_3+2)c_7 L^{-5}$. Choose $c_7 \le c_5$ so that this is below both $c_4 L^{-5}$ and $\frac{1}{2}c_6 L^{-5}$. The contraction proof gives $\|\Theta_V(u,v)\|_\infty \le 2\|t\|_\infty$. Hence

$$(\pi, P_V)(\Theta_V(u,v)) = (u,v), \qquad \mathcal{M}_{\mathrm{an}}(e^{\Theta_V(u,v)}V)(c) = e^{v(c)}W(c).$$

These equations give the inverse coordinates on the image. For real V, u, v with eq. 9.iii, the four-link bound in the proof of Theorem B puts every tested plaquette in the first branch. The identity then holds for the real Borel map $\mathcal{M}$. At $v = 0$ the coordinate is $\Phi_V(u)$. For the fibre graph in Theorem B, the substitution $u = gB$, $g > 0$, gives radius $c_5 L^{-9}/g$ in B.

The first two derivatives of the graph at the reference point are

$$D\Phi_V(0) = \mathsf{C}_V, \qquad D^2\Phi_V(0)[u_1,u_2] = -H_V D^2 P_V(0)[\mathsf{C}_V u_1, \mathsf{C}_V u_2]. \tag{9.xi}$$

Indeed, the second derivative vanishes on the free bonds and is determined on the pivots by twice differentiating the constraint. On smaller uniform balls all fixed-order derivatives of each output row have bounds depending only on L and the order. The Cauchy formula on scalar circles gives these bounds from the holomorphic chart, since each circle can be chosen using the same margin in the supremum radius. At any fixed order, the number of input bonds in a row and the number of rows containing an input bond are bounded in terms of L. This gives volume-independent absolute row and column bounds for the coordinate kernels of the derivatives.

The same differentiation identifies the second variation of a scalar function along a constrained chart. More generally, let $P(0) = 0$, let $K = DP(0)$ have a specified right inverse H, and let C parametrize $\ker K$. For a chart $\Phi(u) = Cu - HN(\Phi(u))$ with $\Phi(0) = 0$ and $N = P - K$, differentiation gives

$$D\Phi(0) = C, \qquad D^2\Phi(0)[u,v] = -HD^2P(0)[Cu, Cv].$$

For any twice differentiable scalar S,

$$D^2(S\circ\Phi)(0)[u,v] = D^2S(0)[Cu,Cv] - DS(0)\big[HD^2P(0)[Cu,Cv]\big]. \tag{9.xii}$$

This is the ordinary second chain rule. If $DS(0) = 0$, the second term vanishes. In a different gauge chart the right inverse H must be supplied and bounded for that chart. The pivot inverse above supplies this formula for the one-step plain constraint.

Suppose instead that eq. 9.i holds in complex gauges s_c whose condition numbers satisfy

$$\|s_c(x)\|\|s_c(x)^{-1}\| \le \kappa, \qquad \kappa \ge 1.$$

Retain $\epsilon \le c_0/L$ for the Taylor estimates and $\epsilon \le c_3 L^{-5}$ for the pivot and fibre graph. Define the continued coarse field in the original frame by

$$W(c) = s_c(c_-)^{-1}\mathcal{M}_{\mathrm{an}}(V^{s_c})(c)s_c(c_+).$$

It agrees with $\mathcal{M}_{\mathrm{an}}(V)(c)$ when the original row chart is also admitted. Compute the relative logarithm in each local frame and conjugate it back by $s_c(c_-)^{-1}$. Two near-identity frames give the same result wherever both admit the calculation. To see this for a contour mean, normalize the family at its first member and start the fixed-point iteration in Lemma 3.1 at zero in both frames. The relative matrices are conjugate at the initial endpoint. The iterates remain conjugate since the logarithm series converges in each frame. The same series argument applies to the closed loops and the output ratio. Thus the continuation is holomorphic on the union of these patches.

Conjugation contributes a factor κ for each input direction and for the output. The radius for the direct Taylor bound becomes $c_0/(\kappa L)$. The operator bounds are

$$\|K_V\| \le C_0\kappa^2 L, \qquad \|DN_V(A)\| \le C_0\kappa^3 L^2\|A\|_\infty, \qquad \|H_V\| \le 2\kappa^2 L^3.$$

The perturbation test for J_c is made inside the near-identity frame, with $\epsilon \le c_3 L^{-5}$. Thus $\|\mathsf{C}_V\| \le C_3\kappa^4 L^4$. The contraction works for $\|t\|_\infty < c_4\kappa^{-5}L^{-5}$. The resulting graph admits the free radius $c_5\kappa^{-9}L^{-9}$ after a fixed decrease of c_5.

The corresponding product chart is also given by eq. 9.x. Its input must satisfy

$$\|\mathsf{C}_V u + H_V v\|_\infty < \min\{c_4, c_6/2\}\kappa^{-5}L^{-5}.$$

Sufficient radii in the original coordinates are $\|u\|_\infty < c_7\kappa^{-9}L^{-9}$ and $\|v\|_\infty < c_7\kappa^{-7}L^{-8}$, after decreasing c_7 so that $(C_3+2)c_7 < \min\{c_4, c_6/2\}$. The bound $\|\Psi_V(t)\|_\infty \le 2\|t\|_\infty$ then places the image in the full uniqueness ball of radius $c_6\kappa^{-5}L^{-5}$.

9.3. **The real fibre density.** Use the real colour inner product $\langle X, Y\rangle = -\frac{1}{2}\operatorname{tr}(XY)$ on $\mathfrak{g}$ and its product Lebesgue measure in the bond variables. For $A \in \mathfrak{g}$, let $r = \|A\|$. Then $A^2 = -r^2 I$ and $e^A = \cos r\, I + (\sin r/r)A$. Under the unit-quaternion identification, normalized Haar measure is round measure on the unit three-sphere divided by its area $2\pi^2$. Its radial factor is $\sin^2 r$, while the Euclidean radial factor is r^2. Consequently, near zero,

$$d(e^A V) = \sigma_0 j(A)\, d^3A, \qquad \sigma_0 = (2\pi^2)^{-1}, \qquad j(A) = (\sin r/r)^2, \tag{9.xiii}$$

with $j(0) = 1$. Right translation by a real V preserves Haar measure.

Proposition 9.3 (Local disintegration of product Haar measure)**.** Let V be real and satisfy eq. 9.iii and the hypotheses of Theorem B. Restrict fine Haar measure to the group-valued image $U = e^{\Theta_V(u,v)}V$ of the product chart in eq. 9.x, with $\|u\|_\infty < c_7 L^{-9}$ and $\|v\|_\infty < c_7 L^{-8}$. Here u is a real raw free-link field and v a real raw coarse-link field. The target is $W' = e^v W$. Relative to Haar measure on this target chart, the unnormalized fibre kernel has density in the u coordinates

$$\mathfrak{d}_V(u,v) = \frac{\sigma_0^{|E_f|}\prod_{b\in E} j(\Theta_V(u,v)(b))}{\prod_{c\in C} j(v(c))\,\prod_{c\in C}\det D_{A(b_c)}P_V(\Theta_V(u,v))(c)}.$$

The density is continuous throughout the real product chart. At the exact target W, $v = 0$ and this yields

$$\sigma_0^{|E_f|}\prod_{b\in E} j(\Phi_V(u)(b))\prod_{c\in C}\det\big(D_{A(b_c)}P_V(\Phi_V(u))(c)\big)^{-1}\prod_{b\in E_f} d^3u_b. \tag{9.xiv}$$

The pivot determinants are positive. Each determinant and each Haar factor at a solved pivot reads only its own row. A free-link Haar factor reads that link. ∎

Proof. The Jacobian of $A \mapsto (\pi A, P_V(A))$, with the free variables ordered first, is block triangular. Its free block is the identity and its pivot block is diagonal in c, with entries $D_{A(b_c)}P_V(A)(c)$. At zero these entries have positive determinant by the invertible straight segment to $L^{-3}I$ in their unitary frames. The frame changes have determinant one. The inequality $\|DN_V(A)H_V\| < 1/2$ preserves invertibility on the whole logarithm ball, so the determinants keep their signs along $A \mapsto sA$, $0 \le s \le 1$.

The real change of variables gives, for every bounded Borel test $F(U, W')$ on this graph,

$$\int_{\text{chart}} F(U, \mathcal{M}(U)|_C)\prod_{b\in E} dU_b = \int_{\text{target chart}} dW' \int_{\text{free ball}} F(e^{\Theta_V(u,v)}V, W')\,\mathfrak{d}_V(u,v)\,du.$$

Here $du = \prod_{b\in E_f} d^3u_b$ and dW' is product normalized Haar on the target links. The fine exponential Jacobian contributes $\sigma_0^{|E|}\prod_b j(A(b))$. The triangular Jacobian contributes the reciprocal pivot determinants. In the coordinates $W' = e^v W$, the target Haar differential is $\sigma_0^{|C|}\prod_c j(v(c))\,dv$, which gives the denominator in $\mathfrak{d}_V$. This construction specifies the continuous density also at individual targets. Setting $v = 0$ uses $\Theta_V(u,0) = \Phi_V(u)$ and $j(0) = 1$, proving eq. 9.xiv.

Let δ_G be the unit Dirac distribution at I for normalized Haar measure, and let $\delta_{\mathbb{R}^3}$ be the unit Dirac at zero for Lebesgue measure. In the real logarithm chart $\delta_G(e^v) = \sigma_0^{-1}\delta_{\mathbb{R}^3}(v)$. Using these factors in the product chart gives the same fibre kernel at W. □

A Borel restriction B of the fine chart multiplies the kernel by $\mathbf{1}_B(e^{\Theta_V(u,v)}V)$. The change-of-variables identity follows by inserting this factor into the test. At a fixed target this convention uses the stated pointwise restriction. It gives a version of the restricted kernel without requiring continuity of the indicator.

Differentiating $(\pi, P_V) \circ \Psi_V = (\pi, K_V)$ gives another expression for the determinant factor in eq. 9.xiv,

$$\det D\Psi_V(\mathsf{C}_V u) \prod_c \det J_c(V)^{-1}.$$

In the convention $Q_V = K_V/L$ for a fluctuation inserted as $e^{A/L}$, the pivot matrix is $J_c(V)/L$. Accordingly,

$$\det J_c(V)^{-1} = L^{-3} \det(J_c(V)/L)^{-1}.$$

The free coordinate substitution $u = gB$ contributes $g^{3|E_f|}$ to eq. 9.xiv.

9.4. **Holomorphic continuation of the density.** Use $z(A) = -\frac{1}{2}\operatorname{tr}(A^2)$ on $\mathfrak{g}_{\mathbb{C}}$. The entire function $(\sin\sqrt{z}/\sqrt{z})^2$, interpreted by its power series at zero, continues j. The reciprocal determinants in eq. 9.xiv are holomorphic on the admitted chart. These continuations give a holomorphic density coefficient. Here a variation of V keeps the relative coordinates fixed: at $v = 0$ the target is the moving value $W(V)$, using the continued coarse field just defined when complex frames are needed. To use a fixed absolute target Y, one must instead admit $v(V) = \log(YW^{-1})$ in the product chart eq. 9.x. The measure and its disintegration in Proposition 9.3 are defined on the real chart.

For local logarithmic densities, each Haar factor has a logarithm with value zero at $A = 0$, from the power series for $\log(1 + (j(A) - 1))$. For the pivot factor normalized at $u = 0$, put

$$B_c(u) = D_{A(b_c)}P_V(\Phi_V(u))(c)\, J_c(V)^{-1}.$$

In the local frame of row c, denote the normalized matrix by $\widetilde{B}_c(u)$. The local Taylor estimate and the pivot inverse give $\|\widetilde{B}_c(u) - I\| \le CL^5 \|\operatorname{Ad}(s_c)\Phi_V(u)\|_{\infty,E_c}$, where the adjoint acts at each fine link's initial point. For unitary frames this is at most $CL^9\|u\|_\infty$. For frames of condition number at most κ it is at most $C\kappa^5 L^9\|u\|_\infty$, by the graph and frame bounds. Thus a fixed decrease of the free radius puts this matrix within $1/2$ of I in either case. The normalized reciprocal determinant has logarithm $-\operatorname{tr}\log\widetilde{B}_c(u)$. Changing frame conjugates this normalized matrix at the coarse initial site, so the trace defines the same logarithm in every admitted frame.

At the reference field, the normalized unperturbed pivot matrix in its local frame is $L^3 J_c = I + S$, with $\|S\| < 1/2$. The imaginary part of $\operatorname{tr}\log(I + S)$ has absolute value less than $3\log 2 < \pi$. The value of this trace is therefore the principal logarithm of the normalized determinant and is independent of the frame. The unnormalized logarithm is $-9\log L + \operatorname{tr}\log(I + S)$. The logarithmic factors normalized at zero have local fixed-order derivative bounds on smaller balls by Cauchy's formula. In unitary frames the bounds depend on L and the order. With complex frame condition number at most κ, the bounds in the original coordinates may also depend on κ. Derivatives in u of a pivot Haar factor or a row determinant read only the free part of that row. A free-link Haar factor reads its own free link, which need not belong to any row. Background derivatives of a row factor may read all of $V|_{E_c}$, including its pivot. The logarithm of each Haar factor has zero derivative at the reference because $j(A) = 1 + O(\|A\|^2)$.

The per-length pivot determinant. Write $m_c(A) = L^{-1}D_{A(b_c)}P_V(A)(c)$ for the pivot derivative in the per-length convention. In the local frame at row c, eq. 9.iv and eq. 9.v give $L^4 m_c(A) = I + S_c(A)$ with

$$\|S_c(A)\| \le C_0 L^5(\epsilon + \|A^{s_c}\|_{\infty,E_c}).$$

Shrink the background and fluctuation ceilings by fixed factors so that the right side is at most $1/8$. The same shrinking is available on the complex-frame balls, since those balls control the increment in each local frame. The product expansion of the three-dimensional determinant, or its three eigenvalues, then gives

$$|L^{12} \det m_c(A) - 1| \le (1 + 1/8)^3 - 1 < 1/2. \tag{9.xv}$$

This determinant is independent of the choice of local frame, since the adjoint changes of colour coordinates at the source and target both have determinant one. Thus

$$\ell_c(A) = 12 \log L - \log(L^{12} \det m_c(A))$$

is holomorphic with the series logarithm about one, $\exp(\ell_c(A)) = (\det m_c(A))^{-1}$, and $|\ell_c(A) - 12 \log L| \le C$. On the real chart this is the positive reciprocal determinant. The coordinate conversions in the real Haar formula are still required when using this per-length factor.

10. Haar normalization of the block gauge

The axial condition may be imposed by changing representatives, or it may be used to weight representatives in a gauge orbit. For the second operation we need the subgroup that acts on the transports. Suppose a family of disjoint blocks has fixed centres. At a selection of sites $x \ne y$ in these blocks allow independent gauge values $g(x) \in G$. The gauge equals I elsewhere. Let $\mathcal{H}$ be this product group with normalized Haar measure. On the invariant domain of real fields where the contour means exist, the selected transports obey

$$T_{U^g}(y, x) = T_U(y, x) g(x)^{-1}.$$

Consequently the map from $\mathcal{H}$ to their new values pushes Haar forward to product Haar. A pinned pair, with both endpoints fixed, has constant transport on each orbit.

Proposition 10.1 (Insertion of normalized weights)**.** Let f_x be nonnegative measurable functions on G of Haar integral one and let $w(U) = \prod_x f_x(T_U(y, x))$ over the selected free sites. Then $\int_{\mathcal{H}} w(U^g)\, \mathrm{d}g = 1$ on the domain above. For a finite measure ν supported on this invariant domain, drawing g with density $w(U^g)$ and replacing U by U^g preserves the integrals of bounded $\mathcal{H}$-invariant functions. If ν is invariant, the resulting measure is $w\,\nu$. ■

Proof. The transport substitutions are independent Haar translations and inversions. For an invariant function the integrand equals its value at U times the density, whose integral is one. For general bounded test functions and invariant ν, change U to U^g at fixed g, then integrate over g. Fubini applies since the weight has orbit integral one. □

10.1. **The quadratic term of the contour weight.** Let $\mathcal{P}$ be the centres of a collection of disjoint blocks. Let W be real with $T_W(y, x) = I$ on the contour pairs used below, and assume their path families have diameter at most $1/80$. This leaves a uniform reserve in the mean chart. The soft contour energy is

$$\mathcal{F}(U) = \sum_{y \in \mathcal{P}} \sum_{x \in B(y) \setminus \{y\}} \left(1 - \tfrac{1}{2} \operatorname{tr} T_U(y, x)\right).$$

For a real raw link increment a and a real parameter t, define the contour differential by

$$C_W a(y, x) = \frac{d}{dt} \log T_{e^{ta} W}(y, x)\Big|_{t=0}.$$

At $W = I$ it is the arithmetic mean of the oriented path sums. In the holomorphic continuation the quadratic coefficient is the bilinear pairing of $C_W a$ with itself. Analyticity of the finite contour map gives, on a fixed smaller ball,

$$\mathcal{F}(e^{ta} W) = \frac{t^2}{2} \sum_{y,x} |C_W a(y, x)|^2 + O_L\left(|t|^3 \sum_{y \in \mathcal{P}} \|a\|^3_{\infty, B(y)}\right). \tag{10.i}$$

The constant is uniform over these real backgrounds and over the number of blocks. To see this, put $Z_{yx}(t) = \log T_{e^{ta}W}(y,x)$. The finite-path estimates give $Z_{yx}(t) = tC_W a(y,x) + O_L(t^2\|a\|^2_{\infty,B(y)})$. For $Z \in \mathfrak{su}(2)$, $1 - \frac{1}{2}\operatorname{tr} e^Z = \frac{1}{2}|Z|^2 + O(|Z|^4)$. Substitution proves the formula; the fixed $L^4 - 1$ contours per block are absorbed in its constant.

For a site field λ supported on the selected free sites (and hence vanishing at the block centres), put $a(b) = \lambda(b_-) - \mathrm{Ad}(W(b))\lambda(b_+)$. Differentiating endpoint covariance gives $C_W a(y,x) = \lambda(y) - \mathrm{Ad}(T_W(y,x))\lambda(x) = -\lambda(x)$. Hence the quadratic form in eq. 10.i is strictly positive on every nonzero such gauge direction. If extra nodes are pinned, restrict to $\lambda = 0$ there as well. This is a positivity statement about the contour weight; identification of all null directions of an action Hessian requires the corresponding action and boundary hypotheses.

The normalized insertion in Proposition 10.1 permits any integrable probability weight on the contour variables. Its validity therefore does not require matching the parameter in that weight to the coupling in the Wilson action.

For instance let $0 < \delta \le 1$ and put

$$z_\delta(s) = \int_G \mathbf{1}_{\{|u-I|<\delta\}} e^{-s(1-\operatorname{tr} u/2)}\,\mathrm{d}u.$$

For real s the integrand divided by $z_\delta(s)$ is such a weight. To control the denominator on complex coupling balls, set $t = 1 - \cos\theta$, where $\theta \in [0,\pi]$ is the class angle of u. The radial density of normalized Haar is $(2/\pi)\sin^2\theta\,\mathrm{d}\theta$, by the unit-quaternion description of G. The cutoff is exactly $t < \delta^2/2$. Hence

$$z_\delta(s) = \frac{2}{\pi}\int_0^{\delta^2/2} e^{-st}\sqrt{t(2-t)}\,\mathrm{d}t. \tag{10.ii}$$

The integral is entire. Fix $s_0 \ge 1$ with $s_0\delta^2 \ge 1$. For $a \in [s_0/2, 3s_0/2]$ it gives $cs_0^{-3/2} \le z_\delta(a) \le Cs_0^{-3/2}$. The lower bound follows by integrating over $0 < t < 1/(2s_0)$, which is inside the cutoff. For the upper bound use $\sqrt{2-t} \le \sqrt{2}$ and extend the resulting integral to infinity.

For $s = a + ib$ with $a > 0$, the ratio $z_\delta(a+ib)/z_\delta(a)$ is the expectation of e^{-ibt} for the probability density proportional to $t^{1/2}e^{-at}w(t)$, where $w(t) = \sqrt{2-t}\,\mathbf{1}_{\{0<t<\delta^2/2\}}$ on $t > 0$, taken to be zero beyond the cutoff. The function w is nonincreasing. For the probability measure μ_a with density proportional to $t^{1/2}e^{-at}$ on $t > 0$, an increasing function f satisfies

$$\int f w\,d\mu_a - \int f\,d\mu_a \int w\,d\mu_a = \frac{1}{2}\iint (f(t)-f(u))(w(t)-w(u))\,d\mu_a(t)\,d\mu_a(u) \le 0.$$

Use $f(t) = t^2$ and divide by the positive integral of w. The resulting second moment is at most $15/(4a^2)$, the second moment of μ_a obtained by two integrations by parts. Since $\cos(bt) \ge 1 - b^2t^2/2$,

$$\operatorname{Re}\frac{z_\delta(a+ib)}{z_\delta(a)} \ge 1 - \frac{15b^2}{8a^2}.$$

If $|s - s_0| \le s_0/2$, then $b^2 \le a^2/3$, since

$$a^2/3 + (a-s_0)^2 - s_0^2/4 = \tfrac{4}{3}(a - 3s_0/4)^2.$$

Thus the ratio has real part at least $3/8$ on this disk. Together with the real lower bound this gives

$$|z_\delta(s)|^{-1} \le Cs_0^{3/2}, \qquad |s - s_0| \le s_0/2. \tag{10.iii}$$

The zero-free disk has a holomorphic logarithm with its real value at s_0. The cutoff is kept at its fixed real value throughout this continuation.

The choice of subgroup affects this identity at a seam. Consider an oriented coarse link leaving a centre y, with target V, and write one determining product as $uSP = V$. Here S is a retained link, P is the subsequent transport to the far endpoint, and u is the contour variable from y to the initial endpoint of S. If that contour variable has weight f, then integration against the Haar constraint

$uSP = V$ gives exactly $f(VP^{-1}S^{-1})$. This follows by right translation in Haar, or by testing the constraint against a bounded continuous function of V. On a line the centered average on a coarse link is its straight transport. The path through any shifted site and back to the terminal centre cancels its backtracks, so every closed loop in eq. 3.iv equals I. For this one-dimensional example, start from product Haar and insert the normalized contour factors. For two completed blocks, take the transports from their centres to their free sites as rooted tree coordinates. The bond joining the two blocks is an independent Haar coordinate. If the coarse output for that internal row is fixed, first integrate this bridge link against its target constraint. The interval product has the bridge link between two fixed transports, so this integration gives one. The only retained bridge is S on the outer seam, where the straight product is uSP. Every unused contour coordinate now contributes the integral of its normalized weight. The remaining constraint therefore gives $f(VP^{-1}S^{-1})$ at fixed V, P, S. Averaging the gauge at y alone sends this factor to one by left Haar translation. Consequently, if the remaining datum measure ν_0 is invariant under this node action, the orbit average of $f(VP^{-1}S^{-1})\nu_0$ is ν_0. This follows by changing the datum by the node gauge in a bounded test integral and then integrating its normalized factor.

A transformation at y alone multiplies this argument on the left. For nonconstant f the resulting factor is in general changed. Using the same gauge value at y and the initial endpoint of S conjugates the argument, preserving it for a class function f. Integrating S against Haar at fixed V, P gives one by the normalization of f.

For an arbitrary finite input measure ν on this invariant domain, let $\overline{\nu} = \int_{\mathcal{H}} (U \mapsto U^g)_* \nu \, dg$. The weighted draw in Proposition 10.1 has measure

$$w\,\overline{\nu}. \tag{10.iv}$$

To verify it, test against a bounded measurable F and write the draw as $\int \nu(dU) \int dg\, (wF)(U^g)$. This is the defining integral of wF against $\overline{\nu}$. The orbit normalization preserves total mass. If some contour pairs are outside their defined domains, omit their weight factors on the invariant domain where this set of pairs is fixed.

The tree slice can have arbitrary prescribed tree transports. On a rooted forest, fix $S_e \in G$ for every tree edge from parent to child. Vary the gauges only at nonroot vertices of the forest, and fix the root and exterior gauges to I. For any real field U, the equations $(U^g)_e = S_e$ have the unique solution $g(x) = S_e^{-1} g(y) U(e)$ along $e = (y, x)$. Let $\pi_T^S U$ denote the resulting field. Conditional on previous tree variables, each equation is a Haar translation and inversion. The Haar delta product therefore gives the point mass at the unique chosen tree gauge. For every bounded Borel test Φ with specified pointwise values, and then for tests integrable against the induced slice measure,

$$\int \prod_{x \neq \text{roots}} dg(x) \prod_{e \in T} \delta_G((U^g)_e S_e^{-1})\, \Phi(U^g) = \Phi(\pi_T^S U). \tag{10.v}$$

There is no determinant from this group-valued tree elimination. For a gauge-invariant Φ its value on the right is $\Phi(U)$. Consequently removing a tree delta gives equality after integration against invariant tests, rather than equality of the two densities as functions on all links.

For non-invariant tests the full averaged density can also be identified. Assume that every tree edge is a variable link. Let dU be product normalized Haar on the variable links, with exterior data held fixed and preserved by the rooted gauges. Write σ_T^S for the measure that fixes each tree link to S_e and keeps product normalized Haar on the off-tree links. For any $\Psi \in L^1(\sigma_T^S)$,

$$\int_{\mathcal{H}_T} (U \mapsto U^g)_* (\Psi\, \sigma_T^S)\, dg = (\Psi \circ \pi_T^S)\, dU, \tag{10.vi}$$

where $\mathcal{H}_T$ consists of the gauges at nonroot vertices, fixed to I at roots and exterior sites. Indeed, each field has a unique representation $U = R^g$, with R in the tree slice. The triangular Haar substitution of Section 5 gives $dU = \sigma_T^S(dR)\, dg$, and uniqueness of tree fixing gives $\pi_T^S(R^g) = R$. Testing against a bounded Borel function proves eq. 10.vi by Fubini, also for complex Ψ. The same

substitution shows that the pullback of a slice-null set is Haar-null, so the density is well defined for the L^1 class of Ψ.

One must also distinguish two uses of a constraint delta. Integrating its Haar target gives $\int_G \delta_G(aY^{-1})\,dY = 1$ for every $a \in G$. Eliminating a fine link through the averaging map instead uses the Jacobian in Proposition 9.3. That derivative is not a Haar translation, and its determinant cannot be discarded.

10.2. **Changes of retained Haar coordinates.** When a constraint is integrated over its output links, it is useful to change the frame of those links while leaving the variables that parametrize the input unchanged. Such a change can be made before any support test is evaluated.

Proposition 10.2 (Change of target frame)**.** Let $V(c)$, $c \in C_{\mathrm{int}}$, be a finite family of target variables with product normalized Haar measure. Hold all other data fixed, including any prescribed targets on a disjoint set C_{fix}. Let u and τ be real site gauges depending only on these fixed data. If those data are subsequently integrated, require this dependence to be measurable. Set $Y = V^u$ on C_{int} and $Y = V$ on C_{fix}, and require u to fix the prescribed target values. The change preserves product Haar measure and rewrites the old and mixed target arguments as

$$V = Y^{u^{-1}}, \qquad V^\tau = Y^{\tau u^{-1}}.$$

For an averaging graph with output $a(c) = \mathcal{M}(v)(c)$, its target delta on each integrated row $c \in C_{\mathrm{int}}$ satisfies

$$\delta_G\big(a(c)(Y^{u^{-1}}(c))^{-1}\big) = \delta_G\big(a^u(c)Y(c)^{-1}\big). \tag{10.vii}$$

In an integrable density on the graph, a factor with old argument $F(V, \xi)$, including its indicator or its domain of definition, becomes $F(Y^{u^{-1}}, \xi)$. A factor with mixed argument keeps $Y^{\tau u^{-1}}$ in that argument. The variables ξ are held fixed in this change. ■

Proof. On each integrated link $Y(c) = u(c_-)V(c)u(c_+)^{-1}$. Left and right translations preserve Haar measure, and composition of gauge transformations gives the order τu^{-1}. The argument of the delta on the right of eq. 10.vii is the conjugate by $u(c_-)$ of the argument on the left. The Haar delta at I is invariant under conjugation. Equivalently, with its Haar target differential, each delta is the point mass at the corresponding output in its chosen frame. On a prescribed target row the same notation is used only for an already defined constraint measure, such as the local fibre kernel of Proposition 9.3. This proves the change for the graph measure at fixed input v. The measurable dependence on any integrated input data then allows Fubini to give the identity after integrating those variables. One may multiply that measure by the full measurable density, including all the tests.

To use this identity as a Haar change of target coordinates in a resolved local chart, additionally require u to be independent of those target coordinates after expressing the input data in that chart. In particular it must not acquire target dependence through a reconstructed pivot. Then substitute $V = Y^{u^{-1}}$ in every reconstructed quantity, including the pivots, τ, and all test and chart domains, while holding the free coordinates fixed. An input-dependent gauge without this extra property still permits the ambient graph identity above, but need not induce an invertible change of target coordinates on the resolved graph. Disjoint target stocks can be changed separately whenever the gauge for each change is independent of the targets being integrated at that step. □

The same rule applies to a prescribed local test when its full domain and its value are gauge invariant. If the test reads all the endpoints of a determining row, its transformed input uses u at both endpoints. Endpoints pinned in the old frame have $u = I$ there. If a second reference and a fluctuating field are written in the same frame, applying the same gauge to both conjugates their relative link at its initial site. The norm of this relative link minus I, and the norm of its logarithm where defined, are unchanged. Consequently the change of frame does not require any of

the tests to pass. It is enough to impose the frame constraints on their actual read sets, including the neighborhoods on which the test is defined.

The fine and target coordinates can also change together. The following identity supplies the corresponding change for the resolved fibre kernel.

Corollary 10.3 (Joint gauge covariance)**.** Let V and its real product chart satisfy the hypotheses of Proposition 9.3. Fix a real fine gauge g, independent of the chart coordinates being integrated, and let $\bar{g}$ be its restriction to the coarse sites. Transform the reference and all input data by these gauges; any data required to remain fixed must be preserved. Put

$$V' = V^g, \qquad u'(b) = \mathrm{Ad}(g(b_-))u(b), \qquad v'(c) = \mathrm{Ad}(\bar{g}(c_-))v(c).$$

The transformed chart has the same radii and satisfies

$$\Theta_{V'}(u', v')(b) = \mathrm{Ad}(g(b_-))\Theta_V(u, v)(b), \qquad \mathfrak{d}_{V'}(u', v') = \mathfrak{d}_V(u, v). \tag{10.viii}$$

The free Lebesgue measure and target Haar measure are preserved. Every extra factor, including a test domain, is pulled back through the inverse change on all of its arguments. ■□

Proof. Replace each row frame s_c by $s_c g^{-1}$. This leaves its smallness bound unchanged. Real branch tests and the supremum balls are preserved because unitary conjugation is an isometry. Endpoint covariance gives $P_{V'}(\mathrm{Ad}_g A) = \mathrm{Ad}_{\bar{g}} P_V(A)$, where each adjoint acts at the tail of its link. Differentiating this identity and using the same pivot stock gives covariance of K_V, H_V, C_V and N_V. The unique fixed point in eq. 9.ix, applied to $t = \mathsf{C}_V u + H_V v$, therefore gives the first identity in eq. 10.viii, including when $v \neq 0$.

The pivot derivative in row c is multiplied on the left by $\mathrm{Ad}(\bar{g}(c_-))$ and on the right by $\mathrm{Ad}(g(b_{c,-})^{-1})$. Both maps have determinant one. All Haar factors in Proposition 9.3 are conjugation invariant. Its density formula proves the second identity. The same determinant-one property preserves free Lebesgue measure, and target links change by left and right Haar translations. Substitution in an integrable test proves the last assertion. □

The identity may be used conditionally when g depends measurably on exterior data held fixed during this change. Its coordinate independence is still required after reconstruction of the fine links.

10.3. **Local reads and plaquette comparison.** Consider a tested level-q plaquette whose links are either given at level q or completed by one step from level $q-1$. The test reads the given links and, for each completed link, the path links and branch-test plaquettes in the projected union of the two level-$(q-1)$ blocks below its endpoints. This includes the links joining the blocks and any opposite-face periodic tests. This follows from Proposition 3.2, including the branch tests. Its whole footprint is within distance Ch_q of each corner, using the directed lift if necessary. In particular, if the test reads a determining row at level at most ℓ and that row is read either directly or by one completion, then $q \leq \ell + 1$.

Here is the distance count used with nested boxes. Suppose that C_i are increasing rectangular boxes, and every carried level-i link in question has one endpoint outside C_i. If the distance from C_{i-1} to C_i^c exceeds h_i, that link cannot have an endpoint in C_{i-1}. Indeed its two endpoints are at distance h_i. At the first outer scale $i = \ell + 1$, the same argument excludes endpoints in the inset of $C_{\ell+1}$ of depth $2h_{\ell+1}$. Assume the separation $\mathrm{dist}(C_{i-1}, C_i^c) > h_i$ at every $i > \ell + 1$. The same inset lies in C_{i-1} at each such level. A cutoff that hides only links reaching this inset can therefore miss only levels at most ℓ. For the distance between two test footprints, set $w = \mathrm{dist}(C_{\ell+1}, C_{\ell+2}^c)$. Suppose their respective points lie within distances a of $C_{\ell+1}$ and b of $C_{\ell+2}^c$. The triangle inequality bounds the length of any connecting path below by $\max\{0, w - a - b\}$. These allowances include any offset of a footprint from its corresponding set, as well as its diameter when the bound is given only for one anchor point. Thus the bound in units of $h_{\ell+1}$ is $w/h_{\ell+1} - C'$ if $a + b \leq C' h_{\ell+1}$. This count concerns the locations read by the tests.

For real fields U, V on the same plaquette, unitarity gives

$$\|U(\partial p)V(\partial p)^{-1} - I\| \le \sum_{b\in\partial p} \|U(b)V(b)^{-1} - I\|. \tag{10.ix}$$

Replace one factor at a time in the ordered product and use unitarity of the other factors. Reversing a link leaves its relative norm unchanged. Hence if the reference plaquette defect is at most ϵ and each of the four relative link defects is at most δ, then the perturbed plaquette defect is at most $\epsilon + 4\delta$. For a link completed by averaging, a raw perturbation $\|a\|_\infty \le \delta$ on its full double block gives a relative defect $C_L\delta$ when the segment $e^{sa}V$, $0 \le s \le 1$, lies in a fixed smaller first-branch chart. Indeed the first-branch formula is uniformly holomorphic in the raw perturbation in the local comb gauge of Lemma 3.3, and integration of its bounded derivative gives the estimate. The real chart must have a buffer on all plaquettes that select the branch. Thus the same four-factor comparison applies when some of the links are completed and some are prescribed. These local bounds apply on a specified support, while a statement about the mass of that support requires estimates for the density.

10.4. **A common tree and the conditional Haar measure.** Let P be an embedded rectangular lift with at most M sites in each direction. Let T be a rooted spanning tree of its literal rectangular graph. Suppose real fields U, V are both equal to I on the tree and their plaquette defects on P are at most ϵ_U, ϵ_V, respectively. For any off-tree bond b of this graph, close b by the two paths from its endpoints to the root. Since the tree links are I, the rooted cycle products are precisely $U(b)$ and $V(b)$. For either field, take its coordinate-comb gauge rooted at the same site. The cycle product is unchanged, and each link in it is within CM times the plaquette defect of I by Lemma 3.3. The rooted cycle has at most $2|P| - 1$ links, counted with their occurrences. Telescoping the unitary product gives

$$\|U(b)V(b)^{-1} - I\| \le C_M(\epsilon_U + \epsilon_V).$$

On the small logarithm branch the same bound holds for $\|\log(U(b)V(b)^{-1})\|$ after enlarging C_M. A prescribed entrance bond common to the two fields may be included in the tree by holding its exterior endpoint fixed. The comparison inside P then uses its internal endpoint as the root.

For a finite forest of variable links with one fixed root in each component, the tree-coordinate substitution of Section 5 gives product Haar in the nonroot site gauges and the variable off-tree links of the slice. A prescribed entrance determines a fixed gauge value at the internal root and supplies no Haar integration variable. Indeed each child gauge is a left translate of the link from its parent, conditional on the preceding tree links. After the tree variables are fixed, all off-tree links undergo independent left and right translations. This remains true conditionally on any fixed exterior data on which the chosen rooted forest depends. If the exterior data are integrated, the forest and the prescribed entrance values must be measurable functions of those data. When the entire integrand is pointwise invariant under these rooted gauges at that fixed exterior, the integrand in the new coordinates is its value on the tree slice. The normalized integrals of the gauge coordinates are then one, giving equality between the unconstrained integral and the slice integral. If the integrand also reads variables attached to other scales, this argument requires invariance of those reads under the lifted site gauges. In particular a gauge which is constant on the fine sites of each coarse pillar induces its prescribed action on the coarse targets by covariance of $\mathcal{M}$. The pins and the support of the gauge must preserve any exterior or auxiliary read that is held fixed.

Appendix A. Index of notation and conventions

Symbol	Meaning	Location
$B(y), B^j(y)$	Centred block and its depth-j version	§2
U^g, F_p	Site gauge action and unscaled curvature	§2
$T_U(y,x)$	Mean of internal transports from y to x	§3
$\mathcal{M}, \mathcal{M}_{\mathrm{an}}$	Real Borel link average and analytic path formula	§3
Q	Flat plain differential in raw logarithms	eq. 3.v
$\varepsilon(U), \mathcal{E}(U)$	Maximum plaquette angle and Wilson energy	§4
$R_W g$	Transported site average of a gauge transformation	eq. 5.i
$F_W(a)$	One-step straightened link coordinate	eq. 5.iv
S_W	Covariant straight-segment row	eq. 5.v
a_j, v_j	Relative link coordinate and accumulated site factor	eq. 5.viii
Ω_j, Γ_j	Separated regions and determining coarse rows	§6.4
ξ	Largest pillar spacing at the endpoints or corners	§6.4
$r_j, \tau_{W,a}$	Effective finite gauge and its infinitesimal site map	eq. 7.ii, eq. 7.iii
D_V	Covariant difference with sign from the gauge action	eq. 7.v
$Q'_j(W)$	Transported site row from depth j	eq. 7.xiv
$\widehat{Q}_j$	Flat straightened link row per unit length	eq. 7.xxvii
X_α, Y_α	Link and site norms controlling differences and Hölder variation	§7.8
E, E_q	Smooth node lift and smooth exact-average lift	§7.8
$\mathcal{C}_U, F_h$	Transported curl and physical plaquette curvature	§8
P_V, K_V	Plain fibre constraint and its differential	eq. 9.ii
$J_c(V)$	Pivot coefficient of a plain row	eq. 9.v
Φ_V	Fixed-average graph over the free links	eq. 9.viii

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